Operations Manual

for the

F20

Thin-Film Analyzer

Revision 8.12.4.0
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Welcome

What is the F20 used for?

The Filmetrics F20 is used to measure the thickness and optical constants (*n* and *k*) of dielectric and semiconductor thin films. Measured films must be optically smooth and within the thickness range set by the system configuration requirements specified in **Performance Specifications**. Commonly measured films include semiconductor process films such as oxides, nitrides, resists, and polysilicon, optical coatings such as hardness and anti-reflection coatings, flat panel display films such as polyimides, resist, and cell gaps, and the various coatings used in CD and DVD manufacture.

Warning: Accurate measurements require suitable recipe settings and a valid baseline. Improper system setup or an aged lamp may also lead to errors. It is the user's responsibility to ensure that this instrument is being used properly for its intended purpose. Please contact **Filmetrics** support for assistance with any questions.

Safety, Maintenance, and Care



This symbol indicates information or instructions that must be read and carefully followed to prevent hazards, injury to the operator, or damage to the instrument.



This symbol indicates a potential shock hazard. Areas marked with this symbol should be serviced by a trained service technician.



This symbol indicates that the product conforms to the WEEE (Waste in Electrical and Electronic Equipment) Directive 2002/96/EC.



This symbol indicates that the product meets the applicable EU safety, health and environmental protection directive requirements.



This label indicates the Model Number and Serial Number of the instrument. This information may be necessary when contacting Filmetrics for assistance.



Disclaimer: Use of this instrument in a manner inconsistent with the information and directions included in this manual may impair the protections designed into the product resulting in danger to the operator or damage to the instrument.



Installation Location: When deciding on a location for the instrument, ensure that the instrument is installed in a well-ventilated location. Enclosing the unit, or blocking the vent holes, may impair the performance of the instrument or damage the internal components.

Cleaning: The need for periodic cleaning of the system is based on the cleanliness of the environment in which the system is installed. Only the external surfaces are to be cleaned. Cleaning and maintenance of all internal components is to be performed by a trained service technician. Do not use aerosol, or spray, cleaners as they may contaminate the sensitive optical surfaces on the instrument.



Before performing any service on the instrument ensure that the main power is disconnected.

Cleanroom, or low lint, wipes should be used to wipe down the system. A mixture of Isopropyl Alcohol (IPA) and De-ionized Water (DI H_2O) at 70% DI H_2O: 30% IPA is to be used for cleaning. Undiluted IPA may be used, if the 70:30 DI and IPA mixture is not available, but extra care must be taken. Avoid using stronger solvents, such as acetone, as these may compromise the surface finish.

Avoid excessive pressure on any surface when cleaning. Excessive force may bend a component, which would damage the system, or cause the system to be out of alignment.

Start with the top of the system and work down. Use slow motions, in a straight line, from the back to the front of the surface being cleaned. Change the cleaning wipe periodically to avoid re-depositing material back onto the surface being cleaned. Allow all cleaned surfaces to dry completely before restoring power to the system.

CE Mandated Warnings



Please read the following instructions carefully to prevent potential shock or fire hazards. This manual should be retained for future use



Bitte lesen Sie die nachstehende Anleitung sorgfältig durch, um Stromschlag und Feuergefahr zu vermeiden. Diese Betriebsanleitung soll für späteren Gebrauch sorgfältig aufbewahrt werden.



Preghiamo di leggere accuratamente, le sequenti Instruzioni, per evitare Prossiomi Incendi e Correnti.



Shock Hazard - Do Not Enter
Achtung Hochspannung - Nicht Berühren
Attenzione Corrent-Forte - Prego non toccare



The front panel's switch is not the power disconnect device. The power cord should be removed after use.
Der Kippschalter an der Vorderseite unterbricht nicht die Stromzuführung. Das Stromkabel soll nach Gebrauch aus dem Gerät herausgezogen werden.
L'interutore nella parete frontale non blocca le Corrente. La Corrente viene.



Never expose the unit to water or liquids. Avoid direct sun.
Bringe das Gerät nicht mit Wasser oder anderen Flüssigkeiten in Berührung. Vermeide direkte Sonneneinstrahlung.
Evitare contatto con acqua oppure liquidi Infiammabili al Macchineggio. Auche entrate di Sole.



Do not insert any objects into the unit.
Keinen Gegenstand in das Gerät einbringen. Do non inseriamo any obietta into gli unità.
Non mettere ogetti dentro la Macchina.



Do not use near open flame or heat.
Das Gerät nicht in der Nähe einer offenen Flamme oder Hitze benutzen.
Non mettere la Macchina vicino a fuochi oppure Riscaldamenti.



The unit should never be enclosed or blocked.
Das Gerät darf nicht eingeschlossen oder blockiert werden.
La Macchina non chinderla per nessun motivo.



Connect unit only to a properly measured supply with a reliable earth ground. Use only the three wire cord which is provided with the unit.
Schließen Sie das Gerät nur an eine ordnungsgemäss vermessene Stromsversorgung an. Verwenden Sie nur ein dreiadriges Kabel, wie es auch mit dem Gerät ausgeliefert wird.
Montare solo con misura normata. Adoperando solo 3 cavi elettrici cosi come e fornita la Macchina.

Table of Contents

1. Software Overview **8**
- 1.1 The Measure Tab 9
- 1.2 The History Tab 12
- 1.3 The Edit Recipe Window 14
- 1.4 Menu Controls 16
- 1.5 Taking a Baseline 31
- 1.6 Setting up the SampleCam 33

2. Editing Film Recipes **35**
- 2.1 Adding, Changing, or Deleting a Structure 35
- 2.2 Film Stack 35
- 2.3 Analysis Options 40
- 2.4 Alarms 45
- 2.5 Acquisition Settings 46

3. Making Measurements **50**
- 3.1 Focusing the Fiber 50
- 3.2 Measuring Thickness 56
- 3.3 Measuring Thickness and Refractive Index 64
- 3.4 Measurement Assumptions 66
- 3.5 Hints for Improved Accuracy 67
- 3.6 Troubleshooting 69

4. Understanding and Evaluating Measurement Results **81**
- 4.1 Display of Measured Data 81
- 4.2 Goodness of Fit 81
- 4.3 Fast Fourier Transform Mode (FFT mode) 82

5. Advanced Features **85**
- 5.1 Creating and Editing n and k Files 85
- 5.2 Contact Probe Template Mode 87

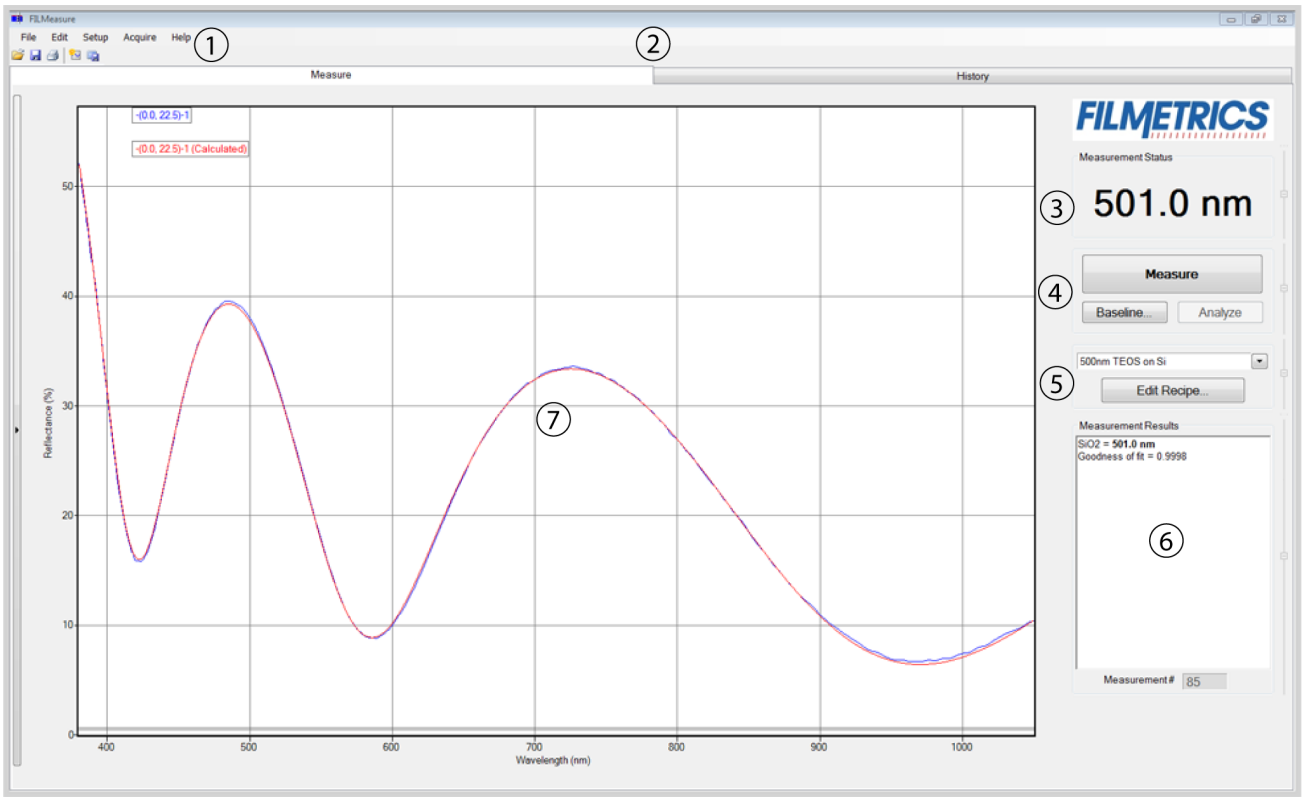
6. Theory of Operation **91**
- 6.1 Measurement Theory 91
- 6.2 Thickness Measurement Details 92
- 6.3 F20 Hardware Operation 92
- 6.4 The Baseline Measurement 93

7. Appendices **95**
- 7.1 Performance Specifications 96
- 7.2 System Configurations 97
- 7.3 Light Bulb Replacement 107
- 7.4 Software Automation 108
- End User Software License 140

1. Software Overview

The way that light reflects off a thin film is determined by the characteristics of the film, such as its thickness, optical constants, and roughness. The F20 is able to determine thin-film characteristics by first carefully measuring the amount of light reflected from the thin film over a range of wavelengths (i.e., by measuring the reflectance spectrum). It then analyzes this data by comparing it to a series of calculated reflectance spectra. Most of the features of the FILMeasure software that runs the F20 can be divided into reflectance acquisition and reflectance analysis functions. The following pages outline the main features of the FILMeasure software.

1.1 The Measure Tab



1. Standard Windows menus: **File**, **Edit**, **Setup**, **Acquire**, and **Help.**

2. Click on these tabs to select between the **Measure** and **History** windows.

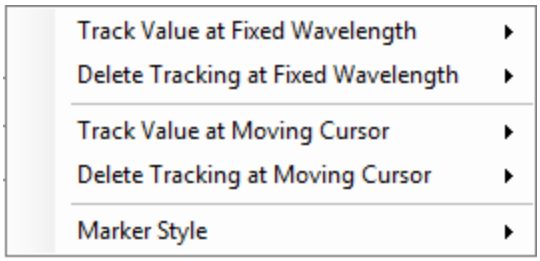
3. Displays the thickness of the film being solved for, as well as whether a sample is good or bad based on the settings selected in the **Alarms** tab.

4.**Measure** a sample, **Baseline** the system, or **Analyze** an already opened spectrum. If the **Measure** button is grayed out, the system will require a baseline before further measurements may be taken.

5. Select saved recipes from the drop-down list, or open the **Edit Recipe** dialog box.

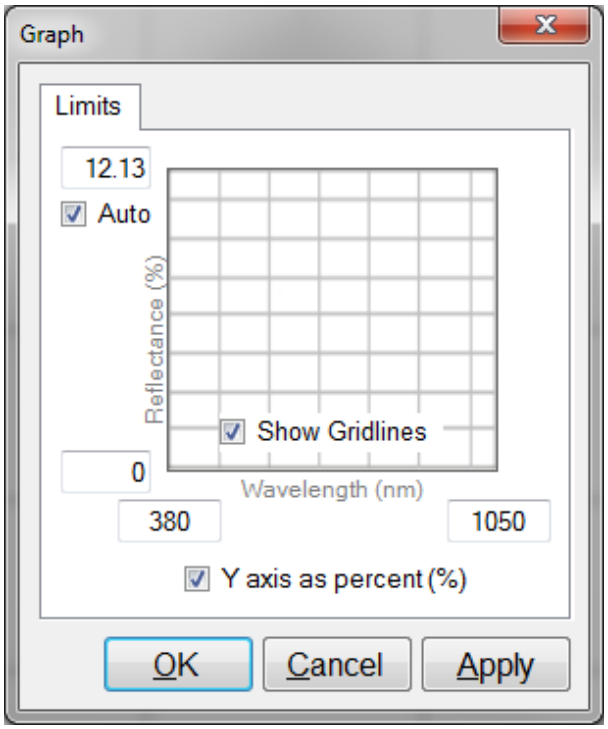
6. Details about the calculated thickness, as well as any additional parameters that were solved for, including **Roughness**, *n* and *k*, and **Nonuniformity**, and the **Goodness of Fit (GOF)** are displayed here.

7. Graphical display for spectra. A click of the right mouse button while the cursor is within the graphical display opens the menu below for easy reading of cursor values in the **Measure Tab**. Use the keyboard up/down and right/left arrows to move the marker to a desired location.



The graph limits can be changed by double-clicking on the graph display. The text boxes along the bottom of the graph control the wavelength range, in nanometers, displayed on the screen. For systems with multiple spectrometers, the software will automatically resize to match anytime the selected spectrometer is changed.

The two boxes on the left represent the *Reflectance (%)* range, with the max being set in the upper box and the min being set in the lower. The boxes on the right are for *Transmittance (%)*(requires the **T-1 upgrade**). Check the **Auto** checkbox to activate y-axis autoscaling. The lower value is always 0 for autoscaling; the maximum y-axis upper limit is 5000. The axes can be set to display in percent or as a decimal value (100% = 1).

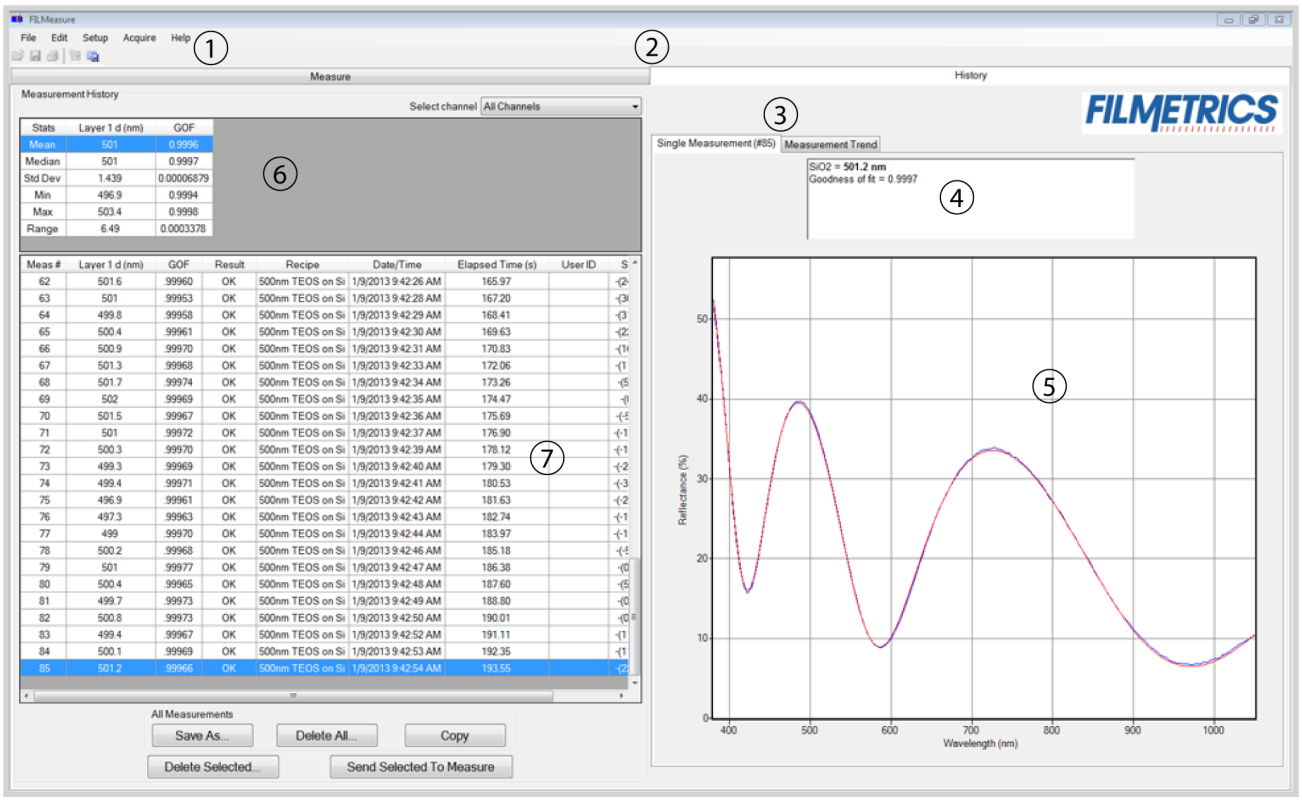


Graph Options can be accessed and edited by double-clicking on the main graph window.

8. Controls and settings for the camera are available here (SampleCam required). By clicking **Show Window**, the video image will launch in separate non-modal window. This allows the user to maximize the graph space available by either minimizing the camera window, or moving it to a seperate monitor if available.


Live Video
SiO2 on Si
Reflectance
Standard
1979.1 A

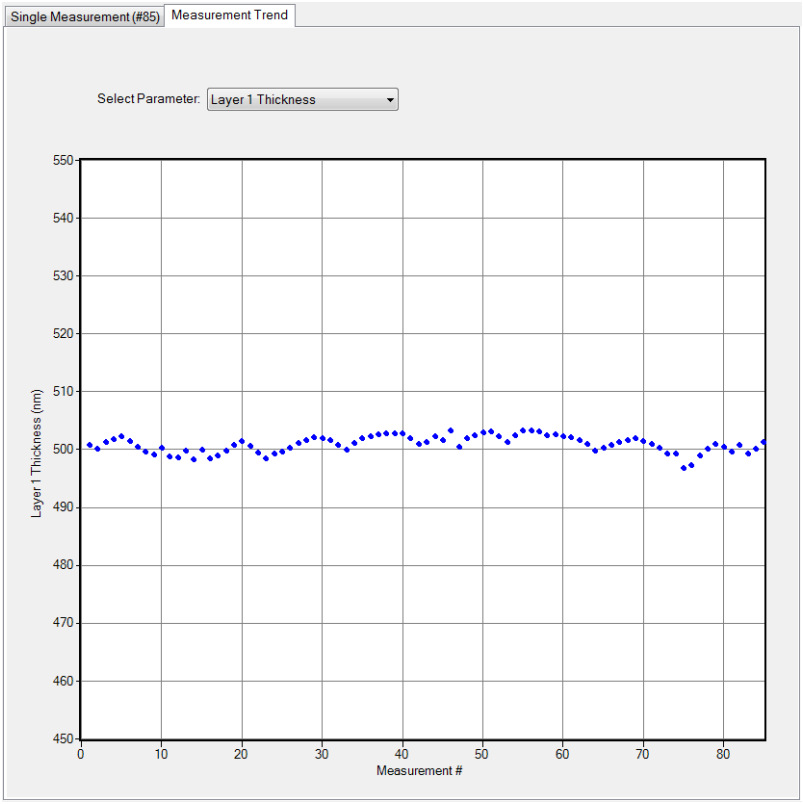
1.2 The History Tab



1. Standard Windows menus, including; **File**, **Edit**, **Setup**, **Acquire**, and **Help.**

2. Click on these tabs to select between the **Measure** and **History** windows.

3. Used to alternate between the **Single Measurement** and **Measurement Trend** tab. Selecting the **Measurement Trend** tab will allow the user to see a plot of the measurement results versus measurement number. A drop-down list allows the user to select between which parameter is displayed.



When the **Measurement Trend** tab is selected in the **History** window, the right side of the screens graphs the results.

4. Shows the measurement results for the selected spectrum (must be in **Single Measurement** mode).

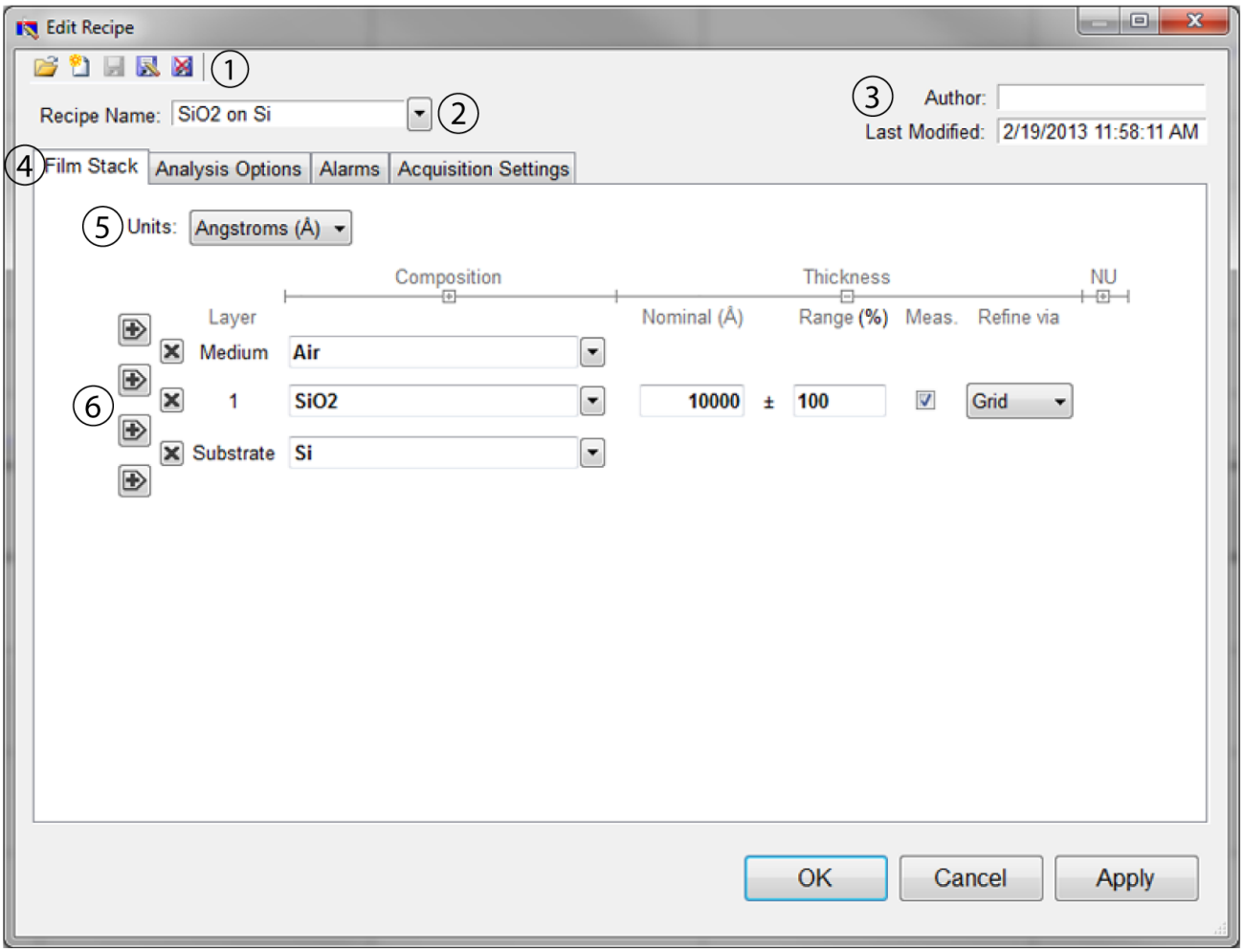
5. Displays the selected spectrum in **Single Measurement** mode or the measurement trend plot in **Measurement Trend** mode.

6. Displays statistics for all measurements included in the history file. This data can be selected for copying and pasting outside of the software.

7. This list contains all spectra currently included in the history file. Left clicking on a column header will re-sort the data in ascending or descending order based on the selected column. Right clicking anywhere is this box will allow the user to open the **Edit Columns** control panel.

1.3 The Edit Recipe Window

The **Edit Recipe** window is used to define the film stack and to set analysis and acquisition parameters.



1. These buttons allow you to, in order from left to right:
    i. Open a saved recipe.
    ii. Start a new recipe.
    iii. Save the current settings.
    iv. Save the current settings as a new recipe
    v. Delete the current recipe.

2. Opens a drop-down list for quick access to saved recipes.

3. **Author**, as defined by **user login name** and recipe modification date is indicated here.

4. Allows the user to select between the **Film Stack**, **Analysis Options**, **Alarms**, and **Acquisition Settings** tabs.

5. The desired thickness units are chosen here. The choices include angstroms (Å, 10^{-10} m), nanometers (nm, 10^{-9} m), kilo-angstroms (kÅ, 10^{-7} m), microns (µm, 10^{-6} m), microinches, mils, millimeters (mm, 10^{-3} m), and **Custom Units**.

6. This is where the materials and parameters to be solved are defined.

1.4 Menu Controls

This section describes the various functions available through the standard Windows File Menus.

File Menu

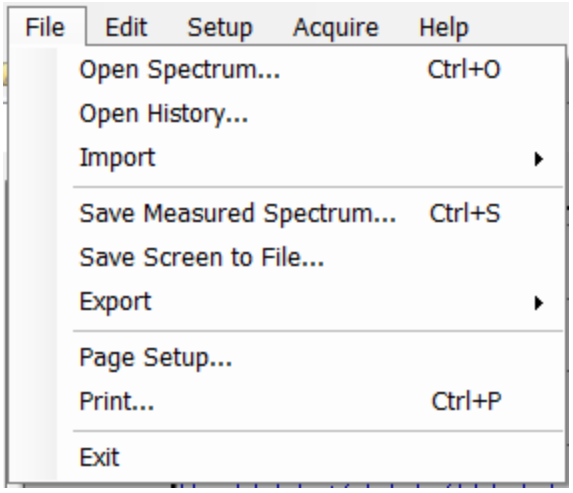
Edit Menu

Setup Menu

Acquire Menu

Help Menu

1.4.1 File Menu



Open Spectrum… (Measure Tab only)
This command is used to open stored reflectance spectra, which are then displayed and can be analyzed for film properties.

Open Measured Template… (Measure Tab Only)
This command is used to open a saved **Contact Probe Template Mode** file.

*Note: Contact probe mode must be enabled in the **Optics Configuration**.*

Open History…
Previously saved **History** data containing measurements and statistics can be loaded with this command.

Import: Recipe…
Import recipes that were previously exported or created with an older version of FILMeasure.

Import: Material…
Import materials not currently in the software either created by the user or provided by Filmetrics.

Import: License…
When the user purchases an upgrade for the system, Filmetrics will send a license file to the user. The license must be imported into the software before upgrades are effective.

Save Measured Spectrum… (Measure Tab only)
This command is used to save spectra for export or later analysis. Files can be saved in four different formats:

*.fmspe: This file can only be opened in FILMeasure 5.0 or greater and is the preferred way to save spectra. Additional information from data acquisition is stored in this file.

*.spe: This file can be opened in all versions of FILMeasure.

*.csv: This is a comma-delimited file that can be opened directly in Excel.

*.txt: This is a tab-delimited file and can be opened with a text editor.

Save Screen to File… (Measure Tab only)
This command takes a screenshot of the FILMeasure window.

Save *n* and *k* to files… (Measure Tab only)
This command is used to save the calculated n and k data to its own material file for use in the **Edit Recipe>Material** list boxes. All data is saved along with the corresponding wavelength data in comma-delimited format.

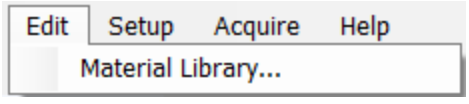
Note: This option is only present when optical constants are being solved for.

Export: Recipe…
Recipes can be exported for use on other computers running FILMeasure. The exported file will also automatically include any material files required by the recipe. To duplicate an analysis, all that is needed is the exported recipe and the spectrum being analyzed.

Export: Recipe for FILMeasure 4…
This command will convert a recipe for use with FILMeasure 4. This will not include material files or recipe features only available in later versions of FILMeasure.

Export: SystemInfo File…
For certain software upgrades , the user will be required to send the SystemInfo file to Filmetrics which can be exported using this command.

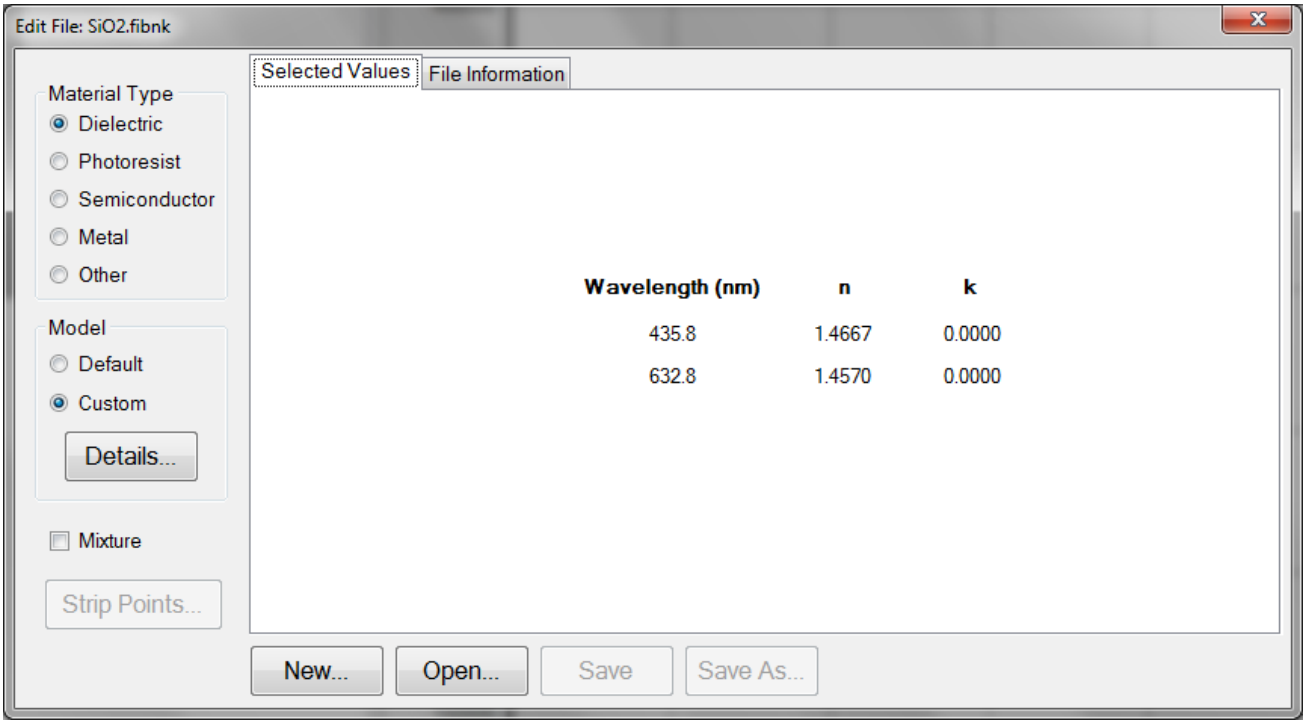
1.4.2 Edit Menu



This menu allows access to the **Material Library**.

Material Library

The material library is used to display information about existing materials as well as create new ones. Material files shipped with FILMeasure have the file extension ".fibnk" and will only display limited information. User created material files, ".fitnk", can be plotted on a graph or viewed in table form.



A new material may be entered into the material library manually if refractive index (*n*) and extinction coefficient (*k*) are known as a function of wavelength for the new material.

To create a new material file:

1. Select **Edit > Material Library…** menu item. The **Edit Material** dialog box will appear.

2. Click **New…** to create a new material.

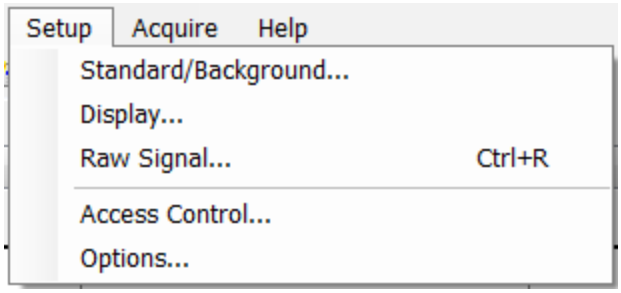
3. Select the type of material you are ready to create (Dielectric, Photoresist, Semiconductor, Metal, or Other).

4. Optimize the functionality of the file by entering index values over the entire wavelength range of the system. The wavelength unit is nanometers. Enter index information every 20 to 50 nm, depending on the complexity of the data. In areas where *n* or *k* is quickly changing, more densely spaced points should be entered.

5. Alternatively, select the **Model > Custom** radio button and click the **Details** button. A **Custom Model Information** dialog box will appear. Select the appropriate model. Enter in the known coefficients and click **OK**. Cauchy coefficients are displayed in microns. If they are entered in Å, FILMeasure will automatically detect this and convert them to microns.

6. Click the save button to save the material. You will be asked for a filename, and you may select one of two file types:
*.fitnk: Creates a unified file with both *n* and *k* information. Compatible with FILMeasure 5 and beyond.
*.nnn: Creates a *.nnn file and, if there is *k* in the material, a *.kkk file to contain material data. For files with both *n* and *k*, the two files must have the same filename, for example xmaterial.nnn and xmaterial.kkk , in order for FILMeasure to properly indentify them. Compatible with all versions of FILMeasure.

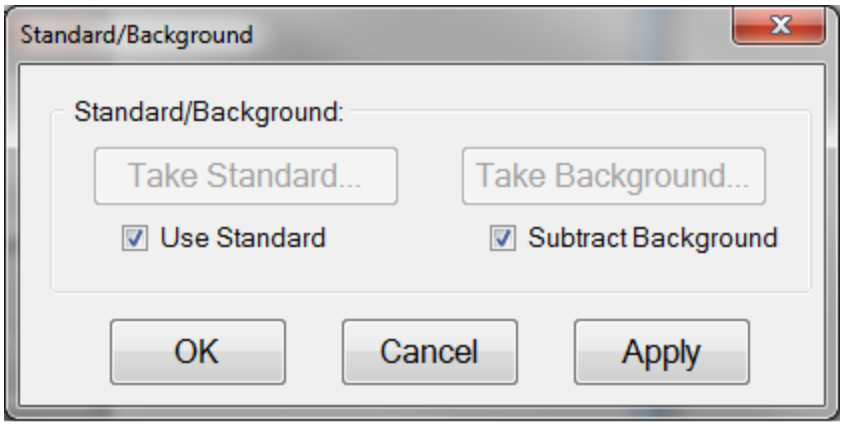
1.4.3 Setup Menu



This menu contains the settings for manually obtaining the **Standard/Background** information, **Display** options, and the **Raw Signal** dialog. This is also where user access can be set through the **Access Control** option, as well as modifying some system **Options**.

Standard/Background

This control is used when manually taking and enabling/disabling the **Reflectance Standard** and **Background** compensation.



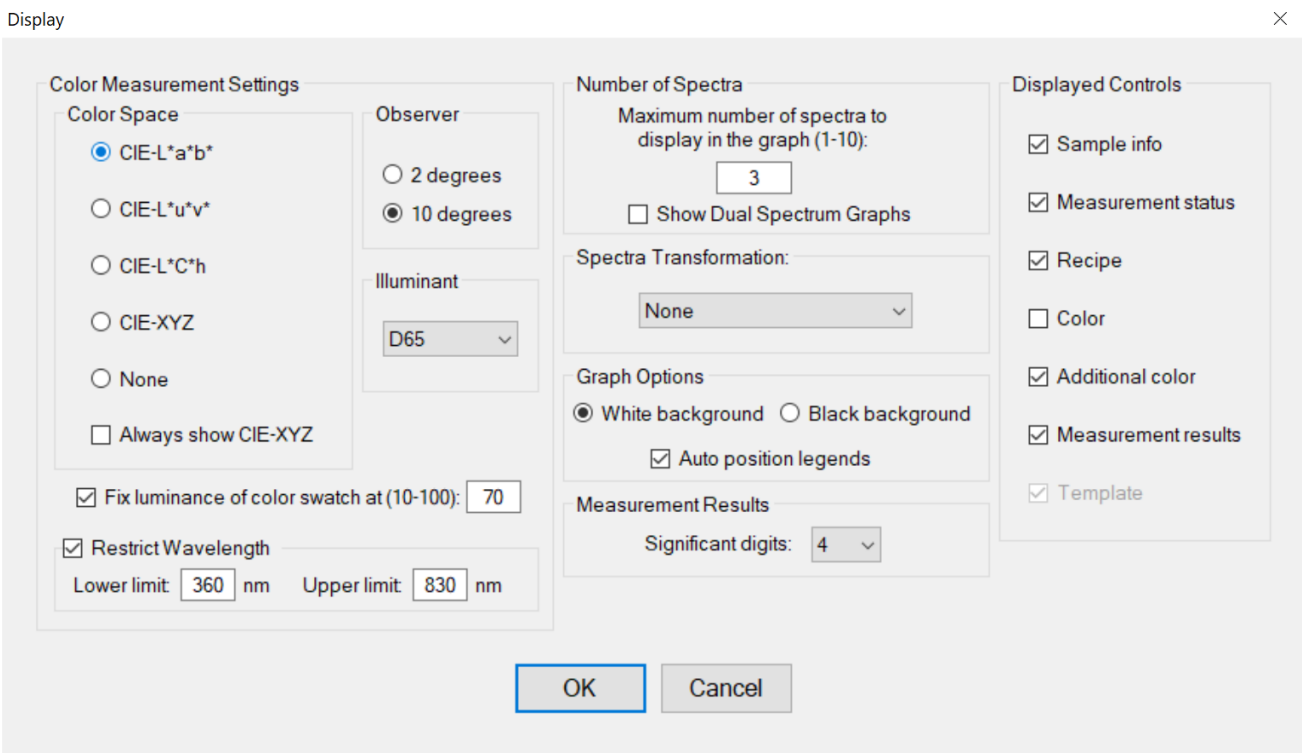
Take Standard
This button allows the reflectance standard portion of the baseline measurement to be taken without having to re-take the background measurement.

Take Background
This button allows the background portion of the baseline measurement to be taken without having to re-take the reference measurement.

Use Standard
This checkbox allows the reflectance standard portion of the baseline correction to be enabled or disabled. This box is only available if a baseline has already been completed.

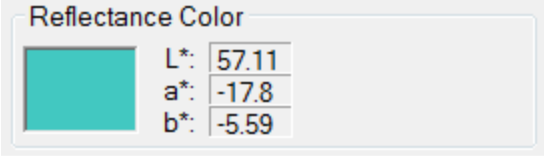
Subtract Background
This checkbox allows the background subtraction portion of the baseline correction to be enabled or disabled. This box is only available if a baseline has already been completed.

Display



Color Measurement Settings: Color Space

The color of a spectrum can be displayed and quantified on the main screen if a color space coordinate system is selected.



When a **Color Space** is selected, the color of the spectrum and the coordinates appear on the **Measure Tab**.

Color Measurement Settings: Observer

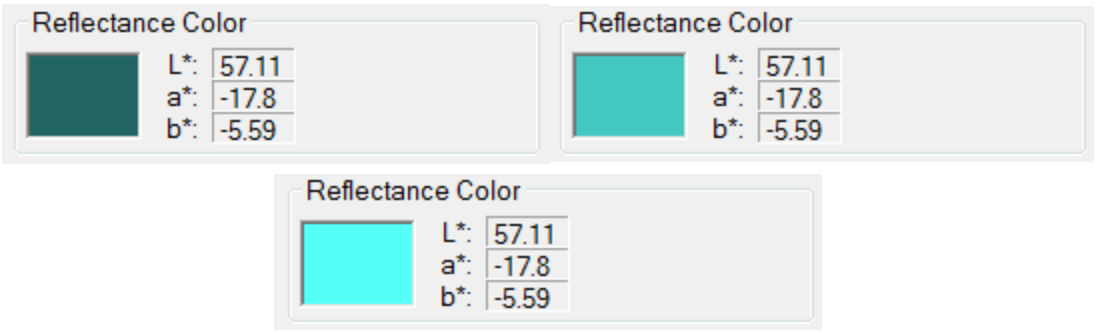
When a specific color space is chosen for display, this option allows the user to choose between the CIE 1931 2 degree observer and the CIE 1964 10 degree observer. Note that this variable is associated with the CIE eye-response convention (see, for example, **http://en.wikipedia.org/wiki/CIE_1931_color_space**) and is unrelated to the illumination or collection angle of the measurement.

Color Measurement Settings: Illuminant

This setting refers to the **standard illuminant** being used by the software to help determine the results of the color analysis.

Color Measurement Settings: Fixed Luminance

The user can fix the luminance of the displayed color. Since antireflective coatings, for example, are inherently transmitting, the spectra have very little luminance and the coating color may appear black on the screen. Adjusting the luminance does not affect the color space coordinate results.



Spectrum colors of the same spectrum with fixed luminance of 10%, 40%, and 70%, respectively.

Color Measurement Settings: Restrict Wavelength

Restrict the wavelength range analyzed when performing color analysis. When disabled the analysis defaults to

Number of Spectra

This sets the maximum number of spectra (up to 10) displayed simultaneously on the main FILMeasure screen. If the maximum number of spectra is reduced to fewer than the number already on the screen, the most recently selected spectra will remain.

You may also use the **Show Dual Spectrum Graphs** option to display two measurement graphs concurrently. The data will remain the same across both graphs (spectra and modeled fit), but you can set the graph options for both windows independently. This can be used to focus on a specific area of interest on a spectra while still seeing the effects on the dataset as a whole.

Spectra Transformation

For spectra with both reflectance and transmittance information present, you can have the software display them as separate spectra (none), added together (sum), or the absorptance (1-R-T).

Graph Options

The background of the graph can be set to white or black using this option, as well as allowing the software to auto-position the legend box to avoid or minimize overlap with the displayed spectra.

Measurement Results: Significant Digits

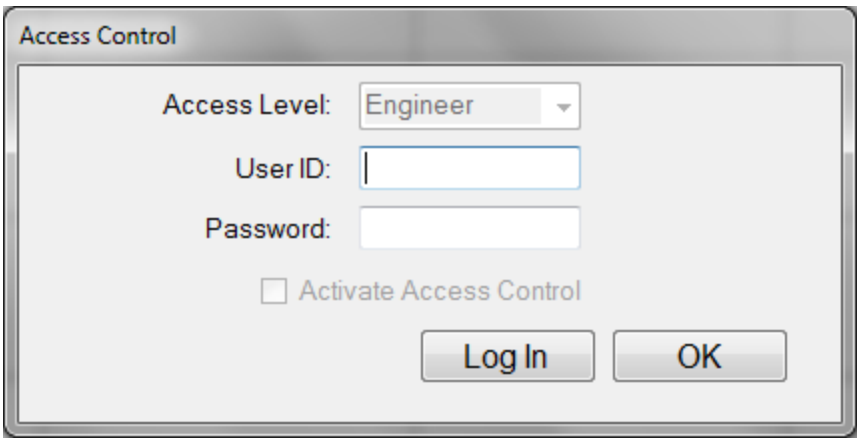
The user can select the number of significant figures, from 1 to 7, that the results are displayed in. For example, with the units set to *Angstroms* and 4 significant digits, The software would report a thickness of 7213 Å. If significant digits is switched to 3, it will instead display the thickness as 7210 Å.

Displayed Controls

These checkboxes are used to determine which controls are shown on the right hand side of the FILMeasure window. Some options may not be available based on current **system configuration**.

Access Control

FILMeasure incorporates password protection to limit access to the measurement software and settings. When the software is initially installed access control is turned off. The software will automatically boot up with *Engineer* level access that enables access to all features of the program except turning on and off access control and adding and deleting users.



Turning on access control requires Supervisor level access. The software is delivered with one user, a supervisor, in the list of authorized users as shown below:

UserID: filmsuper

Password: filmetricsfff

To turn on access control log in as filmsuper using the password shown above. The **Activate Access Control** checkbox should selected enabling access control functions. User ID and Password are case-sensitive.

To add an operator level user, select operator from the **Access Level** list box then type a user name and an initial password for that user. Then click the **Add User** button. To add an engineer or a supervisor user follow the same procedure, but first pick the appropriate access level from the list box before pressing the **Add User** button.

To delete a user, enter the **UserID** and press the **Delete User** button.

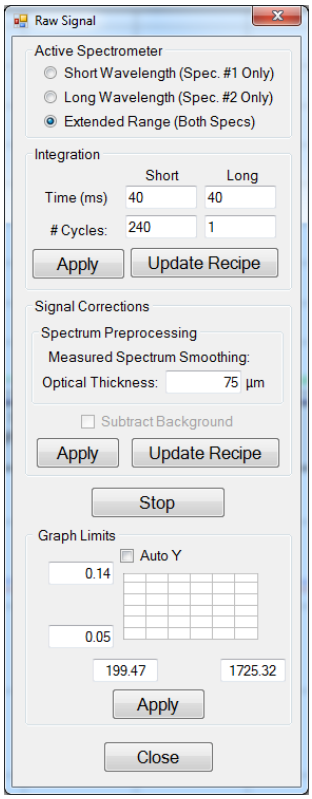
Operator and Engineer level users can change their passwords when they are logged in by entering their password into the password box and pressing the change button. Supervisor-level users can change their password or the password of any other user by selecting the appropriate access level, entering the appropriate **UserID** and the new password and pressing **Add User**. If the program finds that a user already exists, it will delete the old entry for that user and create a new entry.

We recommend that you create a new supervisor level user and delete the filmsuper user for maximum security. If all supervisor level users forget their passwords, it will be necessary to re-install the software and add all the users again.

Raw Signal

Raw Signal

The **Raw Signal** dialog box allows the user to plot the raw (no baseline corrections applied) spectrometer signal on the main graph. This is useful when adjusting the focus or when setting integration times manually.



Active Spectrometer

For systems with multiple spectrometers (EXR and UVX) these radio buttons allow you to select between collecting data from the short wavelength, long wavelength, or both spectrometers. Note that the software will automatically adjust the x-axis of the graph to match maximum wavelength range the selected spectrometer configuration.

Integration

The user can set the integration time and the number of integration cycles for the spectrometer. Increasing cycles can be used to help smooth the data being returned to FILMeasure. Clicking **Apply** will show the changes in the main graph window. Clicking **Update Recipe** transfers the listed integration time and cycles to the current recipe.

Signal Corrections.

Spectrum Preprocessing allows the user to apply a smoothing function to the displayed spectra. See **Editing Film Recipes > Data Selection** for more details.

Subtract Background can be used to remove the inherent background signal from the displayed raw signal spectrum.

Start/Stop

This button starts/stops the acquisition of the spectrometer signal.

Graph Limits

This allows the user to either manually set or autoscale the graph limits. Click **Apply** after any changes to display them on the main FILMeasure window.

Options

The **Options** menu allows the user to enable, disable, or modify several different settings within FILMeasure.

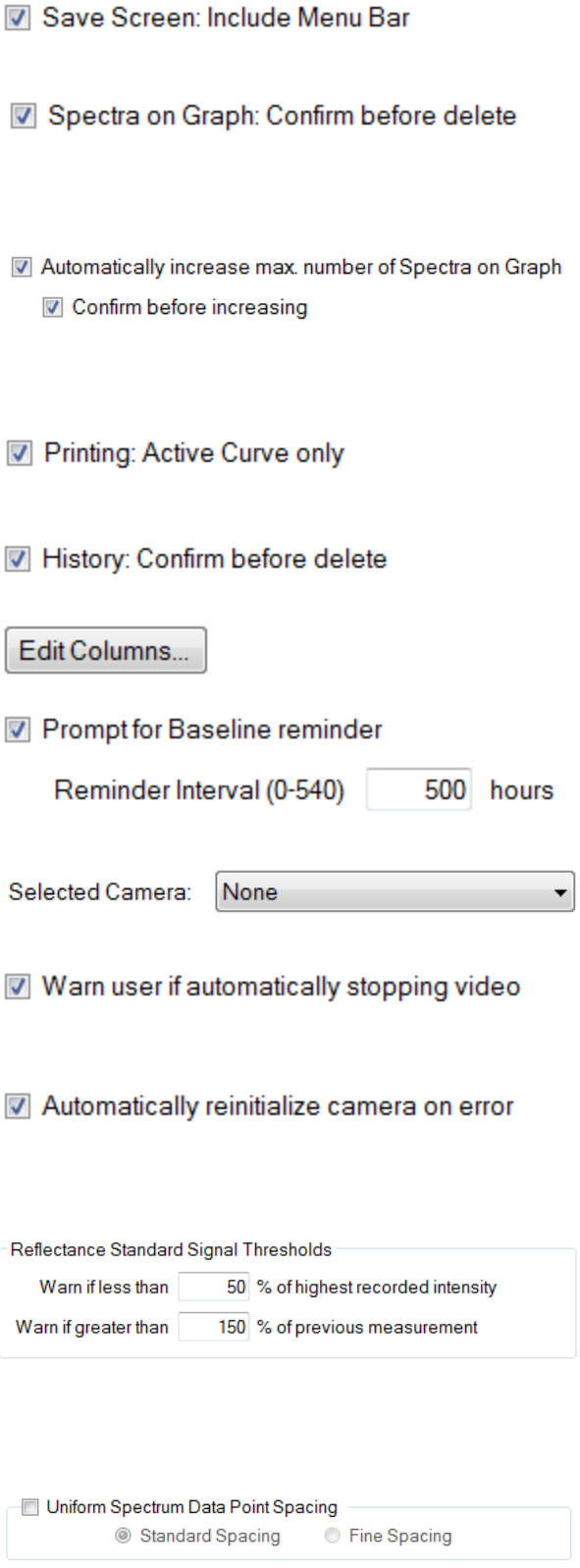
General Settings

Data Recording

Calibration

General Settings

The **General Settings** tab is used to enable, disable, and modify various options in the software.



Hide or show the menu bar when using *Save Screen to File*.

Enable or disable whether the software displays a warning before clearing all spectra on the screen when the *Clear Graph* button is pressed.

Enable or disable automatically increasing the number of spectra displayed on the graph when opening multiple saved spectra.

You can also set whether to show a confirmation message before increasing the number of spectra.

Toggle whether the software prints out all currently loaded spectra, or only the active curve when using the *Print* feature.

Enable or disable whether the warning dialog before you delete entries in the history file.

Open a dialog that allows you to control which columns are visible in the **History** tab.

Enable or disable the baseline reminder feature, and if enabled to set the reminder interval.

Enable any connected SampleCams by selecting their serial number from the dropdown list. Disable the camera by selecting *None*.

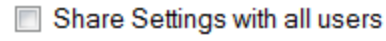
Enable or disable whether the software warns you if an error causes the camera to stop.

Click to enable automatic camera reinitialization should an error occur. If you have a SampleCam it is generally suggested that you keep this feature enabled.

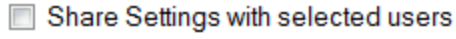
Set your *Reflectance Standard Signal Threshold* values. These determine at what percentage of the initial signal intensity the software will display a warning during the **Baseline** process. This feature is used to track lamp health, proper baseline procedure, and other parameters.

Enable or disable the *Uniform Spectra Data Point Spacing* function. When enabled, saved spectra will have their values set to a multiple of 1, 2 or 5. *Standard Spacing* sets the spacing equal or greater than the smallest interval in the data, while *Fine Spacing* sets the spacing to shorter than the smallest inter-

val.



For systems with multiple user accounts, enable or disable sharing settings across all users. A folder to store the settings file may be selected.



For systems with multiple user accounts, enable or disable sharing settings across any users who have slected this option. A folder to store the settings file may be selected. This option is only available if *Share Settings with all users* isn't enabled.

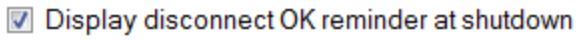


This feature allows the user to select a different location for their *Recipes* folder, for example to a shared network drive. This can be useful for syncing recipes across multiple instruments in varying locations.

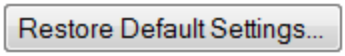
Selecting **Reset** will instruct the software to look in the default location for the *Recipe* folder.



This feature is similar to the one above, only it allows you to choose a different location for your *Materials* folder.



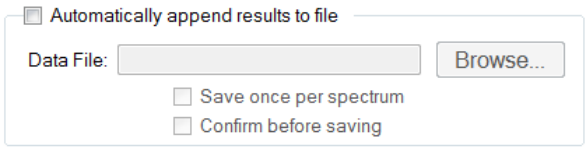
Enable or disable whether the software prompts you when it is ok to disconnect the instrument at software shutdown.



Click to restore all settings to their default values.

Data Recording

The **Data Recording** tab contains controls for the various methods by which FILMeasure may automatically save collected data.

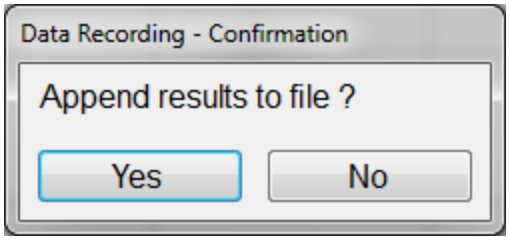


Enabling *Automatically append results to file* will instruct the software to create an excel file at a specified location in which a running tally of measured results are kept.

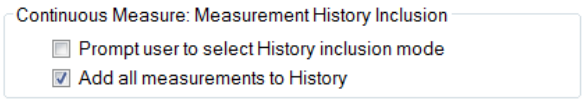
This feature is different from the **History** tab in that it only records results, not the spectra files as well. It does not however have an upper limit on the number of spectra that can be recorded.

Enabling *Save once per spectrum* will only append results when the **Measure** button is pressed or a saved spectra is opened in the software. Any changes in the recipe applied to existing spectra will not be recorded. When this feature is disabled, any time a spectra is analyzed the result is entered into the file.

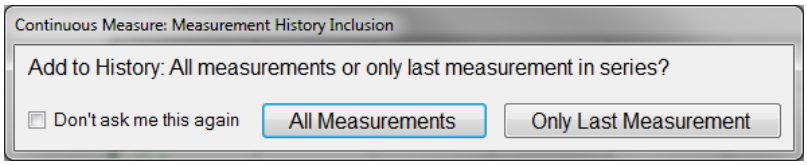
When *Confirm before saving* is turned on, the software will provide a prompt before appending data to the excel file.



Continuous Measure: Measurement History Inclusion controls which spectra are included in the **History** tab when the **Acquire > Measure Continuously** feature is used.

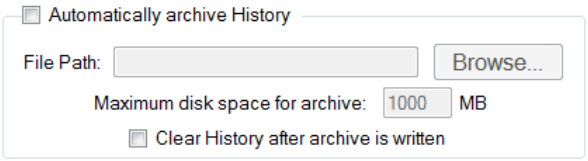


If the software is set to *Prompt user to select History inclusion mode*, a dialog will pop up prior to beginning the continuous measure asking the user whether to save all measurements while measuring continuously, or only the last measurement. The only last measurement feature can be useful on less ideal surfaces (rough, non-uniform, etc.) where the user may have to search around the sample surface for a spectrum capable of being analyzed.



Clicking **Don't ask me this again** will disable the prompt, and the user will have to navigate back to this menu to re-enable the feature.

The user can also select between having the software provide a prompt for which data to include in the history file when using **Acquire > Continuous Measure** or to record all measurements to the **History** tab.



The *Automatically archive History* feature can be used to have the software save a copy of the history file whenever it hits 1000 measurements.

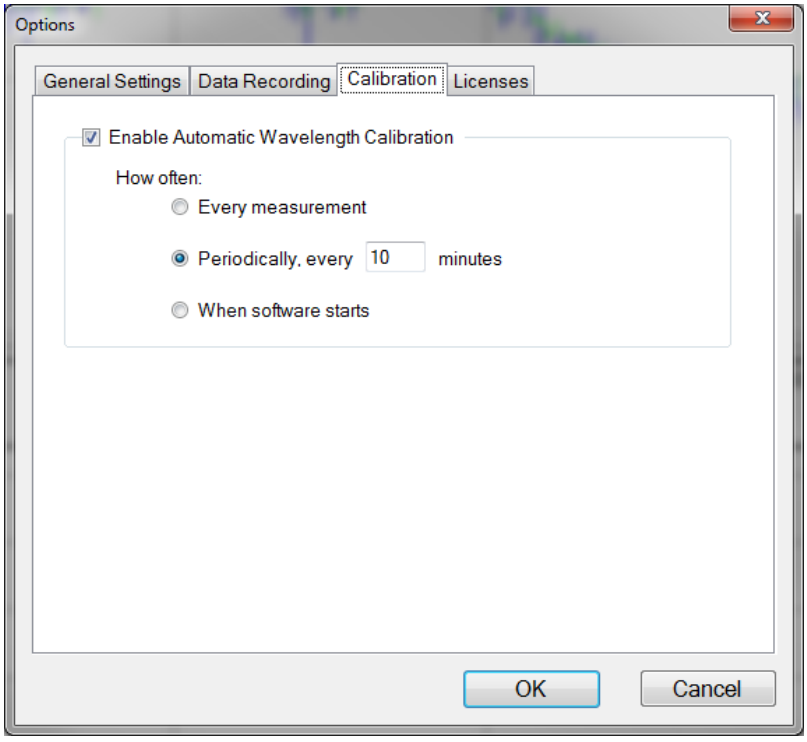
The history files are saved in the following format (YYYY-MM-DD-X) where YYYY is the year, MM is the month, DD is the day, and X is a serialized number based on how many history files have been saved on a given day.

A value can be entered in megabytes for the maximum disk space to be allocated for the saved archives. This will keep the software from using up all available hard disk space if the feature is enabled and then forgotten. Should the software hit the allocated max, it will delete the oldest files in the selected folder until enough space is available for the new history file.

Clicking **Clear History after archive is written** will instruct the software to automatically clear the history results after 1000 measurements have been reached, and then begin counting towards the next group. When it is disabled the history feature will work in the normal mode upon reaching 1000 measurements (newest measurement is added as Meas #1000, Meas #1 is deleted). It will write the next history file upon reaching 1000 measurements again, not each time the *Meas#* counter shows 1000.

Clicking the **Contact Probe Template Mode** enables the template function on the **Measure Tab**. For more information on creating and using templates, see **Contact Probe Template Mode** in the **Advanced Features** section. This check box is only accessible if contact probe is selected under **Optics Configuration**.

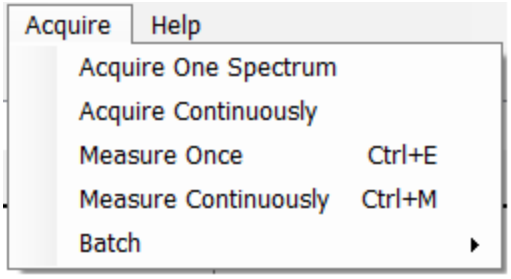
Calibration



The F20 uses an integrated atomic-emission-line light source to update the wavelength calibration of the measurement system's spectrometer. By allowing a system to self-calibrate, automatic wavelength calibration provides significant compensation for calibration drift that could be caused by large temperature changes or by moderate mechanical shock.

The **Wavelength Calibration** tab allows to user to choose how often they would like the system to automatically calibrate the wavelength response of the system. Users can choose to calibrate with every measurement, periodically, or whenever the software boots up. This action is performed in the background and does not require any user input beyond setting the interval.

1.4.4 Acquire Menu



Acquire > Acquire One Spectrum
This function collects a single spectrum and displays it on the screen. No analysis is performed until the recipe is changed or the **Analyze** button is clicked.

Acquire > Acquire Continuously
This feature collects real-time data from the spectrometer. It is useful when searching for a specific location to measure (for example, locating a more uniform area of the sample or one that exhibits interference fringes). Stop continuous data acquisition by selecting **Acquire Continously** again.

Acquire > Measure Once
This feature collects a spectrum, analyzes it and displays it on the screen. This is the same as pressing the **Measure** Button. For systems with the **Sample Cam**, this saves an image to the spectrum file created.

Acquire > Measure Continuously
This feature collects real-time spectrum data and analyzes it – similar to repeatedly pressing the **Measure** button. A measurement interval time can be set in the recipe using the **Advanced** options underneath the **Acquisition Settings** tab. Stop continuous data acquisition by selecting **Measure Continously** again, or by pressing Ctrl+M.

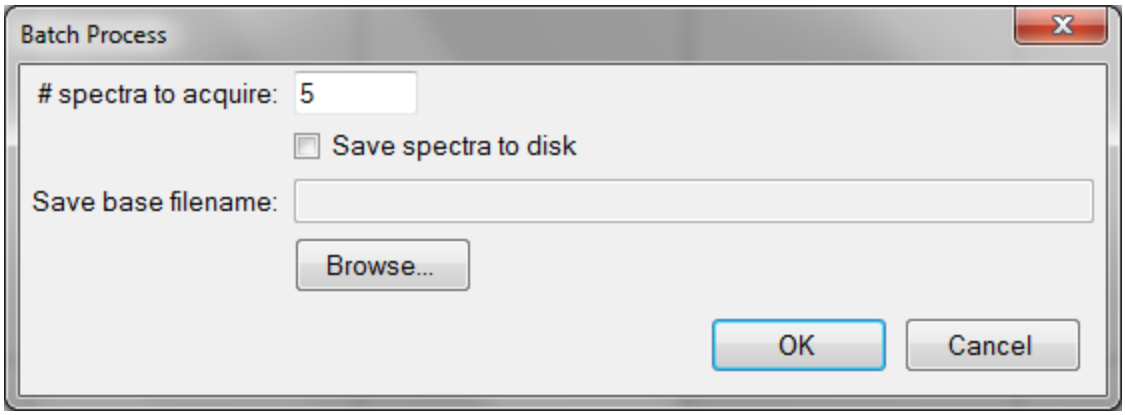
Batch Acquire

The Batch functions allow you to act on groups of spectra, rather than each individual spectrum



Acquire > Batch > Acquire

This option allows you to acquire a pre-selected number of spectra. You can also choose to save the batched spectra as they are acquired.



Acquire > Batch > Measure

This option allows you to acquire and analyze a pre-selected number of spectra. You can also choose to save the batched spectra as they are acquired.

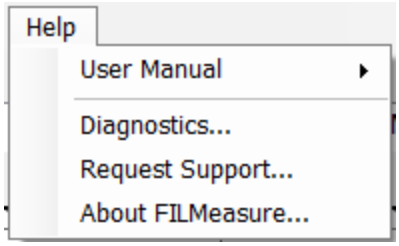
Acquire > Batch > Analyze Spectra

This option allows the user to select multiple spectra for analysis by the current recipe. If there are more spectra to be analyzed than the software is set to **Display**, the analyzed spectra can easily be retrieved from the **History** tab.

Acquire > Batch > Analyze All Spectra in Folder

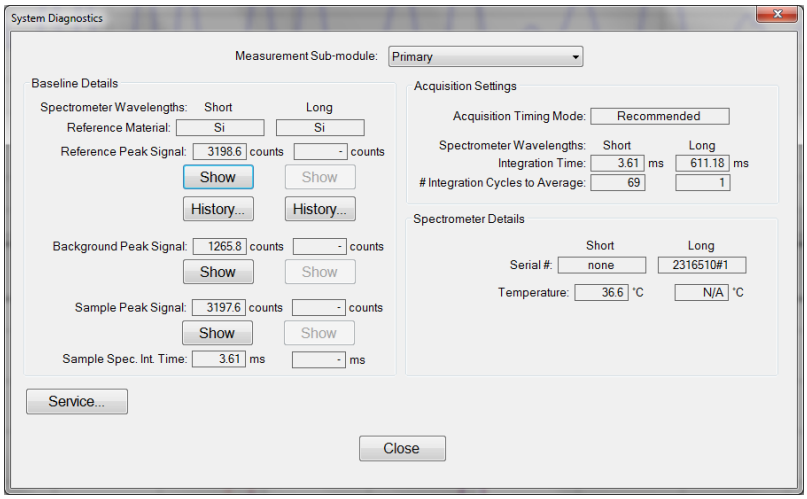
This option will analyze every spectrum in a folder with the currently selected recipe. If there are more spectra to be analyzed than the software is set to **Display**, the analyzed spectra can easily be retrieved from the **History** tab.

1.4.5 Help Menu



Diagnostics

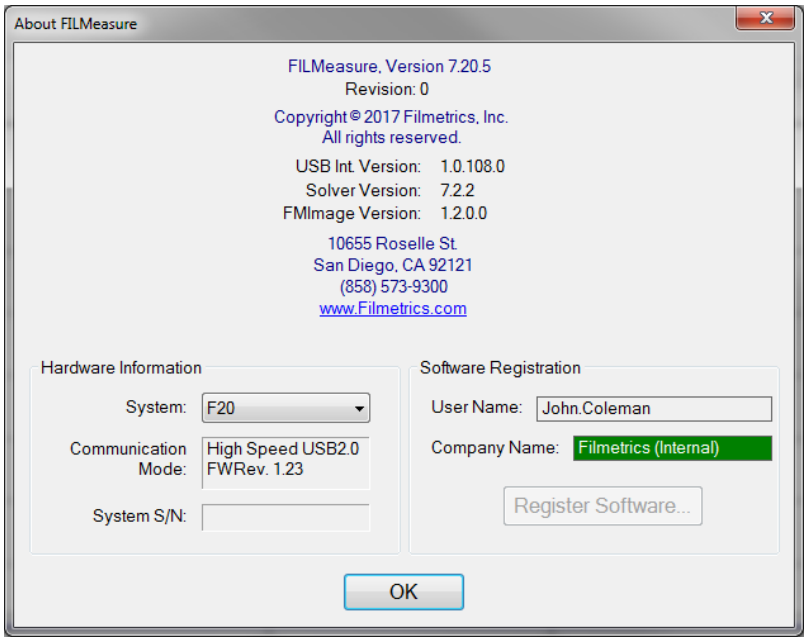
The diagnostics window gives information regarding the data taken during the baseline. Reference and background counts are shown (this is useful to see how much signal the system is getting; good reference signals typically are between 2500-3500 counts).



Help > Diagnostics shows the user information regarding the data taken during the baseline routine.

Help > About FILMeasure

Information about the hardware and software versions and the system serial number can be found here. Please have this information available when calling Filmetrics for technical assistance.



Help>About FILMeasure... window

1.5 Taking a Baseline

The baseline measurement allows the FILMeasure software to take into account the response inherent to the reflectance measurement hardware. It does this by first measuring the sample reflectance, then a reflectance standard, and finally by taking a “dark” reading. The light source should be allowed to stabilize at least five minutes after being turned on before the baseline is taken. When measuring films less than 1000 Å thick, the baseline should be periodically re-taken every 20-30 minutes. For thicker films, a new baseline can be taken less often.

Baseline measurements may be started two ways:

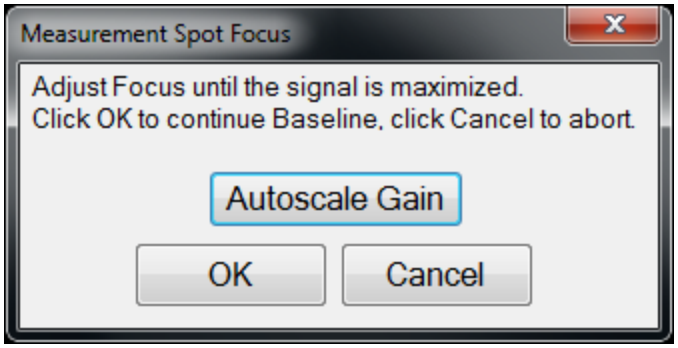
1. Select the **Baseline** button, which will guide the user through the baseline procedure, or

2. Select **Setup>Standard/Background...** from the menu bar for manual access to the baseline functions.

The following instructions assume the user is using the recommended sampling time option, has already focused the system, and clicked the **Baseline** button. For instructions on focusing the system, see **Focusing the Fiber**.

The first step of the baseline measurement is to acquire a spectrum of the sample. Place the sample on the stage under the light path and click **Take Sample Reflectance**. This step determines the best integration times and cycles for measuring the sample.

Systems with manual z motion, such as the F40 or F3-sX series, will prompt the user with the Measurement Spot Focus dialog (pictured below) and display a continuous raw signal measurement in the main graph window. Turn the z-height micrometer on the stage to adjust focus until signal is maximized. If the **raw signal** curve is too low or too high to identify changes in signal, click Autoscale Gain to automatically adjust the integration time and better analyze the raw signal spectra. Once the system has been focused, click **OK** to continue.



The next step is to take a spectrum of a reflectance standard. For systems with no z-motion, use the reflectance standard that is closest in height to your sample. This is to account for the fixed focus point, which you can read more about **here**. For systems with manual z motion, place a reflectance standard of similar reflectance to the test sample on the sample stage.

Select the reflectance standard being used from the **Reflectance Standard** list. For reflectance standards not in the list, reflectance versus wavelength values may be entered into a comma-delimited text file with the extension ".rrr" (stored in the material subfolder).

Click **Take Reflectance Standard**. You will again be prompted with the *Measurement Spot Focus* dialog box for manual z motion systems.

The background measurement method used will be dependent on the optical setup. For the SS-3 or SS-5 in reflection mode, remove the reflectance standard from the stage and click **Take Background**. For transmission mode, the collection optic must be covered using an opaque object. For a contact probe, point the probe from any light source or reflective surface. For in-line or in situ measurements using a lens assembly and mount combination, a specularly reflective sample (glass or bare Si for example) should be placed in the light path at an angle between 10 and 80 degrees to the light path.

After a baseline is acquired, the instrument is ready to make measurements. To verify the system is working properly, select **Acquire One Spectrum** from the *Acquire* menu with the reflectance standard in place. You should see the reflectance spectrum of the reflectance standard. As discussed in the focusing section, when using a stage or lens assembly it is important that the top of the sample and reflectance standard are roughly the same height to insure that the light is properly focused on both.

Autobaseline

When the **Optics Configuration** is set to the contact probe or contact stage, the **Autobaseline** option can be selected. This option will automatically perform a baseline periodically, so that the user does not have to do this routinely. Once selected, the **Baseline** measurement routine described above will only require the user to measure the BK7 reflectance standard and take a dark reading. A user will only have to do a manual baseline once per week.

1.6 Setting up the SampleCam

The SampleCam is an optional accessory that can be used on any system that uses the SS-3 stage. When present it allows for a live video image of the sample to be shown either within FILMeasure or in a seperate window. This can be used for everything from sample tracking, to troubleshooting, or to determine your measurement location on non-scattering samples. Any spectra measured when a SampleCam is enabled will also save an image of the sample when the spectra is saved as a .fmspe file.

Enabling the SampleCam

If you've purchased a SampleCam, please follow the quick start guide included with the system to get the camera set up on the SS-3 stage. Once that is complete, you can then enable and disable the SampleCam by going to the **Setup** menu within FILMeasure. Under the **General Settings** tab there will be a drop-down list labeled *Selected Camera*. Clicking on the list will then show the available cameras, listed by serial number. Click on the serial number that matches the SampleCam you want to use to enable it in the software. To disable the camera select the *None* option from the list. Click **OK** to apply any changes.

The SampleCam Display

With the camera enabled, there should now be a image window visible on the left hand portion of the main FILMeasure window. There are two tabs visible, *Live Video* and *Still Image*. *Live Video* shows the current view from the SampleCam, while the *Still Image* will show either the saved image associated with the currently loaded spectra or a grey box if no spectra is present. Along the top of the image window there is also a slider bar, which controls the zoom settings for the camera, as well as buttons to **Center** the image on the measurement spot, open the **Settings** dialog, or to open the video tab in a separate window.

Once the SampleCam is enabled it is possible to hide it by clicking on the vertical button with the arrow pointing to the edge that is just to the right of the image window. To unhide it, simply click the button again (it will be along the left edge of the FILMeasure window, with the arrow pointing to the right).

Configuring the SampleCam

Clicking on the **Settings** button will open the Image Settings dialog, which contains a variety of options and calibrations for the camera.

The first checkbox allows you to enable or disable display of the measurement spot after the **Set Spot Location** wizard has been completed. You can also change the opacity (0-100%), and color of the measurement spot as well.

The slider underneath is used to adjust the focus of the SampleCam.

The *Microscope Adapter* camera should only be used if the system also has the MA-Cmount upgrade.

Finally, the **Set Spot Location** button will launch a quick dialog to step you through setting up the measurement spot.

2. Editing Film Recipes

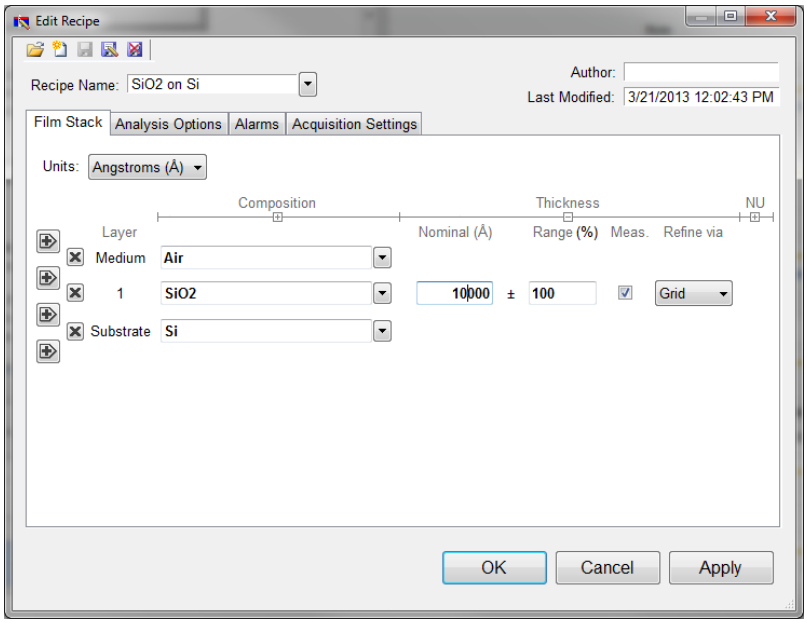
The description of the nominal film structure, as well as the measurement parameters, is specified in the **Edit Recipe** dialog box. Hundreds of different film structures and their measurement specifications may be saved.

The **Edit Recipe** dialog box is accessed with the **Edit Recipe...** button on the **Measure Tab**. The **Edit Recipe** dialog box lists an initial guess at the specifications of the film structure to be measured. These specifications include the number of films in the structure, the layer order of the films in the structure, the specifications of individual films (thickness and index), and the film parameters to be measured.

2.1 Adding, Changing, or Deleting a Structure

When the **Edit Recipe** dialog box is opened, it shows the stored specifications of the structure selected from the **Recipe Name:** list, along with any changes made since the recipe was loaded. Changes to the structure selected can be permanently stored by making the desired changes and then clicking on **Save**. Changes can also be stored as a new recipe by clicking on **Save As**. A structure may be deleted (removed from the **Recipe Name:** list) by clicking on the **Delete** button.

2.2 Film Stack



The **Film Stack** tab is used to define the film structure and desired measurement units, as well as the starting guesses and constraints for thickness, *n*, and *k*. This is also where you determine your analysis method; **Grid**, **FFT**, or **None**, as well as any **Nonuniformity** that may be present in the sample. The [+] and [-] below **Composition**, **Thickness** and **Nonuniformity** can be used to expand or hide solving options.

2.2.1 Setting Up a Film Stack

When measuring a layer, the parameters (thickness, *n*, and *k*) of the known films (including the incident medium and substrate) must be entered into the proper fields in the **Edit Recipe** dialog box, as well as initial guesses for the values to be measured. The refractive index (*n*) and extinction coefficient (*k*) values for common materials can be selected from the drop-down menus on the right-hand side of the listed materials, or by left-clicking on the material and using the **Search** function. Note that if a generic index is selected, the software will instead prompt for a new index value.

If a material is being measured is not present in the material library there are three possible approaches to complete the recipe:

a) Choose a material in the library that is similar.
b) If the material is transparent (an insulator), select **"Enter Refractive Index Value"** from the material list and enter a value for the refractive index (*n* will automatically be varied by FILMeasure to account for dispersion effects, with the entered value being *n* at 632.8 nm).
c) Create a new material file using the **Edit > Material Library...** dialog box. Enter the refractive index values for *n* and *k* as a function of wavelength and save the files so that they may be selected as in the recipe. See **Creating and Editing *n* and *k* Files** for more information.

2.2.2 Choosing the Films to be Measured

To measure a film's thickness, mark the check box on the right-hand side of the **Thickness** field under the **Film Stack** tab in the **Edit Recipe** dialog box. When no boxes are checked, the theoretical spectrum for the specified layer stack and thicknesses entered in the recipe will be displayed.

As with most measurements, the uncertainty of the measured data increases as the number of simultaneously measured values increases. Thus it is best to provide as much information about the film structure as possible.

2.2.3 Constraints

By setting constraints, the user can limit the possible values of the measured film properties. The constraints are set in conjunction with the values entered in the **Range** dialog box for **Thickness**. For example, if the initial guess of the measured thickness of a film is 100 nm and the thickness constraint is set at 50%, FILMeasure will only consider possible thicknesses in the range 50 nm to 150 nm.

Measurement results that are equal or nearly equal to the min or max thickness limit will be marked as invalid, unless the minimum GOF limit is 0 (using the above example, a result of 50 nm would show as invalid). This feature is intended to make sure that the user is alerted to results that may not be accurate as the correct solution is not available in the currently provided range. To clear the invalid result, simply use a larger constraint or change the nominal thickness value.

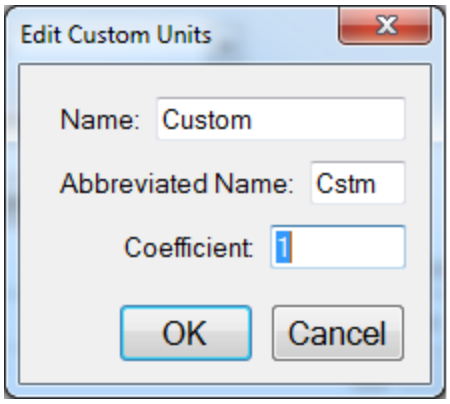
The constraints for *n* and *k* are set using the *n* and *k* dialog boxes under **Composition**. These boxes are only visible when the **Thickness** or **Composition** options are expanded.

You can select between constraining by a percent of the thickness guess, or by absolute thickness units by clicking on the % or measurement unit above the constraints dialog box. Constraining the measurement range can speed up analysis and can also help exclude non-physical solutions.

2.2.4 Units

The **Units** drop down menu allows the user to easily switch between their desired measurement unit, ranging from angstroms up to inches. The software will automatically convert between units when a different unit is selected (e.g. 1 micron will be converted to 1000 nm).

The **Custom Unit** function allows you to create your own custom unit for measurements if the metric you'd like isn't already available in the drop-down list.



The **Edit Custom Units** dialog box.

To determine the correct coefficient to use, you must first know the relationship between your unit and nanometers. For example, if your units are mg/ft^2 and it's been determined that 1 mg/ft^2 works out to a film thickness of 150 nm, then you would calculate your coefficient as follows, where X is the coefficient.

$$1\frac{mg}{ft^2} * X\frac{nm}{\frac{mg}{ft^2}} = 150nm$$

$$Xnm = \frac{150nm}{1nm}$$

$$X = 150$$

Custom units can be saved as part of the **Recipe** file, and only one custom unit can be saved per recipe.

2.2.5 Composition

When measuring the *n* and *k* of a film, the general dependence of these values upon wavelength must be specified. This dependence is determined by the type of material to be measured. For example, insulators, semiconductors, and metals all have a unique type of *n* and *k* wavelength dependence.

Dozens of models for these different dependencies have been proposed and used over the years. FILMeasure uses a few of the most versatile and accepted models. For insulators the Cauchy model is used, for semiconductors either the Amorphous or Bridge-Lorentzian model, and for metals the Drude model.

There are also a number of special-purpose models for specific applications. These models and starting coefficients values are selected automatically when a material is chosen from the **Material** lists in the **Edit Recipe** dialog box. Other models may also be specified by selecting them from the ***nk* Model** drop-down menu under the **Composition** tab which is accessed by right clicking on the **Meas** checkbox when solving for optical constants is enabled.

2.2.6 Grid Search for Thickness

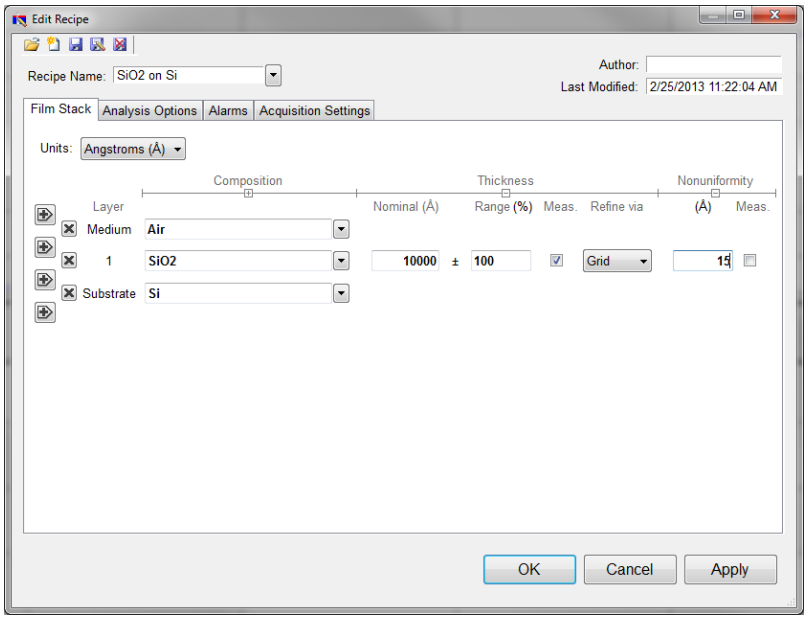
There are a number of methods that FILMeasure can use to determine thickness. Each is a different trade-off between speed, accuracy, and robustness (i.e., the ability to find the best solution among many that are nearly as good). Because thickness can vary over many orders of magnitude and many near-solutions may exist, it is often best to use a very robust method to get close to the best solution, and then let a more accurate method take over.

One very robust method is the **Grid** method, which can be activated by selecting the **Grid** option in the **Refine via** drop-down menu. The **Grid** method searches the entire allowed thickness range (as defined by the initial guess and the constraints) to find the best initial thickness. However, on some very complex multilayer spectra, it is possible for the Grid method to give the wrong answer. In such cases it is best to use the **Fourier Transform(FFT)** method to determine initial thicknesses, or to provide them manually.

2.2.7 Fourier Search for Thickness

The **FFT** option in the **Refine via** drop-down list is an alternative option to let FILMeasure choose an initial thickness for analysis. The Fourier Transform method analyzes the oscillations present in the spectrum and determines the film thicknesses based on the periodicity of those oscillations. It is generally somewhat less robust than the **Grid** method, but is better at finding the correct thickness in cases where the shape of the initial theoretical spectrum is different than the measured data (i.e., the reflectance spectrum is non-ideal in some way) or occasionally in cases where there is more than one film thickness is being measured.

2.2.8 Nonuniformity

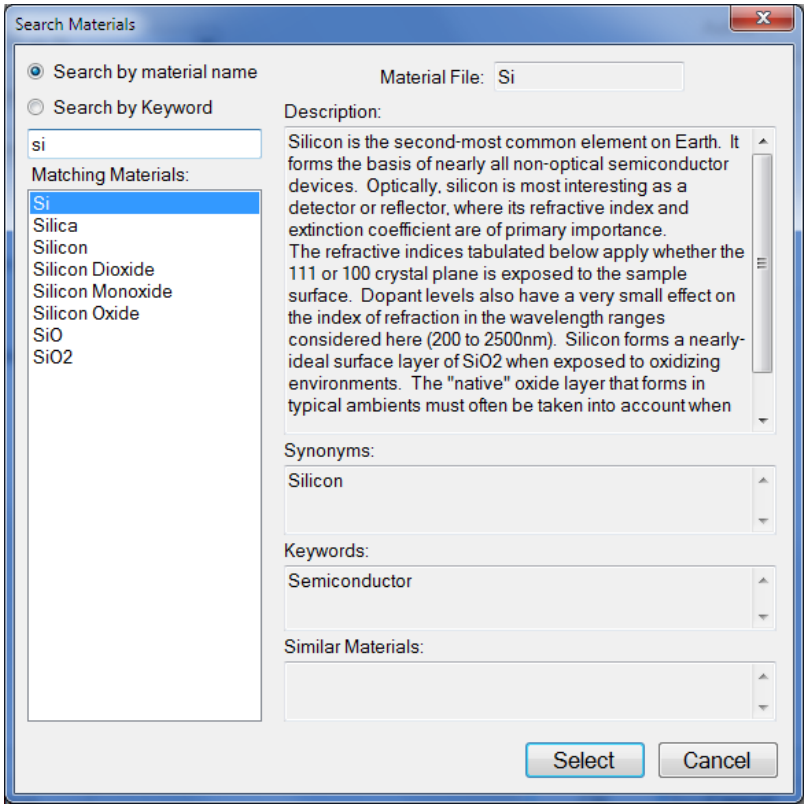


The **Edit Recipe > Film Stack** tab with the **Nonuniformity** option expanded.

Selecting this option enables modeling of thickness nonuniformity within the measurement spot. An initial guess for **Nonuniformity** must first be made, and then solving can be enabled or disabled by toggling the **Vary** box. If solving is not enabled, the software will assume the value provided for nonuniformity.

Note: Solving for nonuniformity can greatly increase solving time.

2.2.9 Search



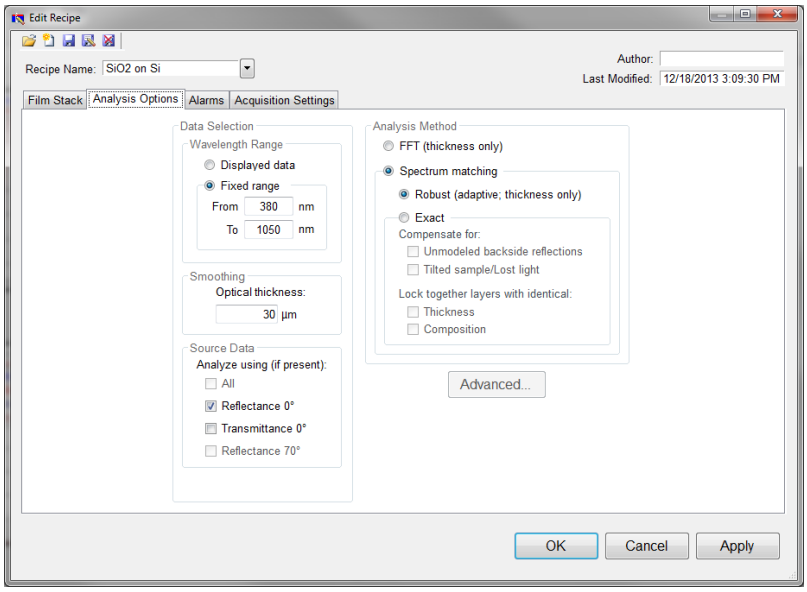
The **Search Materials** dialog box is accessible by left-clicking on a material name in the **Edit Recipe > Film Stack** tab, or by selecting **Search...** from the material selection drop-down list.

Note: When clicking on a generic index file, the software will instead provide you with a dialog to enter a new index value.

This feature allows the user to search for materials by using either the material name, or a keyword for a certain class of materials. The search function will also show a brief **Description** of the material selected, as well as any **Synonyms** or **Keywords** related to the material file. When files have a **Synonym**, that means they are referencing the same material file under different names. For example, if Silicon is selected the file that will be shown in the **Film Stack** is Si, not Silicon, as that is the file both names are referencing. The **Similar Materials** box will list any materials that may have similar optical properties.

2.3 Analysis Options

Correct setting of the following options will help ensure accurate measurements. Many of the options are set automatically when film information is supplied in the **Edit Recipe** dialog box, and all of them can be saved so that subsequent measurements can be made as quickly and easily as possible. If **Robust (Adaptive; Thickness Only)** or **FFT (Thickness Only)** is checked, only the **Data Selection** settings will be accessible.



2.3.1 Data Selection

Wavelength Range

This sets the wavelength range that is to be analyzed. It may be set as either the entire range displayed on the graph or a fixed range as defined by the user. Wavelength range may also be changed by clicking and dragging the ends of the gray bar near the bottom of the graph area in the main FILMeasure window. Moving the bar will cause the software to reanalyze the data in the newly selected wavelength range.

Smoothing

This function performs wavelength-dependent boxcar averaging on the measured spectra. This can improve signal-to-noise levels and can filter out oscillations that correspond to optical thicknesses (defined as thickness * refractive index) equal or greater than the specified value. This value should be lowered to increase smoothing, and raised to decrease smoothing. To eliminate smoothing, enter a very high value (1000 µm, for example).

Source Data

If more than one data set is present, this option tells the software whether to analyze the 0 degree reflectance data, 0 degree transmittance data, 70 degree reflectance data, or some combination of the three.

*Note: While every Filmetrics system is capable of opening and analyzing spectra with 70 degree reflectance data, only the **F10-RTA** is capable of collecting this data.*

2.3.2 Enable FFT (Thickness Only)

The **FFT (thickness only)** solver is one of three analysis methods available in the FILMeasure software.

This solver is best suited for films greater than 250 nm thick, as well as multi-layered films. It is also an effective option when the index of refraction of the film isn't well known. While this feature is selected there will be fewer options under the **Analysis Options** tab. You cannot solve for index, roughness, or nonuniformity with the **FFT (thickness only)** solver enabled. More information on the solver can be found **here**.

2.3.3 Enable Robust (Adaptive; Thickness Only)

The **Robust (adaptive; thickness only)** solver is the second of three analysis methods available within FILMeasure. It is best suited for measuring the thickness of single-layer films greater than 150 nm thick.

This method can oftentimes successfully measure films when the data is affected by non-ideal properties, such as thickness nonuniformity, grading, and birefringence. Much like the **FFT (thickness only)** solver, enabling **Robust (adaptive; thickness only)** will limit the options available in the **Analysis Options** tab. You also cannot solve for roughness, nonuniformity or index with the **Robust (adaptive; thickness only)** solver enabled.

2.3.4 Exact Spectrum Matching

The **Exact Spectrum Matching** is the third of three analysis methods available in FILMeasure. This solver can be used to measure thickness, thickness nonuniformity, and roughness of single and multilayer films, and can additionally solve for index.

Unmodeled Backside Reflections

When measuring films on transparent substrates, reflectance off of the backside of the substrate may occur. Selecting **Compensate for: Unmodeled backside reflections** allows the software to shift the intensity of the calculated reflectance curve to account for the additional light. This feature should not be used when measuring very thin films (< 100 nm).

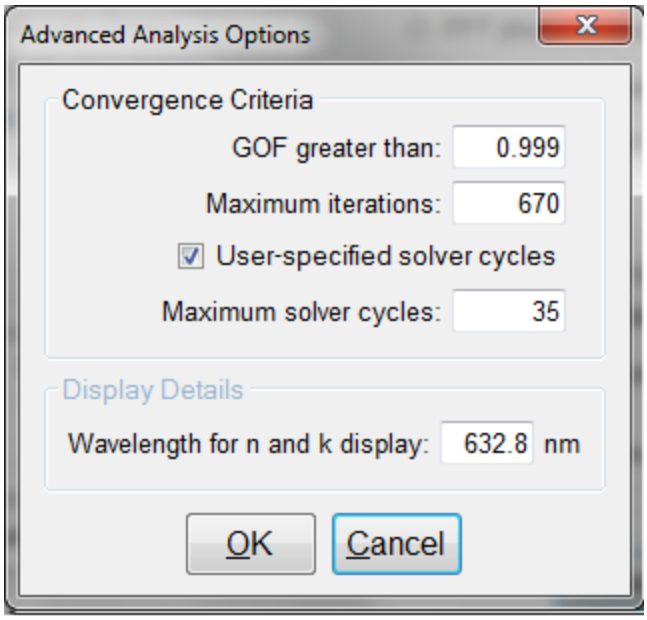
Tilted Sample/Lost Light

Samples with non-coplanar surfaces scatter some of the light away from the collection optics. This option automatically compensates for the light lost due to nonplanarity.

Lock Identical Layers

Samples comprised of repeating layers can be more accurately measured by activating this option. By activating these options, all properties of any layers that initially have the same material or thickness are locked together.

2.3.5 Analysis Options: Advanced



Advanced Analysis Options for Spectrum Matching.

Convergence Criteria

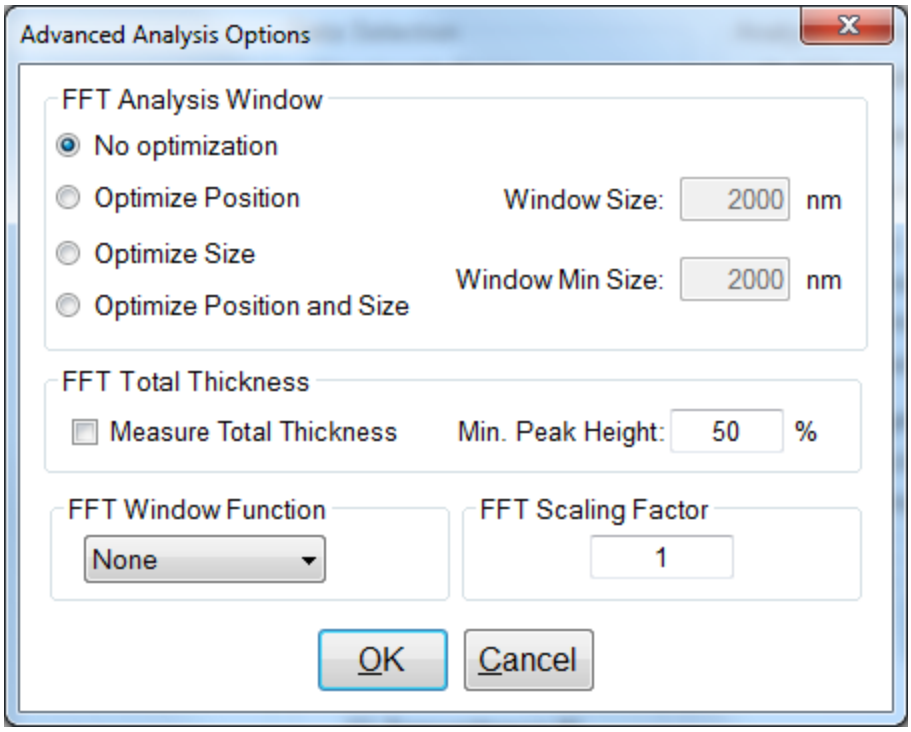
This sets the minimum **GOF** between the measured and calculated spectra that is attained before the measurement routines consider the solution final. In most cases a value of 0.999 is sufficient. For cases where the desired **GOF** is not attainable, the value in the **Maximum Iterations** field limits the number of iterations performed by the analysis routine.

Solver Cycles

This option will enable multiple solver cycles, with the number of cycles being defined by the number entered in the text box. When using this option, the software will run through the model for a solution once, and then try again starting from the solution previously found. This can help find the best solution amongst a number of near-best solutions, which can be the case when solving for a large number of parameters (several thicknesses and/or refractive index). A greater number of cycles will lead to a longer solving time.

Display Details

This feature controls the wavelength at which *n* and *k* are displayed in the **Measurement Details** portion of the main screen. These values are only displayed if solving for *n* or *k* is selected in the **Film Stack** tab.



Advanced Analysis Options for FFT Analysis.

FFT Analysis Window

When the FFT solver is selected, the advanced options give you access to various settings to help optimize analysis performance. When **Optimize Position** is selected, the solver will check for the best possible GOF value in different areas across the spectrum in a wavelength range defined by the entered window size. For example, if a window size of 200 nm is selected, the solver will look at six different 200 nm sections of the spectrum, and then select the range the provides the best GOF.

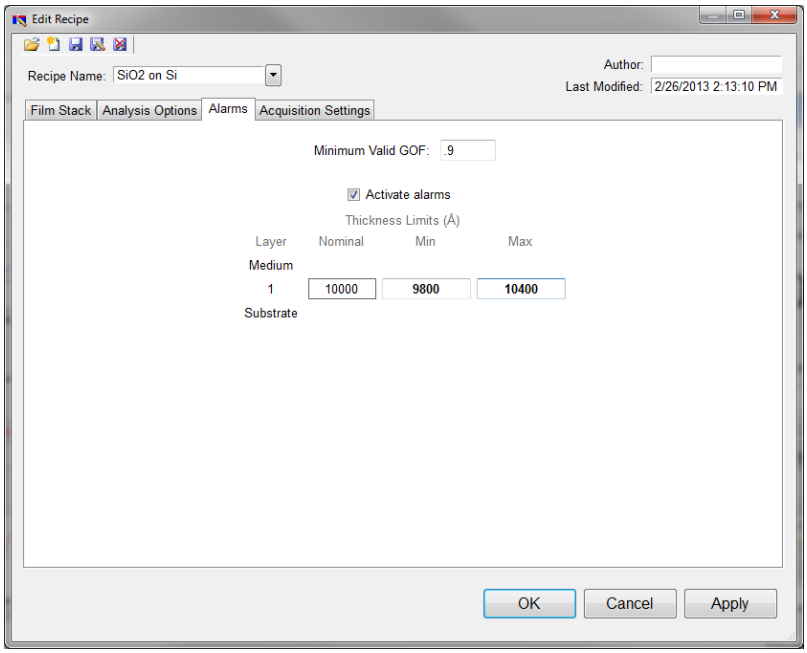
If **Optimize Size** is selected, the solver will instead begin with a minimum window size as defined by the user. It will then start at the center of the analysis range, and then extend equally in both directions six times to find the best GOF. When **Optimize Position and Size** is selected, the software does both functions, resulting in measuring thirty-one different combinations of location and analysis range.

The **FFT Total Thickness** is used when only the total thickness of a multi-layer stack is of interest. In these cases the user would define a single layer recipe, and enable the total thickness option. The solver will then perform a special analysis and look for the peak that corresponds with the thickest measured value within the current parametrs of the recipe, checked against a minimum peak height. This value is used to insure that only a true thickness peak, and not a side lobe of another peak is identified as the correct result.

The **FFT Window Function** allows you to select between four different window functions to apply to the FFT analysis. By default this feature will be disabled (*None*). Descriptions of different mathematical functions can be found **here**

The **FFT Scaling Factor** is used for non-ideal samples, where there maybe an alignment in the x-axis of the FFT plot, but not the y-axis. This may occur for many reasons such as a very rough, non-uniform, or in some other way non-ideal film. By adjusting the scaling factor, you can increase or decrease the amplitude of the modelled fit to improve your GOF values.

2.4 Alarms



Minimum Valid GOF

Constraining the **Minimum Valid GOF** will exclude measurements where the calculated **GOF** does not reach the specified limit. A warning or **"Invalid"** result will appear if the GOF constraint is not met. Specify 0 to have all solutions reported.

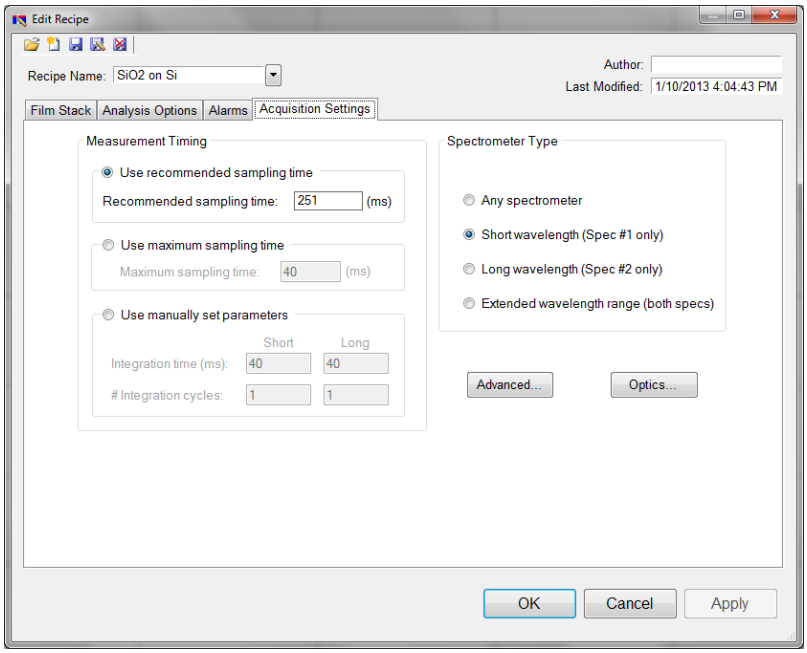
Activate Alarms

When **Activate Alarms** is selected, an alarm will trigger if the measured thickness of the sample is outside the specified **Thickness Limits**. Measurements that fall within the thickness limits will be indicated by a green box surrounding the measurement result in the **Measurement Status** window under the **Measure Tab**. A measurement that falls outside of the selected limits will instead show a red box surrounding the measurement result.

Thickness Limits

Setting the minimum and maximum allowable thickness limits will trigger an alarm if the measured thickness is outside the limits when **Activate Alarms** is selected.

2.5 Acquisition Settings



This menu lets the user control **Measurement Timing**, **Spectrometer Type** (on EXR and UVX systems),**Optics Configuration**,AutoFocus settings, and **Advanced Acquisition Settings**.

2.5.1 Measurement Timing

Use Recommended Sampling Time

Selecting this option uses the sampling time that has been automatically calculated by the software during the baseline procedure. The displayed value is the total sampling time and not necessarily the integration time. To see the integration time and number of integration cycles go to the **Help > Diagnostics** menu.

Use Maximum Sampling Time

Users can set a maximum sampling time by selecting this option. By increasing sampling time the user may see a decrease in signal noise, but at the trade-off of a longer acquisition time. Integration time and cycles will still be determined by the FILMeasure software.

Use Manually Set Parameters

Users can set both the integration time per acquisition cycle and the number of integration cycles over which to average. When using this option, the total sampling time is equal to the integration time multiplied by the number of integration cycles.

To set the integration time manually, set the sample on the stage and select **Setup>Raw Signal…** If the proper integration time is unknown, an initial value of 40ms is a good starting guess. The vertical scale of the spectrum extends from zero to 4000. Thus, a good working level is when the peak height of the sample being measured, or the reflectance standard, whichever is greater, is 3000-3500 counts. If the maximum signal is too low, increase the integration time. Too large an integration time will result in excessive dark current, so it is sometimes necessary to accept a smaller signal level to limit the dark current.

Notes:

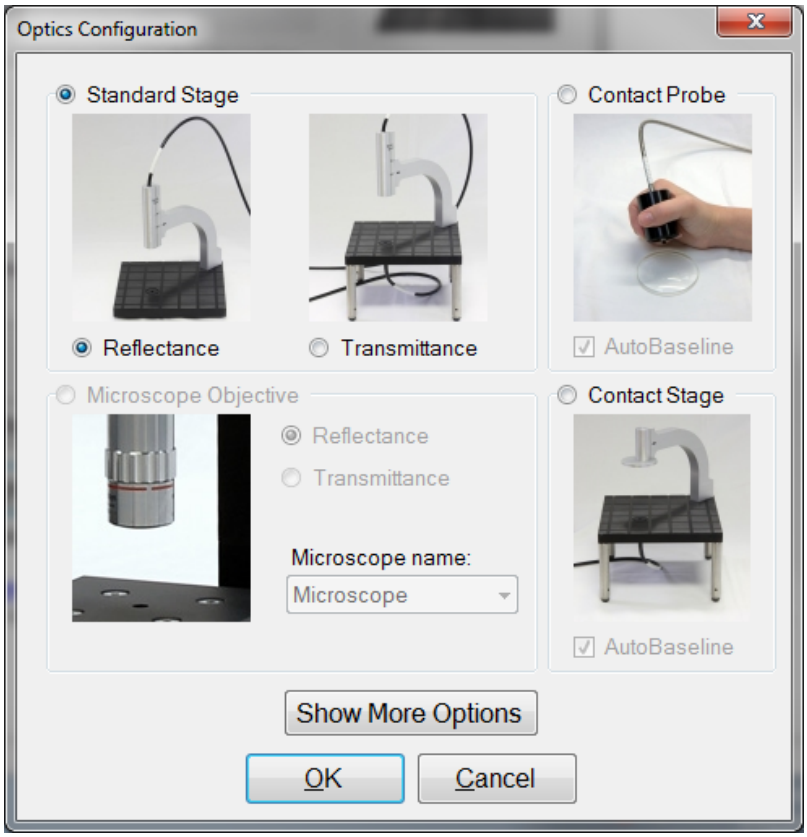
1. If the reference measurement saturates, the software will display a detector saturation error. Analyzing a saturated spectrum will result in spurious final measurements. Take a new baseline with a lowered integration time.

2. If the integration time is changed, it will be necessary to perform the baseline procedure, or re-acquire the reflectance standard and background scans.

2.5.2 Spectrometer Type

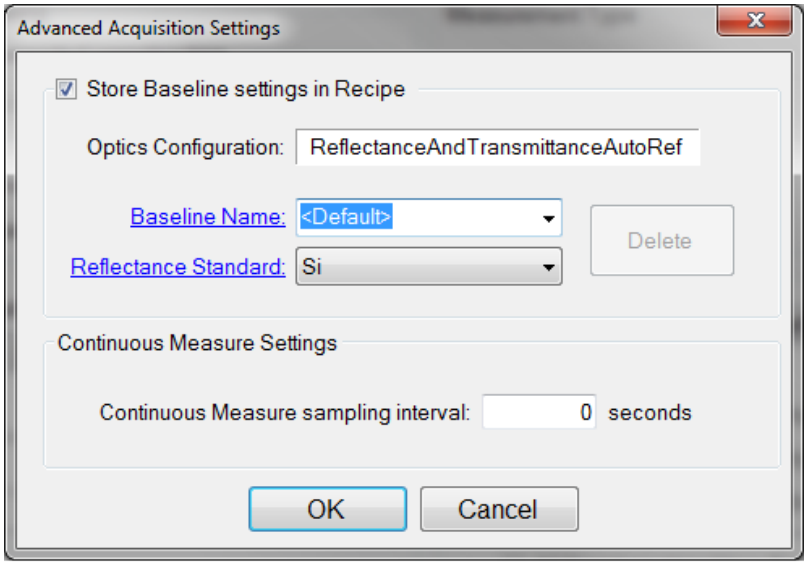
EXR and UVX systems have two spectrometers(VIS-NIR and UV/VIS-NIR respectively). Here the user can choose to use only one or both spectrometers.

2.5.3 Optics Configuration



The **Optics Configuration** dialog box is accessible through the **Acquisition Settings** tab under the **Edit Recipe** window. It is important to make sure that you remember to select the proper configuration for your measurements in order to obtain the best possible results. Not all options are available with all systems.

2.5.4 Advanced Acquisition Settings



Store Baseline settings in Recipe

This option allows the user to save the selected **Optics Configuration**, **Baseline Name**, and **Reflectance Standard** used with each individual recipe.

Continuous Measure Settings

This option allows the user to set an interval, in seconds, between measurements when using the **Measure > Continuous Measure** command from the **Acquire Menu**. By default this is set to 0.

3. Making Measurements

Accurate measurements with the F20 rely on using the proper measurement setup. The basic steps for any F20 measurement are:

1) Selecting and/or editing the **Film Stack** to be measured.
2) Taking a **Baseline** measurement.
3) Clicking on the **Measure** button to acquire and analyze the measured spectrum.

Each example below will take you through this sequence of steps. In each example it is assumed that the hardware has been set up as described in the **Quick Start Guide**, and that you have first read through the **Software Overview** section to familiarize yourself with the basic controls.

The**Focusing the Fiber**, **Measurement Assumptions**, **Hints for Improved Accuracy**, and **Troubleshooting** sections describe techniques that should be followed for the most accurate measurements.

3.1 Focusing the Fiber

In order for the system to properly make measurements, the light must first be properly focused to the height of the reflectance reference to be used. How this process is completed is dependent on the optical setup being used. Instructions for each setup are included in this section.

SS-3 Users
SS-5 Users
Kinematic Mount (KM-GL25) Users

For systems with automated z-motion controls, see the instructions **here**.

3.1.1 Focusing the Fiber (SS-3)

When the fiber-optic is fully inserted into the lens tube, the SS-3 is designed to provide optimal focus for samples the same thickness as the silicon reflectance standard. If the top surface of the object to be measured is a different height than the reflectance standard to be used, please follow the procedure below to focus the fiber-optic.

A. Turn on the light source

B. Start FILMeasure.

C. Go to the **Setup>Raw Signal…** dialog box, set the integration time for 20 milliseconds, and click **Apply**.

D. Place a reflective sample, such as a silicon or BK7 reflectance standard, on the sample stage. You will now see the raw signal from the spectrometer. If the signal exceeds 4000 counts, reduce the integration time (step C).

*** * Note * * Focus position is dependent on sample thickness. Make sure the surface you intend to measure is at the same height as the reference surface of the reflectance standard. It may be necessary to place a flat object, a washer for example, under the sample or reflectance standard in order to match surface heights.**

E. Move the fiber-optic cable slightly up the lens tube until maximum signal is obtained. Integration time may also need to be reduced as the signal increases.

The recessed set screw on the back of the lens tube is actually a spring-loaded ball plunger and does not require tightening. Users should be aware that the fiber may slip out of focus if the stage or cable is moved. If one focus height is often used, the set screw can be tightened to reduce the chance of inadvertent slipping once an optimal focus position is found.

F. Close the **Setup > Raw Signal…** dialog box. Focusing is now complete.

There is also a video demonstration of this procedure on YouTube, which can be found **here**. An active internet connection will be needed to view the video.

3.1.2 Focusing the Fiber (SS-5)

Focus height on the SS-5 is controlled by the micrometer on top of the stage arm. Rotating the micrometer in a clockwise direction will move the focus height closer to the stage, while turning it counter-clockwise will raise it. If the top surface of the object to be measured is a different height than the reflectance standard to be used, please follow the procedure below to focus the fiber-optic.

A. Turn on the light source

B. Start FILMeasure.

C. Go to the **Setup>Raw Signal…** dialog box, set the integration time for 20 milliseconds, and click **Apply**.

D. Place a reflective sample, such as a silicon or BK7 reflectance standard, on the sample stage. You will now see the raw signal from the spectrometer. If the signal exceeds 4000 counts, reduce the integration time (step C).

*** * Note * * Focus position is dependent on sample thickness. Make sure the surface you intend to measure is at the same height as the reference surface of the reflectance standard. It may be necessary to place a flat object, a washer for example, under the sample or reflectance standard in order to match surface heights.**

E. Rotate the micrometer on top of the stage arm until maximum signal is obtained. Depending on sample thickness and previous focus point, it may be necessary to rotate the micrometer in both directions before finding the proper focus point. Integration time may also need to be reduced as the signal increases.

F. Close the **Setup > Raw Signal…** dialog box. Focusing is now complete.

3.1.3 Focusing the Fiber (KM-GL25)

This procedure assumes that the lens assembly has been mounted on a KM-GL25 with normal-incidence optical access to the deposition chamber, and the "read" return fiber from the lens assembly to be adjusted is connected to the spectrometer input.

A. Turn on the light source.

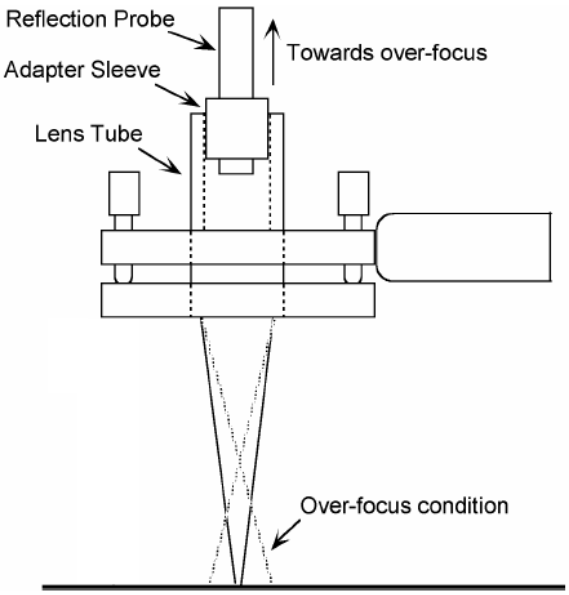
B. Start FILMeasure.

C. Make a rough focus adjustment by placing a diffuse surface (a business card works well) at the sample position and then focus the light beam on the sample position by sliding the fiber-optic cable inside the lens tube. (The diffuse reflector will allow you to see the beam.) When 6 individual spots can be seen, slide the fiber-optic cable farther into the lens assembly until the spots merge into the center of the illuminated area.

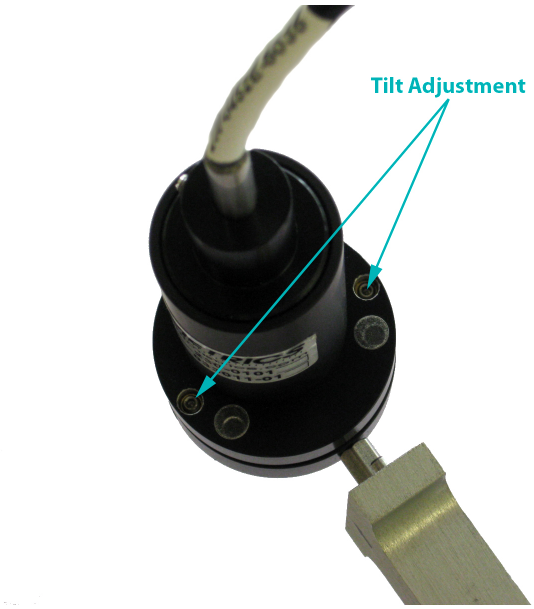
D. Go to the **Setup>Raw Signal…** dialog box, set the integration time for 20 milliseconds, and click **Apply**.

E. Place a reflective sample, such as a silicon or BK7 reflectance standard, on the sample stage. You will now see the raw signal from the spectrometer. If the signal exceeds 4000 counts, reduce the integration time (step D).

F. Optimize the focus by adjusting the fiber-optic cable position in the lens assembly until you reach maximum signal on the raw signal trace. It can be helpful to set a maximum y value of 4100 counts and disable the auto scale function on the graph in the raw signal dialog, to more easily identify when you've reached max signal. When finished, tighten the socket head set screw that secures the fiber. If the signal exceeds 4000 counts, it may be necessary to reduce the integration time. (Step D)



G. With your focus position optimized, use the tilt adjustment screws to further maximize the returning signal. Alternate making small adjustments to the two screws until the signal is optimized. If the signal exceeds 4000 counts, it may be necessary to reduce the integration time (Step D).



H. Repeat steps F and G to make certain the adjustments are optimal.

I. Close the **Setup>Raw Signal…** dialog box. Focusing is now complete.

Focusing for Systems With Automated Z-Motion

Focusing on systems with the automated z-motion controller is very similar to that of the fiber with lens assembly or microscope based systems, only with the controls for height adjustment included in the software.

1. Connect the system to the SS-XY8-Auto using the shorter supplied USB cable.

2. Connect the SS-XY8-Auto to the computer using the longer supplied USB cable.

3. Start FILMeasure.

4. Go to the **Setup>Raw Signal…** dialog box, set the integration time for 20 milliseconds, and click **Apply**.

5. Place a reflective sample, such as a silicon or BK7 reflectance standard on the sample stage. You will now see the raw signal from the image sensor. If the signal exceeds 4000 counts, reduce the integration time .

6. Adjust the focus to maximize the measured signal using the focus controls in the software.

7. If micropositioners are present on the optical assembly, F3-XXT systems for example, they may be adjusted at this time . Alternate making adjustments between the two knobs until signal is maximized. As the system is very sensitive to the tip/tilt adjustments being made with the micropositioners, it is important that adjustments are made slowly and in small increments.

8. Close the **Setup>Raw Signal…** dialog box. Focusing is now complete.

You will be prompted to focus the system during the Baseline during both the sample and reference steps. For the most accurate results, it is important to refocus to your sample after the **Baseline**.

Displaying the Measurement Spot Using the SampleCam

On systems with a SampleCam, FILMeasure can add a measurement spot with varying opacity to the live image displayed in the software. This is meant to aid users in locating the measurement spot on the sample. This feature is especially helpful on samples that don't scatter light, SiO2 on Si for example, where it can be hard to determine the measurement spot with the naked eye.

Enabling the SampleCam

The first step in displaying the measurement spot is to enable the SampleCam in the software. With the SampleCam plugged into the computer and FILMeasure launched, go to the **Setup Menu** and select **Options**. In the **Options** dialog box there will be a drop down list, showing any SampleCams that are attached to the system. Select the serial number matching the SampleCam you'd like to use, and then click OK.

You should now see a vertical bar with an arrow pointing to the right on the left side of the main graph in FILMeasure. Click on this bar to expand the SampleCam window. You will be able to choose between the **Live Video** tab, which is a live stream from the SampleCam, or the **Still Image** tab, which displays the image that was captured when the selected spectrum was taken. If there is no image associated with a spectrum, FILMeasurewill instead display a blank gray box.

Setting the Measurement Spot

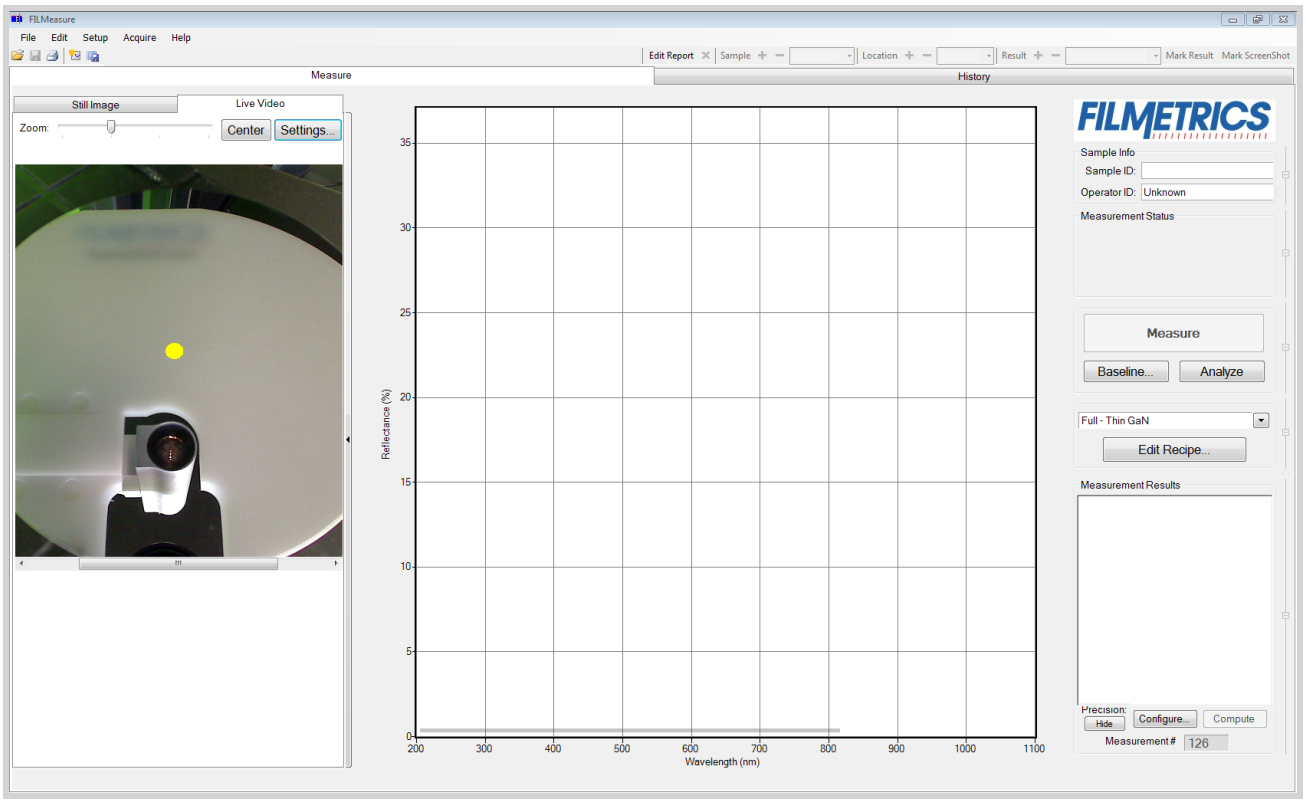
Once the camera has been enabled and you are looking at the **Live Video** tab, click on the **Settings…** button to access the **Video Settings**. From this dialog box you can begin the process to display the measurement spot by clicking on **Set Spot Location**.

When you click on **Set Spot Location**, the software will prompt you to confirm that your fibers are connected correctly. If the fiber you are using has an end labeled light source, you'll want to disconnect that end of the fiber from the light source and connect the other, unlabelled leg into the light source input. For all other fibers, you can leave the system as normally configured. You will then want to place a flat, uniform, and diffuse sample underneath the light path, a blank sheet of paper is a good example. You can then click the **OK** button to set the measurement spot.

Displaying the Measurement Spot

Once you've clicked OK, you will be brought back to the **Video Settings** dialog box. Make sure that the check box next to **Show Measurement Spot using Opacity of:** has been selected. Opacity of the measurement spot is determined by the value entered into the text field on the right, with 0% being completely transparent, and 100% being completely opaque. It is up to the user to determine what level of opacity they would prefer. Click **OK** to save any changes.

At this point, the measurement spot should be fully calibrated and ready to use.



3.2 Measuring Thickness

Examples that describe the most common measurement setups are described in this section.

Example 1: Measuring Single Layer Films.

Example 2: Measuring Multi-Layer Films.

Example 3: Measuring Films on Transparent Substrates.

Example 4: Transmittance Measurements.

Example 1: Measuring Single Layer Films

For this example we will demonstrate the measurement of SiO2 on Silicon using the TS-SS3-SiO2-8000 thickness standard provided with the system. This type of measurement does apply to an extremely broad range of applications, including hardcoats, polysilicon, and LCD cell gaps to name just a few.

Hardware:

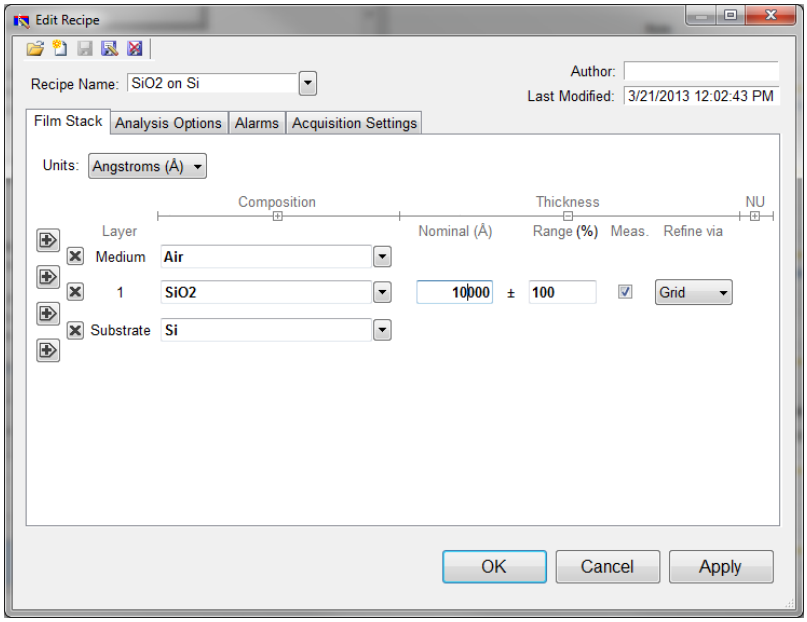
If the sample to be measured is flat, any standard or rotating sample stage will work fine. If the sample is curved, or is transparent with films on its backside (such as ophthalmic lens applications), a hand-held contact probe measurement device (e.g., **CP-1-1.3**) may be necessary.

Step 1: Select the film recipe

Select the film structure to be measured, in this case “SiO2 on Si”, from the **Recipe**: list box on the main screen. If the structure to be measured does not exist, a new structure must be defined. See **Editing Film Recipes** for more details.

Step 2: Edit the film recipe

To edit the film stack, click the **Edit Recipe...** button to open the dialog box. Check to see that the film stack matches that of the actual sample. If not, different films can be selected using either the drop down menu next to the material or the **Search** function. This is also where you can enter a starting guess for thickness as well as the desired constraint range.



Example Edit Recipe>Film Stack window

For this example, we'll use 10000 angstroms as the **Nominal** thickness, with a constraint range of 100%. Make sure that the SiO2 layer thickness is being solved for by clicking the checkbox to the right of the nominal thickness field. The substrate should be set to Si. Underneath the **Analysis Options** tab, set the **Analysis Method** to **Exact**, with nothing else enabled. Click **OK** to confirm the changes to the recipe.

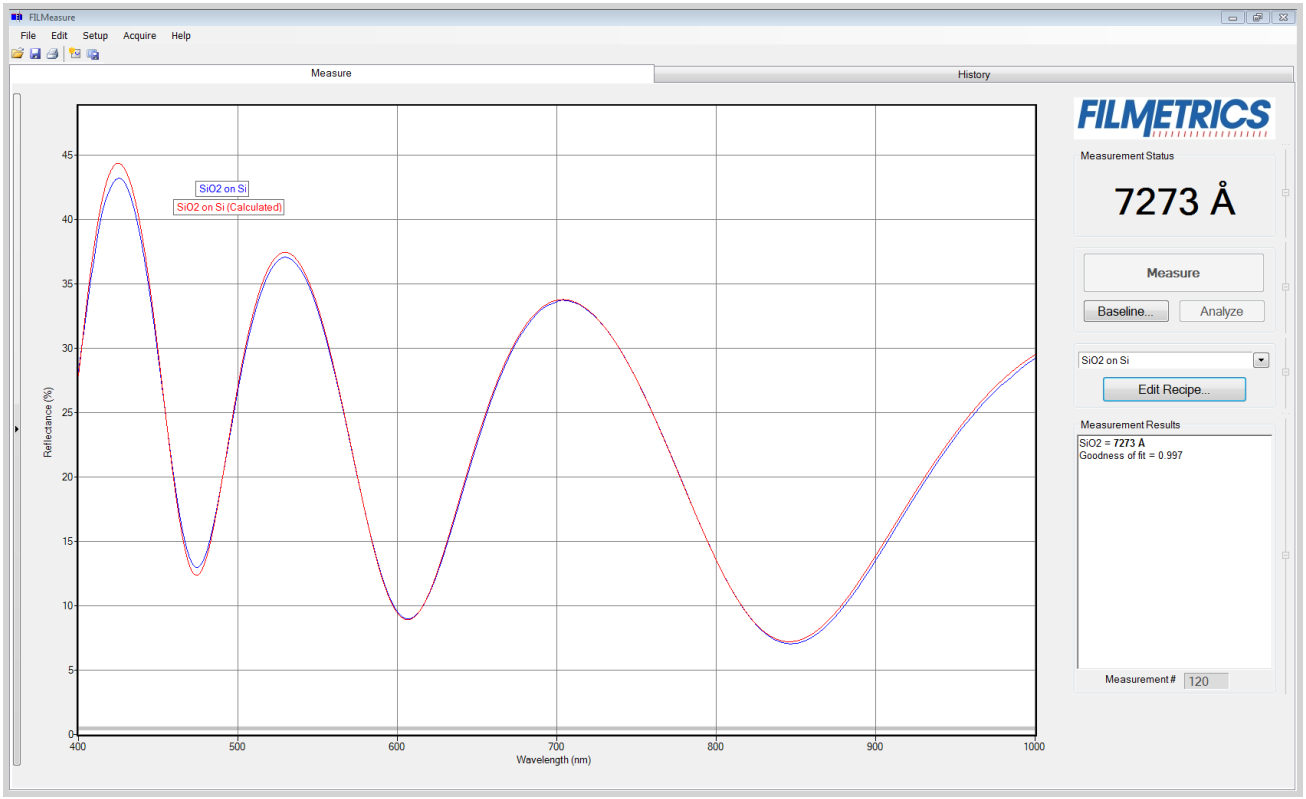
Step 3: Take a Baseline Measurement

Take a baseline measurement by first clicking on the **Baseline** button on the main screen. A dialog box will appear to step you through the process. Follow the steps on screen, and be sure to match the material selected with the reflectance standard that will be used. For this example use the Si reflectance standard.

Step 4: Make the Measurement

Make the measurement by placing your sample on the stage (or the contact probe on your sample) and click on the **Measure** button. If you're finding it difficult to align the test spot under the light path, a diffuse reflector like a piece of paper or business card can be used to help.

FILMeasure will then acquire the reflectance spectrum and calculate the corresponding thickness. If the measurement was successful, the minima and the maxima of the calculated reflectance (the red line on the graph) will coincide in wavelength with the minima and the maxima of the measured reflectance (the blue line on the graph). In most cases they will not overlap, but will be separated in amplitude.



Measured and calculated reflectance spectra when measuring the thickness of SiO2 on silicon.

If the calculated (red) and measured (blue) minima and maxima do not coincide, then the measurement was not successful. There are several possible causes of an unsuccessful measurement. The most common for this type of measurement are described in **Cases #1**, **#2**, and **#3** in the **Troubleshooting** section.

Example 2: Measuring Multi-Layer Films

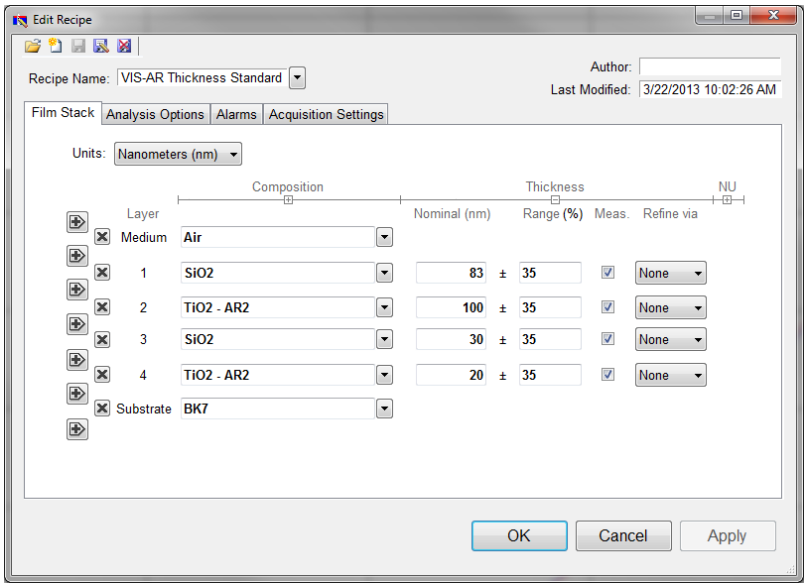
This example demonstrates the measurement multi-layer films using the VIS-AR reflectance standard. This type of measurement has a broad range of other applications, most commonly optical coating and flat-panel display process films.

Hardware:

If the sample to be measured is flat, any standard or rotating sample stage will work fine. If the sample is curved, the hand-held contact probe (e.g., **CP-1-1.3**) may be necessary.

Step 1: Select the film recipe

Select the film structure to be measured, in this case “VIS-AR Thickness Standard”, from the **Recipe:** list box on the main screen. If the structure to be measured does not exist, a new structure must be defined see **Editing Film Recipes** for more details.



Example Edit recipe>Film Stack window for measuring the VIS-AR Reflectance Standard.

Step 2: Edit the film recipe

To edit the film stack, click the **Edit Recipe...** button to open the dialog box. Check to see that the film stack matches that of the actual sample. If not, different films can be selected using either the drop-down menu next to the material or the **Search** function. If there are fewer layers present in the recipe than there are in the film stack, additional layers can be added with the + buttons on the left hand side. The X buttons can be used to remove additional layers if there are too many present. This is also where you can enter a starting guess for thickness as well as the desired constraint range.

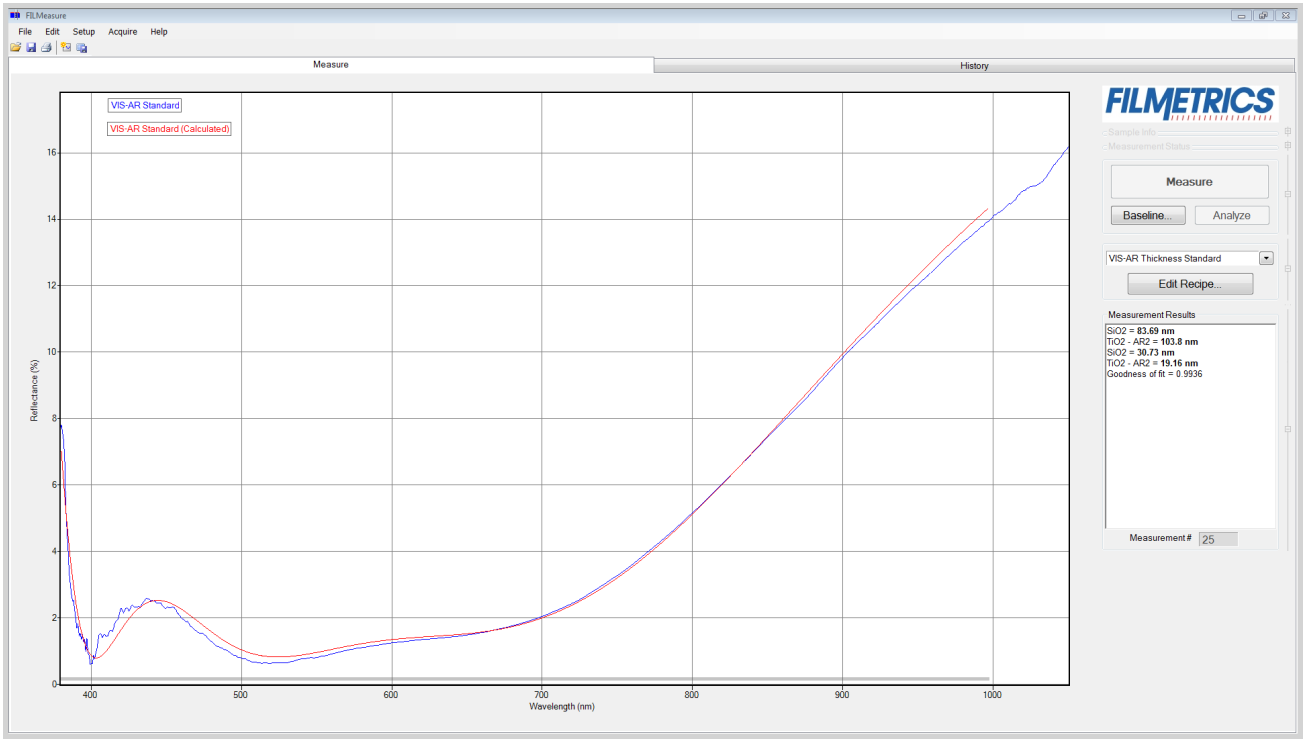
For this example, we'll use 83 nm as the **Nominal** thickness for the first layer, 100 nm for the second, 30 nm for the third and 20 nm for the forth. When solving for multiple layers, it is suggested to use smaller constraints ,35% in this case, to help improve solver speed and accuracy. Make sure that the layer thicknesses are being solved for by clicking the checkboxes to the right of the nominal thickness fields. The substrate should be set to BK7. Underneath the **Analysis Options** tab, set the **Analysis Method** to **Exact**, with nothing else enabled. Click **OK** to confirm the changes to the recipe.

Step 3: Take a Baseline Measurement

Take a baseline measurement by first clicking on the **Baseline** button on the main screen. A dialog box will appear to step you through the process. Follow the steps on screen, and be sure to match the material selected with the reflectance standard that will be used. For this example use the BK7 reflectance standard.

Step 4: Make the Measurement

Make the measurement by placing your sample on the stage (or the contact probe on your sample) and click on the **Measure** button. FILMeasure will then acquire the reflectance spectrum and calculate the corresponding thickness. If the measurement was successful, the minima and the maxima of the calculated reflectance (the red line on the graph) will coincide in wavelength with the minima and the maxima of the measured reflectance (the blue line on the graph). In most cases they will not overlap, but will be separated in amplitude.



Measured and calculated reflectance spectra when measuring the thickness of the VIS-AR Reflectance Standard.

If the calculated (red) and measured (blue) minima and maxima do not coincide, then the measurement was not successful. There are several possible causes of an unsuccessful measurement. The most common for this type of measurement are described in **Cases #1** and **#2** in the **Troubleshooting** section.

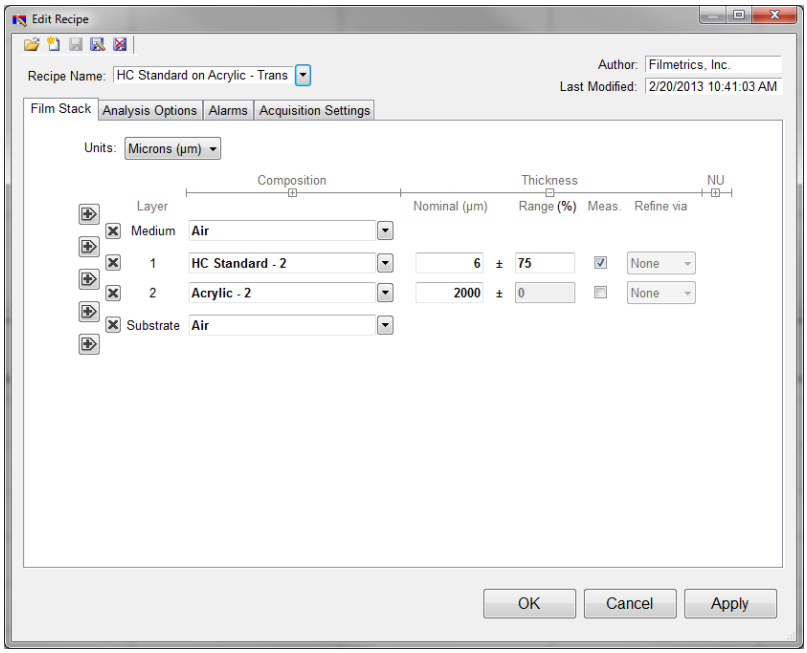
Example 3: Measuring Films on Transparent Substrates

In this example we will demonstrate the measurement of hardcoat on acrylic using the TS-Hardcoat-Trans thickness standard. This type of measurement applies to an extremely broad range of applications, including packaging films

Step 1: Select the film recipe

Select the film structure to be measured, in this case “HC Standard on Acrylic - Trans”, from the **Recipe:** list box on the main screen. If the structure to be measured does not exist, a new structure must be defined. See **Editing Film Recipes** for more information.

Step 2: Edit the film recipe



Example **Edit Recipe>Film Stack** window for measuring the thickness of a film on a transparent substrate

To edit the film stack, click the **Edit Recipe...** button to open the dialog box. Check to see that the film stack matches that of the actual sample. If not, different films can be selected using either the drop-down menu next to the material or the **Search** function. This is also where you can enter a starting guess for thickness as well as the desired constraint range.

For this example, we'll assume the measurements are being taken on a SS-3 stage. Select HC Standard - 2 as your layer 1 under the composition settings. Materials may be selected using either the drop-down list, or the **Search** function. Set the **Nominal** thickness at 6 microns, with a constraint range of 75%. Make sure that the HC Standard - 2 layer thickness is being solved for by clicking the checkbox under to the right of the nominal thickness field.

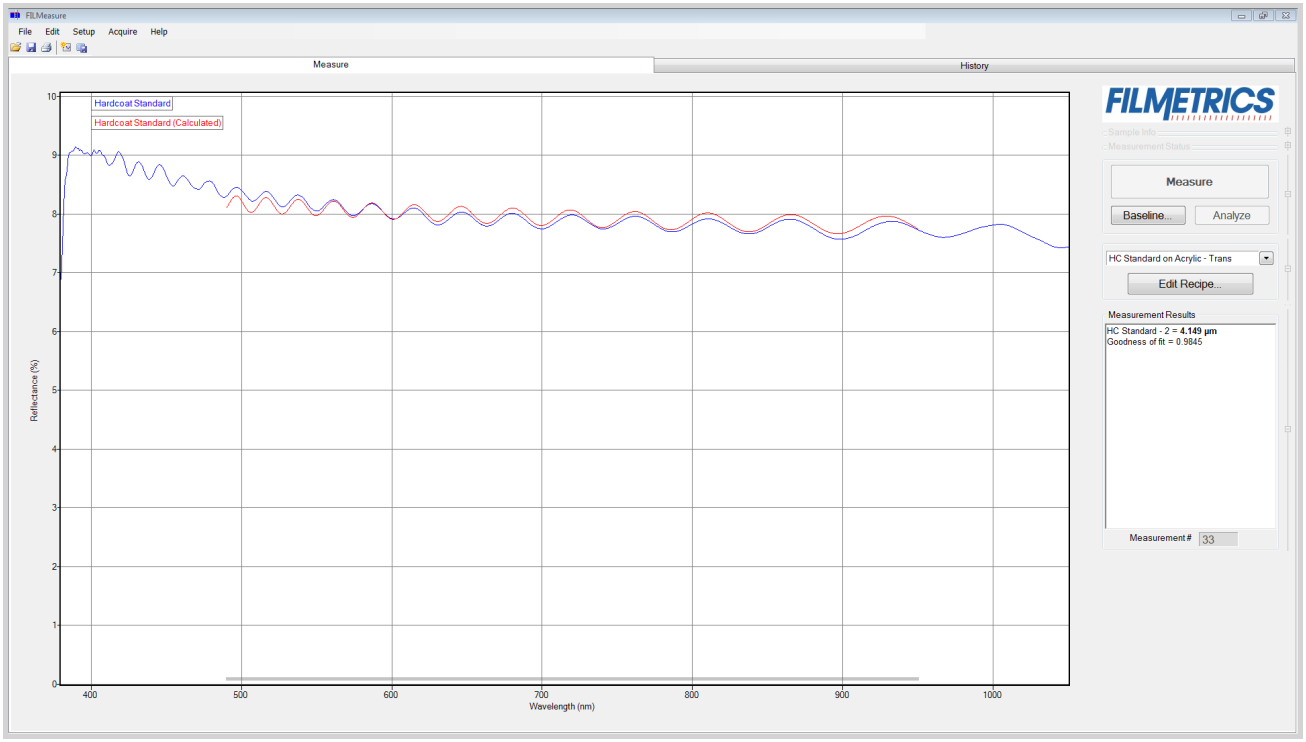
The second layer should be Acrylic - 2 with a thickness of 2000 µm, or 2 mm, with Air as the substrate. Do not enable solving for the Acrylic - 2 layer. This lets the software know that there is a finitely thick layer of acrylic, so that it will accurately account for back reflections. If you're using the **contact probe** or **contact stage** to take measurements, you can simply set the substrate Acrylic - 2 and enable account for backside reflections under the **Analysis Options** tab. Click **OK** to confirm the changes to the recipe.

Step 3: Take a Baseline Measurement

Take a baseline measurement by first clicking on the **Baseline** button on the main screen. A dialog box will appear to step you through the process. Follow the steps on screen, and be sure to match the material selected with the reflectance standard that will be used. For this example use the BK7 reflectance standard.

Step 4: Make the Measurement

Make the measurement by placing your sample on the stage (or the contact probe on your sample) and click on the **Measure** button. FILMeasure will then acquire the reflectance spectrum and calculate the corresponding thickness. If the measurement was successful, the minima and the maxima of the calculated reflectance (the red line on the graph) will coincide in wavelength with the minima and the maxima of the measured reflectance (the blue line on the graph). In most cases they will not overlap, but will be separated in amplitude.



Measured and calculated reflectance spectra when measuring the TS-Hardcoat-Trans.

If the calculated (red) and measured (blue) minima and maxima do not coincide, then the measurement was not successful. There are several possible causes of an unsuccessful measurement. The most common for this type of measurement are described in **Cases #1**, **#2**, and **#3**, in the **Troubleshooting** section.

Example 4: Transmittance Measurements

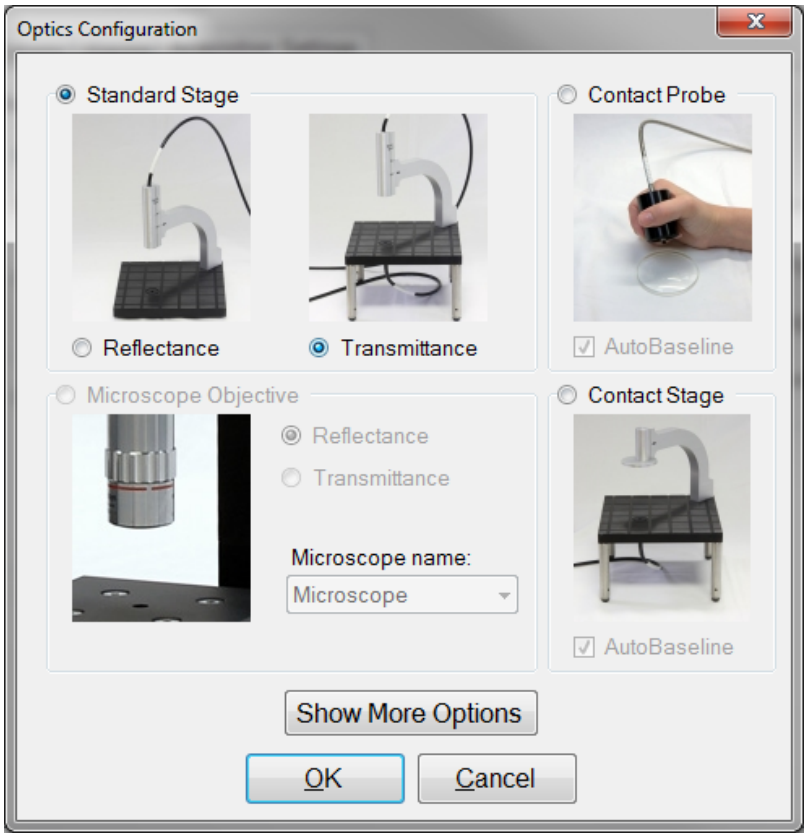
In this example we will demonstrate the acquisition of a transmittance spectrum. Modeling the data remains the same as with reflectance measurements, though analyzing the transmittance data needs to be enabled under **Analysis Options > Source Data** in the recipe.

Hardware:

For this type of measurement the sample must be flat, with the front and backsides parallel. A standard sample stage equipped with the **T-1 transmission option** is required. When measuring very thin films, use of a vacuum plugged into the vacuum port is recommended.

Step 1: Specify Transmittance Configuration

Set the **Optics Configuration** to **Standard Stage - Transmittance** in the recipe under **Acquisition Settings>Optics Config**.



Optics Configuration dialog box.

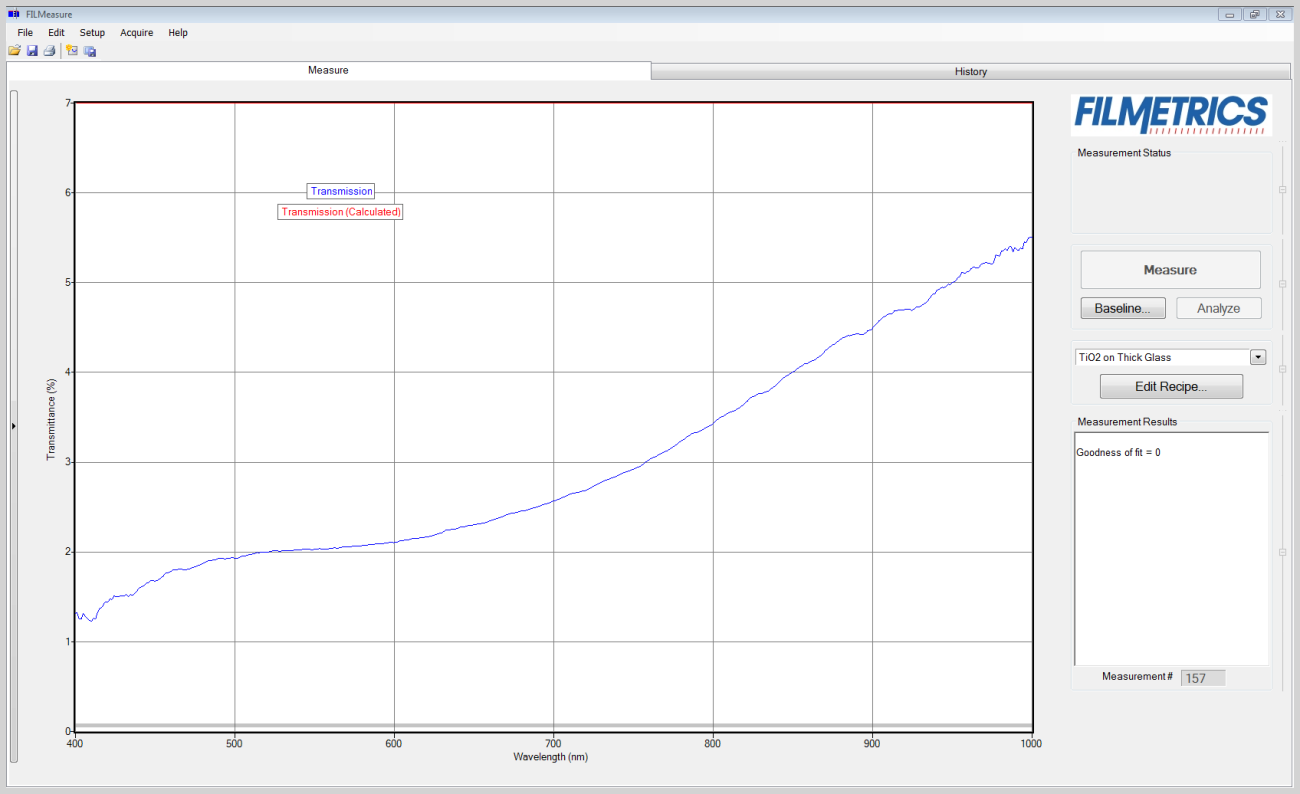
Step 2: Take a Baseline Measurement

Take a baseline measurement by first clicking on the **Baseline** button on the main screen. A dialog box will appear to guide you through the process. The first step will be a measurement of the reference material. Air will be used as our reference material. Select “Enter constant…” from the drop-down list, enter a refractive index of “1” and remove all objects from the light path. The second step is a background measurement. Place an opaque object over the bottom fiber on the state. No light should enter the fiber during the measurement.

Click “Finish” to complete the baseline procedure.

Step 3: Make the Measurement

Make the measurement by placing your sample on the stage. Select the **Acquire One Spectrum** option under the **Acquire** menu to make the measurement.



Single Acquire collects the measured spectrum (blue curve) of the sample.

3.3 Measuring Thickness and Refractive Index

The following example describes the most common setup for index and thickness measurement.

Example 1: Measuring Thickness and Optical Constants of a Film.

Example 1: Measuring Thickness and Optical Constants of a Film

For this example we will demonstrate the measurement of Si3N4 on silicon but this type of measurement has an extremely broad range of applications, including the measurement of oxides, polysilicon, and optical coatings.

Hardware:

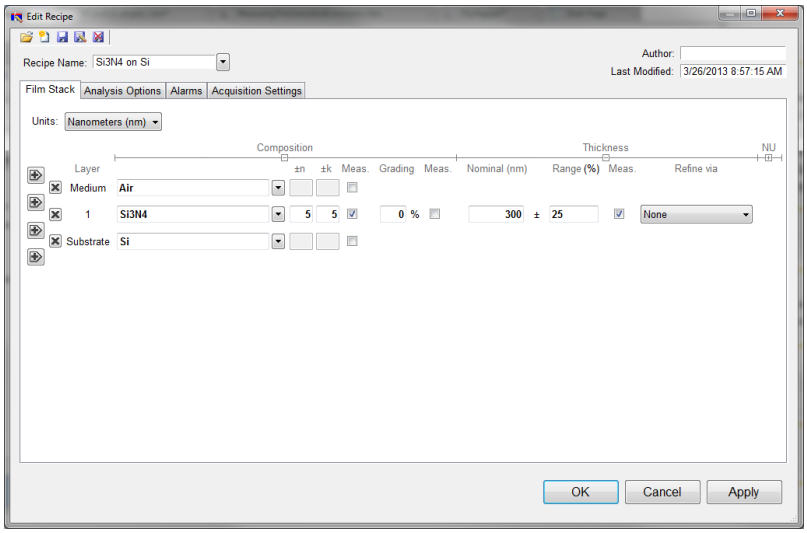
Any standard or rotating sample stage may be used. The sample must be flat. If the sample is not flat or the backside is not parallel with the front, then the contact stage or contact probe must be used.

Step 1: Select the film recipe

Select the film structure to be measured, in this case “Si3N4 on Si”, from the **Recipe:** list box on the main screen. If the structure to be measured does not exist, a new structure must be defined. See **Editing Film Recipes** for more details.

Step 2: Edit the film recipe

To edit the film stack, click the **Edit Recipe...** button to open the dialog box. Check to see that the film stack matches that of the actual sample. If not, different films can be selected using either the drop-down menu next to the material or the **Search** function. This is also where you can enter a starting guess for thickness as well as the desired constraint range.



Example **Edit Recipe>Film Stack** window for measuring the thickness, n, and k of films on an opaque substrate.

For this example, we'll use 300 nanometers as the **Nominal** thickness, with a constraint range of 25%. **None** should be selected underneath the **Refine via** drop-down list. Make sure that the Si3N4 layer thickness is being solved for by left clicking the checkbox under **Meas**.

We'll also have to enable optical constants solving under the **Composition** options. These options should be visible by default, but if they are not they can be accessed by clicking on the [+] underneath **Composition** in the **Film Stack** tab. Left click on the **Meas.** button to enable optical constants solving, and set the constraints for both n and k at 5. We'll also want to confirm that the correct model is selected. Right click on the **Meas.** button to access the nk model drop-down list, and select the Si3N4 model. For more information on optical models, see **nk Model**.

The substrate should be set to Si. Underneath the **Analysis Options** tab, set the **Analysis Method** to **Exact**, with nothing else enabled. Click **OK** to confirm the changes to the recipe.

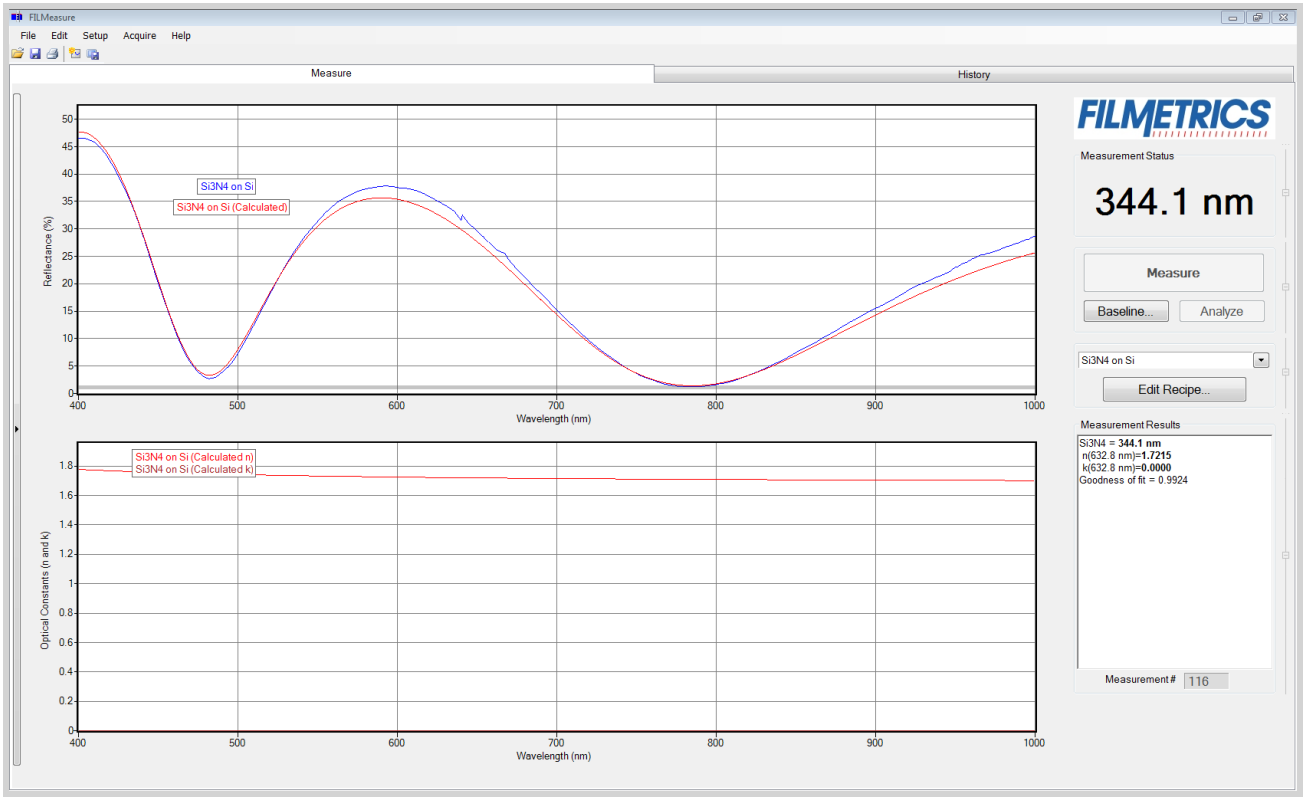
Step 3: Take a Baseline Measurement

Take a baseline measurement by first clicking on the **Baseline** button on the main screen. A dialog box will appear to step you through the process. Follow the steps on screen, and be sure to match the material selected with the reflectance standard that will be used. For this example use the Si reflectance standard.

Step 4: Make the Measurement

Make the measurement by placing your sample on the stage and click on the **Measure** button. FILMeasure will then acquire the reflectance spectrum and calculate the corresponding thickness and optical constants. If the measurement was successful, the calculated reflectance (the red line on the graph) will coincide with the measured reflectance (the blue line on the graph).

If the measured and calculated spectra do not fall on top of each other, the resulting thickness, n, and k values are incorrect. If the mismatch between measured data and calculation is only slight, the results reported will only be off by a small amount. If the measured and calculated spectra match, but the results are implausible there may be a problem with the sample positioning and light collection. Causes and corrective actions to improve the measurements are listed in **Cases #4** and **#5** in the **Troubleshooting** section.



Measured and calculated reflectance spectra when measuring the thickness, *n*, and *k* of Si3N4 on silicon

3.4 Measurement Assumptions

The following assumptions must be valid if accurate measurements are to be made with the F20:

1. Every film present in the structure is specified in the **Edit Recipe** dialog box. This includes every film present in the sample including so-called adhesion films, oxide films (unless they are less than 20 Å thick), and films on the bottom surface of the substrate if the substrate is transparent.

2. If the film is **Nonuniform**, it has been accounted for in the recipe.

3. The light source has been allowed to warm up for at least 5 minutes if measuring thickness of films greater than 250nm, or 15 minutes if measuring thickness of films less than 250nm, refractive index, or reflectance.

In addition, the following assumptions are made if optical constants and/or very thin films (<500 Å) are to be measured:

4. If any grading is present (i.e. the refractive index and extinction coefficient are constant as a function of depth and constant over the entire spot being measured), it has been accounted for in the recipe.

5. When using the SS-3 or SS-5 stage, the top surface of the sample under test is at the same height as the reflectance standard that was used to make the reference calibration measurement.

6. The sample is flat. (The vacuum port may help in this case when samples are warped.)

7. No changes to the measurement system (such as fibers being moved) or light source have occurred since the acquisition of the most recent baseline.

8. No significant changes in room temperature (> 5 degrees F) have occurred since acquisition of the most recent baseline.

If any of the above assumptions are not true, it may still be possible to make a measurement, but accuracy may be degraded.

3.5 Hints for Improved Accuracy

This section contains various tips to help improve accuracy in your measurements.

Roughness.

Restricting the wavelength range of the analyzed reflection spectra.

Measuring thickness when the approximate thickness is not well known.

Measuring photosensitive films.

3.5.1 Roughness

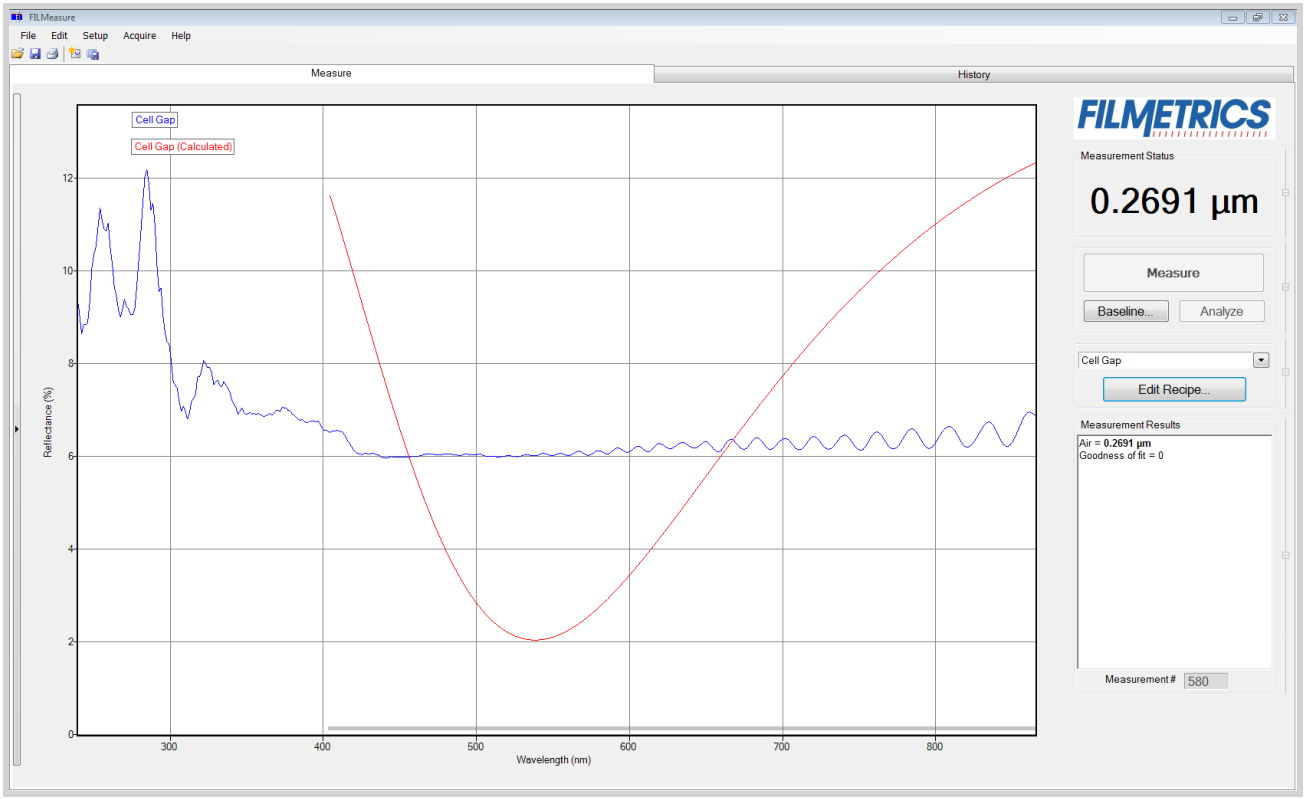
Slight amounts of surface or interface roughness that may be present can decrease the **GOF** value of a measurement. Entering a value or solving for roughness can partially account for this roughness so that a better **GOF** may be achieved. Generally, roughness is only present when the sample surface looks hazy at the measurement spot. The fact that haze can be seen means that there is scattered light. (A perfectly smooth surface will scatter no light, and thus the measured spot will not be visible.) Usually roughness less than 2.5 nm will not be visible, while roughness greater than about 25 nm will be extremely hazy.

FILMeasure assumes a Gaussian distribution of the surface height irregularities about the mean and it assumes that the roughness is small compared to the wavelength. FILMeasure reports the rms roughness value.

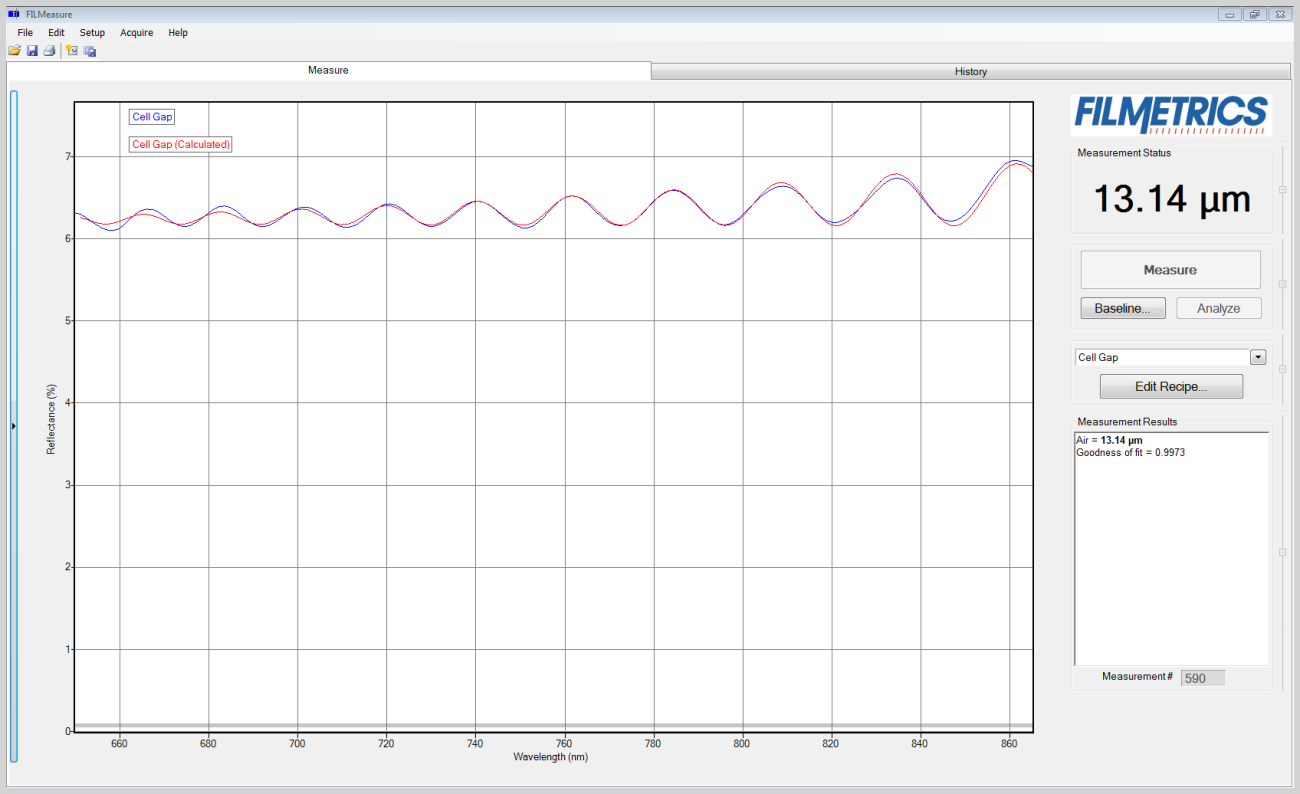
3.5.2 Restricting the wavelength range of the analyzed reflectance spectrum

Occasionally spectra from measured films are adversely affected by factors such as absorbing dyes, birefringence, or nonuniformity – all of which are difficult to model properly. It is often still possible to make accurate thickness measurements of these films by analyzing only unaffected portions of the spectra. The portion of the spectrum that is used to calculate film properties is determined in the recipe underneath the **Analysis Options** tab.

You can also adjust the wavelength analysis range in the main FILMeasure window using the gray bar at the bottom of the graph. Click and drag either end of the bar to a new position on the x-axis to adjust the analysis range.



Example of a non-ideal film (spectrum with no oscillations at lower wavelengths) that requires reduced wavelength range for accurate measurement.



Example of a reduced wavelength range for measuring thickness of non-ideal films.

3.5.3 Measuring thickness when the approximate thickness is not well known

Unless otherwise specified, FILMeasure will determine a film's thickness by finding the best answer within about 1000 Å of the initial (user supplied) value. If the approximate thickness of the film is not known to better than about 1000 Å, then the **Grid** method or the **Fourier Transform** method of determining approximate thickness may be applied. See **Editing Film Structures** for more details.

Measuring photosensitive films

For layers such as unexposed photoresist that cannot be exposed to short-wavelength light, a filter may be inserted in the slot on the light source.

3.6 Troubleshooting

For error messages or hardware problems encountered with your Filmetrics instrument, please use the chart below to best diagnose your problem. For help with modeling, see **Troubleshooting - Thickness Modeling**.

Question 1: Are you able to start FILMeasure without an error message?

No: Continue to **Question 2**.
Yes: Continue to **Question 1-1**.

Question 1-1: What sort of problem are you seeing?

1: The lamp doesn't turn on or the fan isn't working properly. See the **Non-Operating Box, Lamp, or Fan Troubleshooting Guide**.
2: I get an error message when I click on something. Continue to **Question 1-1-2**.
3: Measurement results are wrong or unstable, after they had been good. See the **Measurement Results are Incorrect or Unstable Troubleshooting Guide**.
4: Measurement and/or baseline takes longer than it used to. See the **Slow Operation Troubleshooting Guide**.

Question 1-1-2: If you're still seeing an error after you've power cycled and installed new software, then select the description which most closely describes your error.

1: The error occurs during the baseline. See the **Error Message During Baseline Troubleshooting Guide**.
2: Some other error. See the **Error Messages -Miscellaneous Troubleshooting Guide**.

Question 2: Are you able to launch FILMeasure but it shows an error?

Yes: Continue to **Question 2-1**.
No: Then you need to install the software that come with your new system, or you can request new software. See the **Software Reinstallation Guide**.

Question 2-1: Does the error message start with "No Filmetrics instruments detected...."?

Yes: See the **Non-Operating Box, Lamp, or Fan Guide**.
No: See the **Error Messages - Miscellaneous Troubleshooting Guide**.

3.6.1 Power Cycling

Over the years we've found that about a third of support issues can be resolved by simply power cycling the Filmetrics system and computer. Follow the simple steps below to power cycle your instrument. Further assistance may be received by contacting us using the methods described in the **Contact Information** section.

Step 1: Shut down FILMeasure (it's OK in the rare case that you can't).

Step 2: Unplug all electrical connectors from the back of your Filmetrics box.

Step 3: Restart your computer, giving it time to load all drivers.

Step 4: Reconnect the electrical connections to your Filmetrics box. (Make sure that the power cord is getting power, and that any switch near the power connector is turned on.)

Step 5: Restart FILMeasure

3.6.2 Software Reinstallation Guide

Follow the steps below to reinstall FILMeasure. Further assistance may be received by contacting us using the methods described in the **Contact Information** section. *Warning: Upgrading software versions may result in slightly different (<1%) measurement results due to continual improvement of our solving algorithms and material files.*

Step 1: Shut down FILMeasure (it's OK in the rare case that you can't).

Step 2: Unplug all electrical connectors from the back of your Filmetrics system.

Step 3: Are you installing this software on a computer which already has FILMeasure installed?
Yes: Go to Step 4.
Yes: Jump to **Step 6.**

Step 4: Make a backup copy of your recipe and material files:
FILMeasure versions 1 thru 4: Backup the directory `C:\Program Files\FILMeasure`
FILMeasure versions 5 and up:
On Windows XP: Backup the directory `C:\Documents and Settings\All Users\Application Data\Filmetrics`
Note: The application Data folder may be hidden by default.
On Windows 7/8/Vista/10: Backup the directory `C:\Program Data\Filmetrics\`
Note: The Program Data folder may be hidden by default.

Step 5: Uninstall the current version of software:
On windows XP: Click Start > Control Panel > Add or Remove Programs > FILMeasure > Uninstall
On Windows 7/Vista: Click the Start orb > Control Panel > Programs and Features > FILMeasure > Uninstall
On Windows 8: Control Panel > More settings > Programs and Features > FILMeasure > Uninstall

Step 6: Run the software installer and follow installer instructions.
Note: you will need administrative privileges to install the software.

Step 7: Reconnect your Filmetrics system.
*Note for Windows XP: The "New Hardware Found" wizard may appear if your instrument has a USB connection. You **must** click "Next and complete the driver installation process. This dialog box may appear multiple times depending on the system configuration.*
Note for Windows 7/8/Vista: An Information balloon may appear over the taskbar while drivers are being installed. Please wait until the balloon indicates the drivers have been installed before starting FILMeasure.

Step 8: Run the software to verify the installation was successful.

3.6.3 Non-Operating Box, Lamp, Or Fan

Follow the steps below to troubleshoot symptoms of a non-operating spectrometer, light source, or fan. These are usually caused by having too much or not enough light. Further assistance may be received by contacting us using the methods described in the **Contact Information** section.

Step 1: Does the green LED on the right side of the front panel illuminate?

Yes: Jump to **Step 3**.
No: Go to Step 2.

Step 2: Is the power cord fully inserted into the back of the unit and a powered wall outlet? If your unit has a rear power switch, is it on?

Yes: Go to Step 3
No: Plug into a powered outlet or switch on the unit

Step 3: Does the LED light on the light source switch illuminate when switched on?

Yes: go to Step 4
No: Possible faulty switch or faulty indicator lamp, contact **Filmetrics**.

Step 4: Is the cooling fan spinning? Determine this by listening to the cooling fan which is located directly below the light source. (You'll need to get close to hear the fans on systems built after 2010.) For F50 systems the chassis cooling fan is louder than the light source cooling fan. You'll need to get close to the light source to hear the light source cooling fan.

Yes: Go to Step 5.
No: The cooling fan needs replacement, contact **Filmetrics**. *Note: a bad cooling fan can draw enough power to prevent the light source and box from operating properly.*

Step 5: Does the light source illuminate?

Yes: Go to Step 6
No: The lamp has burned out. Please replace the lamp following the instructions **here**. A spare lamp is included with most systems and additional lamps may be purchased on our website **www.filmetrics.com***Note: users can confirm a bad halogen lamp by removing it and checking it for electrical continuity.*

Step 6: The following steps are to troubleshoot a box that is not being detected by the FILMetrics software. This could be a software driver or (rarely) a hardware problem. To see what hardware your computer is detecting, look in the windows device manager. (instructions on how to access the Device Manager are included at the end of this document.)

In the Device Manager, which device do you see?

"Filmetrics Measurement Instrument" or similar: In this case the system should be operation correctly. If not, please contact **Filmetrics**.
"Unknown Device" or similar: The hardware appears to be operating properly, but the drivers for the hardware are not installed. If you have already power cycled and installed the most current Filmetrics software, please contact **Filmetrics** for further assistance.
Neither of the above: The hardware is not working properly. Go to Step 7

Step 7: Try a new USB cable and plug it into a different USB port. Does this fix the problem?

Yes: Either the USB cable or port was bad.

No: The USB controller inside the FILMetrics box may be bad. Please contact **Filmetrics**.

Accessing Device Manager in Windows 7:

1. Click on the Start Orb.
2. In the Star Search box type: device manager. Press enter.
3. Select Device manager from the list.
4. You should see a Filmetrics Measurement Instrument tab. Expand the tab and the appropriate driver should be listed for the Filmetrics instrument.

Accessing Device Manager in Windows XP:

1. Click Start.
2. Click on Control Panel
3. In the Control Panel double-click the Systems icon.
4. In the Systems Properties window click on the Hardware tab.
5. In the Hardware tab click the Device Manager button.
6. You should see a Filmetrics Measurement Instrument tab. Expand the tab and the appropriate driver should be listed for the Filmetrics instrument.

3.6.4 Error Message during Baseline

Follow the steps below to troubleshoot error messages encountered while taking a baseline. These are usually caused by having too much or not enough light. Further assistance may be received by contacting us using the methods described in the **Contact Information** section.

Step 1: Which error message are you seeing while taking a baseline?

Received light signal is too bright: If you are manually setting the integration time (see **Measurement Timing**) then you need to reduce the integration time. Otherwise you can reduce the light source intensity by using a flattening or neutral density filter. Optical filters can be purchased on our website, **www.filmetrics.com**.

Baseline Failure. Reference and background spectra are almost the same: This error is often caused by using the reference sample during the background step or vice versa, and is easily corrected by taking retaking the baseline while providing the proper material for the proper step. If this is not the case, go to Step 2.

Spectrometer peak intensity is x% of previous reference intensity: This error is often caused by using the incorrect reference sample and is easily corrected. Jump to **Step 5**.

Step 2: Does the light source illuminate?

Yes: Go to Step 3.

No: Check the **Non-Operating Box, Lamp, Or Fan Troubleshooting Guide**.

Step 3: Check to see if there is a broken or misconnected fiber. With the light source on, confirm that the illumination fiber is screwed into the light source securely, and that plenty of light is exiting the fiber at the far end near where the sample is placed. The light can be seen by placing paper (e.g., business card) near the sample location (do not look directly into the fiber if you are using a UV source). If plenty of light is seen, next check the return fiber by removing it from the spectrometer input and connecting it to the light source (removing the other fiber first). You should see light at the far end as before, but it may be 3-4 times smaller and dimmer. Did you see light at the sample end using both fibers in the light source?

Yes: The fiber appears to be good. Replace the fibers securely in their original positions (there will be a white band on the light source fiber if the two ends are different) and go to Step 4.
No: The fiber optic may be broken. Replacements can be purchased on our website, **www.filmetrics.com**

Step 4: Does your system have multiple spectrometers (i.e. EXR or UVX)

Yes: Check the **Acquisition Settings** tab under **Edit Recipe** to make sure that your fiber optic configuration matches the enabled spectrometers. Do this by making sure that each enabled spectrometer has a fiber attached to it (or disable a spectrometer that is not being used).
No: If you are still experience problems, please contact us using the **Contact Information** section.

Step 5: Was the correct reference selected from the drop-down list?

Yes: Go to Step 6.
No: Re-baseline using the correct reference.

Step 6: Has the light source lamp recently been changed?

Yes: This error is an expected response, no further action required
No: Go to Step 7.

Step 7: Has the optical configuration been altered (i.e. switching from standard spot size to small spot optics or a contact probe)?

Yes: This error is an expected response, no further action required
No: If you are still experiencing problems, please contact us using the **Contact Information** section.

3.6.5 Measurement Results are Incorrect or Unstable.

Follow the steps below to troubleshoot when measurement results are incorrect or unstable after the results had been good for some time. Additional help troubleshooting general measurement issues may be found in the **Troubleshooting: Thickness Modeling** section. Further assistance may be received by contacting us using the methods described in the **Contact Information** section.

Step 1: First confirm that you have selected the correct recipe for the sample that you are measuring (i.e., the same recipe that previously was giving good measurements). *Note that this troubleshooting guide does not cover instances where recipe parameters have been changed.*

Step 2: Confirm that the light source is turned on and that the fibers are securely tightened into the light source and the spectrometer input. Confirm that the fiber is properly inserted into the stage (See **System Configurations** if you are uncertain how to do this). Place a piece of paper at the sample measurement location and confirm that you can see a light beam. Can the light beam be seen?

Yes: Jump to **Step 4**
No: Go to Step 3

Step 3: Is there light coming out of the light source? (If you are checking a UV light source, put on the goggles that were supplied with the light source before checking.)

Yes: It appears the fiber may be broken. Replacements can be purchased on our website **www.filmetrics.com**
No: Replace the lamp following the instructions **here**. A spare lamp is included with most systems and additional lamps may be purchased on our website **www.filmetrics.com***Note: users can confirm a halogen bad lamp by removing it and checking it for electrical continuity.*

Step 4: Perform a new baseline, paying particular attention to make sure that you are using the proper reference and that it is clean with the correct side up.

Step 5: Put a sample that is known to be good on the stage and perform a measurement. If the measurement is still not correct, go to Step 6.

Step 6: Your lamp may be weak and needs to be replaced. The best way to tell is to replace it with the instructions found **here**. A spare lamp is included with most systems and additional lamps may be purchased on our website **www.filmetrics.com**.

Step 7: It may be that the stage optics have become unfocused. Please refer to the **Focusing The Fiber** section.

If you are still experiencing problems with an unstable or incorrect measurment after the preceding steps, please contact us using the methods described in the **Contact Information** section.

3.6.6 Slow Operation Troubleshooting Guide

Follow the simple steps below to troubleshoot slow system operation. Further assistance may be received by contacting us using the methods described in the **Contact Information** section.

Step 1: When did you experience the slow operation?

During a baseline: Go to Step 2.
When I click Measure: Jump to **Step 7**.

Step 2: Slow baselines are generally caused by a low light level, which can have several causes. Does the light source illuminate (be sure it is switched on, and that the shutter is open on UV systems)

Yes: Go to Step 3.
No: Check the **Non-Operating Box, Light, or Fan Troubleshooting Guide**.

Step 3: Check to see if there is a broken or misconnected fiber. With the light source on, confirm that the illumination fiber is screwed into the light source securely, and that plenty of light is exiting the fiber at the far end near where the sample is placed. The light can be seen by placing paper (e.g., business card) near the sample location (do not look directly into the fiber if you are using a UV source). If plenty of light is seen, next check the return fiber by removing it from the spectrometer input and connecting it to the light source (removing the other fiber first). You should see light at the far end as before, but it may be 3-4 times smaller and dimmer. Did you see light at the sample end using both fibers in the light source?

Yes: The fiber appears to be good. Replace the fibers securely into their original positions on the spectrometer box and light source (there will be a white band on the light source fiber if the two ends are different) and go to Step 4.

No: The fiber optic may be broken. Replacements can be purchased on our website, **www.filmetrics.com**

Step 4: Does your system have multiple spectrometers (i.e. EXR or UVX)

Yes: Check the Acquisition Settings tab under Edit Recipe to make sure that your fiber optic configuration matches the enabled spectrometers. Do this by making sure that each enabled spectrometer has a fiber attached to it (or disable a spectrometer that is not being used).

No: Go to Step 5.

Step 5: It may be that your lamp is old and needs to be replaced. The best way to tell is to replace it with the instructions found **here**. A spare lamp is included with most systems and additional lamps may be purchased on our website **www.filmetrics.com**.

Step 6: It may be that the stage optics have become unfocused. Please refer to the **Focusing The Fiber** section. Otherwise, go to Step 7.

Step 7: Unless the slow operation during measurement is a new problem, please confirm that your computer meets the minimum system requirements as listed in the **system specifications**. If it doesn't, please upgrade your computer. Otherwise, go to Step 8.

Step 8: A measurement that takes a long time generally means that there is a large amount of processing required to find a solution, which is usually caused by a non-optimal recipe setting. Please first confirm that you have selected the correct recipe for the film stack you are measuring. Otherwise, go to Step 9.

Step 9: Open the recipe and make sure that you are not solving for too many parameters. Solving for one or two thicknesses or a thickness and index should only take a second or two, but in some circumstances the solving time can be increased 5-10x for each additional parameter. Solving for nonuniformity (or adding it to increase accuracy of other parameters) can be especially time consuming.

Step 10: If you are solving for more than one thick layer (i.e., >10 um), you can speed up the solving process by reducing the thickness constraint range (which can be safely reduced even more by using an accurate nominal thickness guess)or by switching from using the **Grid** solver to the **FFT** solver in the recipe.

If you are still experiencing problems with a slow measurement after the preceding steps, please contact us using the methods described in the **Contact Information** section.

3.6.7 Error Messages - Miscellaneous Troubleshooting Guide

This guide provides extended information on error messages a user may encounter on rare occasion while using our software. Further assistance may be received by contacting us using the methods described in the **Contact Information** section.

Error during spectrum analysis/ unable to analyze spectrum.

This error is generally caused by a problem with the recipe constraints. Try adjusting the solver wavelength range, thickness constraint range, or starting thickness value. Also make certain that the starting thickness values have been entered using the correct units of measure specified for the recipe.

Spectrum saturated.

This error indicates that the sample measured is more reflective than the spectrometer was expecting. Reflectance data may not be accurate as a result. Take a new baseline to fix this problem.

3.6.8 Troubleshooting: Measurement Results

In all cases it has been assumed that the user has read the **Measurement Assumptions** section, and that each criterion has been met.

Case #1: No oscillations, or portions of oscillations, are present in the measured reflectance spectrum.

Case #2: The measured reflectance spectrum has periodic oscillations across the entire screen, but its minima and maxima do not match up with the calculated reflectance.

Case #3: Two or more different, but nearly the same, thickness readings are obtainable from the same measurement location.

Case #4: Poor matches between the measured and calculated reflectance spectra when measuring thickness and optical constants.

Case #5: Several different answers, or one unreasonable answer, are found when measuring thickness and optical constants, even though a good match is found between the measured and calculated reflectance spectra.

Case 1: No oscillations, or portions of oscillations, are present in the measured reflectance spectrum.

There can be several possible causes in this case;

1. The film may just be very thin (<200 Å), thus presenting no discernible oscillations or only one identifiable minima or maxima. The FFT solver will not be able to accurately analyze films without at least one discernible full oscillation, in general meaning a film less than 0.2 microns thick. Use one of the **Spectrum Matching** solver methods (**Robust**, **Exact**) to analyze very thin films.

2. If the film is very rough, non-uniform in thickness over the measurement area, or has a graded interface it may not support coherent optical interference and thus it may not be able to be measured. Try moving to a different measurement spot for a better signal.

3. An improper baseline measurement was taken, for example a thickness standard is presented as a reflectance standard. Re-baseline the system while taking care to select the proper reflectance standard and reference file.

Case 2: The measured reflectance spectrum has periodic oscillations across the entire screen, but its minima and maxima do not match up with the calculated reflectance.

The most common cause is that the initial thickness guess was considerably different than the actual film thickness, and due to the constraints on possible thicknesses, FILMeasure was unable to find the correct answer.

To better understand this, it helps to know that the number of oscillations on the screen is proportional to the film thickness. For example, if the measured spectrum has roughly twice the number of the oscillations that the calculated spectrum has, then the measured film is roughly twice the thickness of the calculated thickness. Using this information, the initial thickness guess and the thickness constraints can be set more appropriately in the **Edit Recipe** dialog box. Another possible cause is found in **Case #3** (incorrect dispersion).

Case 3: Two or more different, but nearly the same, thickness readings are obtainable from the same measurement location.

When this occurs usually the measured and calculated spectra match somewhat, but not very well across the entire spectra (i.e., the measured and calculated spectra match exactly only over a small wavelength range and then gradually walk off one another outside this wavelength range). When thickness is the only value being measured, this is normally caused by the refractive index of the measured layer not matching that used by FILMeasure, especially when the film is greater than one micron thick. Unless more accurate index values can be obtained, the best way to solve this problem is to restrict the wavelength range used in the analysis.

Case 4: Poor matches between the measured and calculated reflectance spectra when measuring thickness and optical constants.

There can be many causes of this problem, including those listed in **Case #1** and **Case #2** above. Most commonly when a poor fit between the spectra occurs, it is because:

1) The components of the film structure are not all included in the **Edit Recipe > Film Stack** dialog box. Adjust the recipe accordingly and reanalyze the spectrum.

2) Very inaccurate initial guesses for the film thickness(es) have been listed. Try a new nominal starting guess, or expand the thickness constraint range.

3) The optical constants listed in the **Edit Recipe > Film Stack** dialog box are far from the actual optical constants in the material. Try selecting a different but similar material file if one is present, for example switching from Acrylic to Acrylic - 2.

4) The film being measured has properties that are not taken into account by FILMeasure. Examples of these properties are graded interfaces and non-uniform films. You can enable solving for these conditions in the recipe.
Note: Enabling Nonuniformity solving will increase solving time.

Case 5: Several different answers, or one unreasonable answer, are found when measuring thickness and optical constants, even though a good match is found between the measured and calculated reflectance spectra.

This normally occurs when a large number of properties are being measured on a very thin film. In general, the thinner a film is the less unique information that can be obtained from it. To understand this, see **Theory of Operation**. To solve the problem, you can either:

A. Decrease the number of variables solved for.
B. Increase the amount of data provided to the solver (analyzing reflectance and transmittance at the same time, or using the multi-spectral solver).
C. Enable multiple **Solver Cycles** under the **Edit Recipe > Analysis Options > Advanced** dialog.

4. Understanding and Evaluating Measurement Results

This section covers information on the **Display of Measured Spectra** and a more in depth look at the **Goodness Of Fit** value.

4.1 Display of Measured Data

Once film stack and measurement information have been entered and a baseline has been taken, measurements may be made by clicking on the **Measure** button. After measurement, the measured and calculated reflectance spectra are displayed on the graph, and the thicknesses of the films are listed in the results box. If any of the thickness values were measured, they are displayed in bold numbers.

The quality and accuracy of the result is determined by how well the measured spectrum matches with the calculated spectrum. In general, thickness information is determined by the number of oscillations, and index information is determined by the amplitude of the oscillations. In the case of the **Robust (adaptive; thickness only)** solver and **Exact Spectrum Matching** solver, a good result is one where the minima and maxima of the measured and calculated spectrum align. For the **FFT (thickness only)** solver, this is when the measured intensity peaks overlap. See the **Goodness of Fit** and **Troubleshooting - Thickness Modeling** sections for more information.

4.2 Goodness of Fit

The accuracy of a calculation fit, and thus the reliability of the measurement, can be judged by the match between the measured and calculated spectra, which is quantified by the GOF value. GOF is a number between 0 and 1.

A perfect match between measured and theoretical spectra will result in a GOF of 1. To ensure accurate results the user may set a minimum acceptable GOF value by entering a number between 0 and 1 into **GOF Error** on the **Alarms Tab** in the **Edit Recipe** dialog box. A warning message will be displayed after each measurement where the GOF falls below the user-specified **GOF Error**. If the GOF is less than 1 there are several possible causes:

a) **Nonuniformity** – Any variation of layer thickness, or presence of roughness, within the measurement spot will result in a reduction in reflectance. In severe cases the interference peaks may disappear entirely.

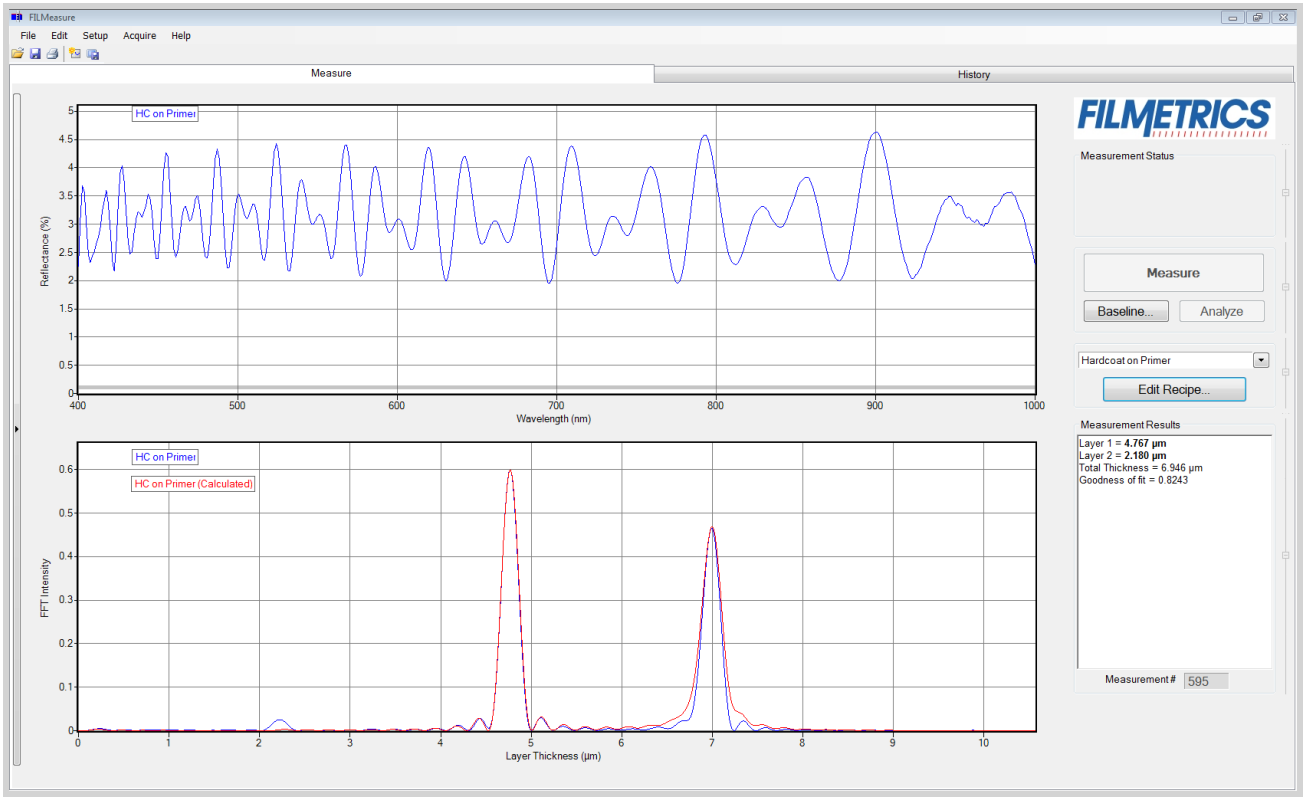
b) **Incorrect Structure Definition** – If the refractive index of one or more layers in the structure is different from what is described in the recipe, then there will be some mismatch between the blue curve and the red curve.

c) **Graded Interfaces** – The amplitude of the peaks in the spectrum depends on both the size of the change in refractive index at each interface in the layer structure and the sharpness of the interface. For example, in many hardcoat samples there is some degree of layer intermixing at the interface, leading to a weaker than expected reflectance at that interface. In severe cases the peaks may disappear entirely.

4.3 Fast Fourier Transform Mode (FFT mode)

The FFT mode tends to be more robust when measuring very thick films or thick multi-layer stacks, where the total thickness is greater than 5 microns.

To activate FFT mode, select the select **FFT (thickness only)** button located on the **Analysis Options > Analysis Method** tab in the **Edit Recipe** box.



The **Measure Tab** shows a split screen with the measured spectrum on the top graph and the FFT spectrum on the bottom graph.

In FFT mode, instead of matching the measured reflectance spectrum with a calculated reflectance spectrum, FILMeasure identifies the peaks in the FFT Spectrum and uses them to compute the layer thicknesses. The number of possible FFT peaks is n*(n+1)/2 where n is the number of layers. Thus, for a two layer structure, we expect to see three peaks in the FFT spectrum.

The image above shows the screen just after measurement of a two layer stack using FFT mode. The upper graph shows a blue curve representing the reflectance spectrum of the sample. The lower graph shows a blue and red FFT spectrum. The blue FFT spectrum is computed from the measured reflectance. The red FFT spectrum is computed from the theoretical reflectance spectrum of a layer stack with the measured layer thicknesses. The small side lobes visible on either side of the two tall peaks are artifacts of the FFT process and can generally be ignored. The only time the side lobes will cause a problem is if a weak peak from a multi-layer sample is very close to a much stronger peak. If this case occurs FILMeasure may lock onto one of the side lobes by mistake and it may not be possible to measure the layer thickness associated with the weak peak. Reducing the **Constraint** and/or increasing the guess for the thickness for the layer with the weak peak may be helpful in getting FILMeasure to lock onto the weak peak. If the peak is too weak to reliably measure it is best to change the layer structure definition so that the layer corresponding to the weak peak is eliminated.

Along with the measured layer thicknesses, the **Results** text box displays important information about the measurement. This important information includes the **Goodness Of Fit (GOF)** and a description of which peaks were used to determine the measured layer thicknesses. In the figure above layer 1 thickness is determined based on the position of the peak located at 4.780 microns and layer 2 thickness is determined using the peak corresponding to layers 1+2 (7.011 microns). Notice that the thickness of layer 2 is not exactly equal to 7.011-4.780 = 2.231 microns. FILMeasure computes a thickness for layer 2 of 2.113 microns because it precisely accounts for the fact that layer 2 has a different refractive index than layer 1.

Discussion of GOF:

A perfect match between measured and theoretical FFT spectra will result in a **GOF** of 1. To ensure accurate results the user may set a minimum acceptable **GOF** value by entering a number between 0 and 1 into **GOF Error** on the **Alarms** settings in **Edit Recipe**. A warning message will be displayed after each measurement where the **GOF** falls below the user-specified **GOF Error**. If the **GOF** is less than 1 there are several possible causes as follows:

a) **Nonuniformity** – Any variation of layer thickness within the measurement spot will result in a reduction in the FFT peak heights. In order to permit easy visual interpretation of the FFT results FILMeasure always scales the amplitude of the red curve so that it matches the blue curve. However, if the scaling factor is not 1.0, FILMeasure reports a reduced GOF even if the red curve appears to exactly match the blue curve on the screen.

b) **Incorrect Structure Definition** – If the refractive index of one or more layers in the structure is different from what is described in the layer structure, then the FFT peak amplitudes in the blue curve will not match the peak heights in the red curve. The software will attempt to scale the red curve, but if there is more than one FFT peak present, the scaling will only result in one correct peak height. If you wish to view the impact of changing the refractive index of one or more layers, simply change the refractive index in the **Edit Recipe** dialog box and click **Analyze**. If there is only a single layer you will notice that the GOF will change, but the red curve should not change much. If there are multiple layers in the structure you will notice that the relative peak heights in the red curve will change when you click **Analyze**.

c) **Graded Interfaces** – The amplitude of the peaks in the FFT spectrum depends on both the difference in refractive index at each interface in the layer structure and the sharpness of the interface. In many common samples you will find that there is some degree of layer intermixing at the interface, leading to a weaker than expected reflectance at that interface. The FFT peak(s) associated with the intermixed interface will suffer a reduced amplitude and in severe cases the peaks may disappear entirely.

5. Advanced Features

This section describes how to create and edit material files within Windows Excel, as well as how to use the **Contact Probe Template Mode**.

Creating and Editing *n* and *k* Files

Contact Probe Template Mode

5.1 Creating and Editing *n* and *k* Files

Creating new *n* and *k* files

New *n* and *k* files can be created by going to **Edit > Material Library…** Clicking the **New** button will allow the user to define a new material file. Once clicked, while the **Default Model** is selected, the user can manually enter wavelength versus n or wavelength versus k values. It is also possible to copy and paste values into the columns if the data is available.

The user can also select **Custom Model**. While this is selected, clicking on the **Model…** button allows the user to create *n* and *k* files using predefined mathematical models.

Once completed, clicking **Save As…** will save the new file as a custom material for later use.

Creating *n* and *k* files from a calculated spectrum

Files for *n* and *k* can be created directly from a calculated spectrum, where *n* and *k* are measured. Once a spectrum is analyzed, go to **File>Save Measured *n* and *k*…** This will save your *n* and *k* files as a custom material which can be loaded into the user recipe.

Editing *n* and *k* files

Existing material files can be edited by going to **Edit > Material Library…** There, the user can load a material file by clicking **Open**. Values can then be changed manually. Clicking **Save** will save over the original file for later use. Use **Save As** to save the edited file under a new name.

Converting Microsoft Excel files into FILMeasure *n* and *k* files

For wavelength versus *n* or wavelength versus *k* data saved in Excel (*.xls) format, the data can be directly saved as material files that can be read by FILMeasure.

For .nnn and .kkk files.

For both *n* and *k* files, the data should follow the format shown below, where the wavelength values, in units of nanometers, are in the first column, and the *n* or *k* values are in the second. Note that the version number (cell B1) refers to version of the file. For this cell, you can use 1. The material number (cell B2) refers to what material category the file corresponds to. The material categories (and their corresponding numbers) are: Metal (1), Semiconductor (2), Dielectric (3), Other (4), and Photoresist (6). By putting the correct value for the material number, the file will automatically be sorted into the correct category when selecting the material in the **Edit Recipe** window.

To save the file created by Microsoft Excel, go to **File>Save As…** For **Save as Type:** choose **CSV (Comma delimited) (*.csv)**. For n files, save the file as "**FILENAME.nnn**", and for k files, save the file as "**FILENAME.kkk**" (FILENAME represents the name you wish to call the file; note that quotation marks must be used to enclose the filename in Excel to force the extension to **.nnn** or **.kkk**). The file can then be saved to the desktop, and **Imported** using the **File > Import > Material File** option

	A	B
1	Version	1
2	Material	3
3	206.6	1.543
4	226.7	1.5228
5	248.3	1.5084
6	265.2	1.5
7	280.3	1.494
8	302.2	1.4872
9	330.3	1.4805
10	361.1	1.4751

For .fitnk files.

The data should follow the format of shown below, where the wavelength values, in units of nanometers, are in the first column, index of refraction (*n*) in the second, and extinction coefficient (*k*) if present, in the third.

The first few columns of the file should be identical to the figure below the only exception being cell B3 which should contain a number describing the material category. The material categories (and their corresponding numbers) are: Metal (1), Semiconductor (2), Dielectric (3), Other (4), and Photoresist (6).

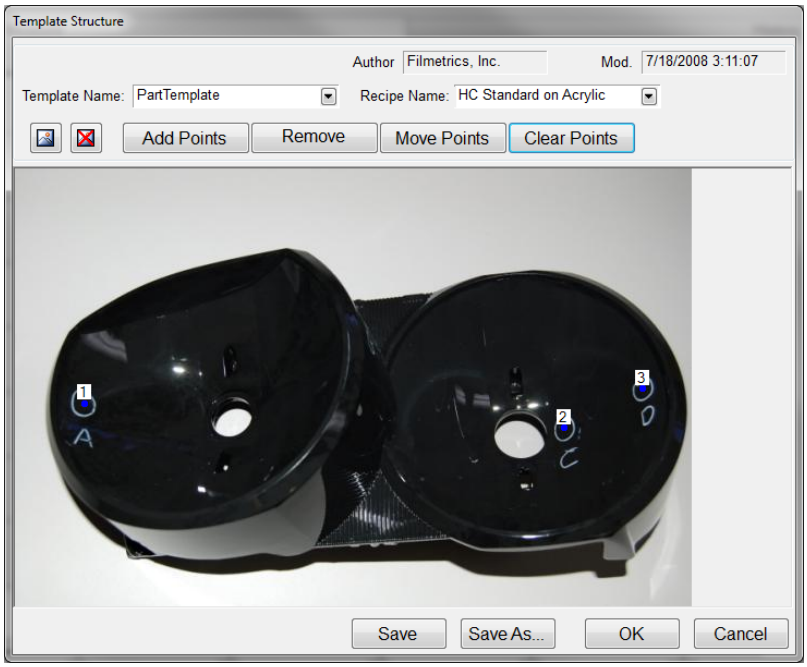
To save the file created by Microsoft Excel, go to **File > Save As…** For **Save as Type:** choose **CSV (Comma delimited) (*.csv)**. Save the file as "FILENAME.fitnk" (FILENAME represents the name you wish to call the file; note that quotation marks must be used to enclose the filename in Excel to force the extension to .fitnk). The file can then be saved to the desktop, and **Imported** using the **File > Import > Material File** option

	A	B	C
1	Version	1	
2	Compat. Version	1	
3	Material	3	
4	File Information		
5	Wavelength (nm)	n	k
6	206.6	1.543	0
7	226.7	1.5228	0
8	248.3	1.5084	0
9	265.2	1.5	0
10	280.3	1.494	0
11	302.2	1.4872	0
12	330.3	1.4805	0
13	361.1	1.4751	0

5.2 Contact Probe Template Mode

This mode allows users to create and define template files for use with contact probe measurements. These templates can be used to ensure that measurements are taken from the same area of a given part. To enable this function, you must first set the **Optics Configuration** to contact probe, and then enable **Contact Probe Template Mode** underneath the **Setup>Options>Data Recording** menu.

With **Contact Probe Template Mode** enabled, a new option is available in the **Measure** tab on the main FILMeasure screen. This menu allows you to make measurements to the selected template file, select a new template file, or edit a template file. The following sections will demonstrate how to create and use a template file.



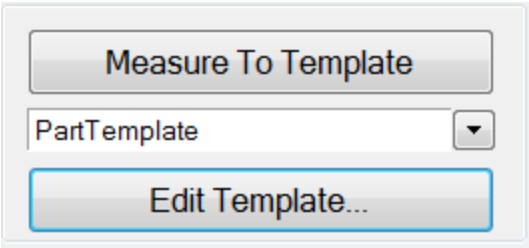
Example Template Structure.

[Creating a Template for Contact Probe Template Mode.]

[Using a Template File to Take Measurements.]

5.2.1 Creating a New Template

Before creating a new template file, **Contact Probe Template Mode** must first be enabled. The [Optics Configuration] must be set to **Contact Probe** mode, and the checkbox next to **Contact Probe Template Mode** underneath the [Options > Data Recording] menu must be checked. Once **Contact Probe Template Mode** is enabled, a new option will be available in the main FILMeasure window. Click **Edit Template...** to open the **Template Structure** dialog box.



Within the **Template Structure** dialog box there are several options to build your template file.

Template Name:

This drop-down list allows you to select between saved templates.

Recipe Name:

Shows the recipe currently associated with a given template. This drop-down list may also be used to select a new recipe.

Insert Background Image  **:**

This allows the user to select an image of the sample to be measured as a background image over which the template will be displayed. Images must be in the JPEG, Bitmap, or GIF format.

Delete Background Image  **:**

For templates with a generic shape but no specific sample in mind (i.e. a simple x,y grid), you can use this option to remove a background image and set the template points against a gray field.

Add Points:

Allows the user to add points to the Template by left clicking anywhere on the image or field. These points will automatically be numbered by the order they are added.

Remove:

Removes previously added points by clicking on them.

Move Points:

Click and drag points already included in the template to move them to a new location.

Clear Points:

Removes all points from the template at once.

Once you've finished editing the template file you can save the template file using one of the three buttons at the bottom of the dialog box. The **Save** button will save the changes to the template selected underneath the **Template Name:** box. The **Save As** button will prompt you to save the current template under a new name. The **OK** button will save the changes for immediate use, however any changes not saved with either the **Save** or **Save As** button will be lost if another template is selected, or the software is shut down.

With a new template defined, you can now use the file you've created to make measurements. See **Using Contact Probe Template Mode** for more information.

5.2.2 Using Contact Probe Template Mode

With a template already defined, you are now ready to use **Contact Probe Template Mode** to make measurements. See **Creating a New Template** for details on creating templates.

Step 1:

Select the desired template from the drop down list in the main FILMeasure window. If the template menu isn't available, see **Setup > Options > Data Recording**.

Step 2:

Click on the **Measure To Template** button to open the Template file. If the button is grayed out, the system will first require a **Baseline**.

Step 3:

A new window will open showing the selected template file. The software will be ready to acquire data, so place the contact probe on the sample to measure the first spot. Lift the contact probe off the sample to save the result. The currently active spot is indicated by a blinking dot. Repeat this process until all measurements have been taken.

Note: If a measurement is taken incorrectly, it can be removed by clicking the undo button.



Step 4:

Save the output. By default the results will be placed in the history file. An image of the template with thicknesses can also be saved with the **Save To File** button, or printed out using the **Print** button. A working printer must be connected for the Print button to work. Clicking **Accept** will close the template window.

6. Theory of Operation

This section describes how the instruments work and includes the theory the measurements are based on, specifics about the hardware, and more detail about the baseline measurement.

Measurement Theory

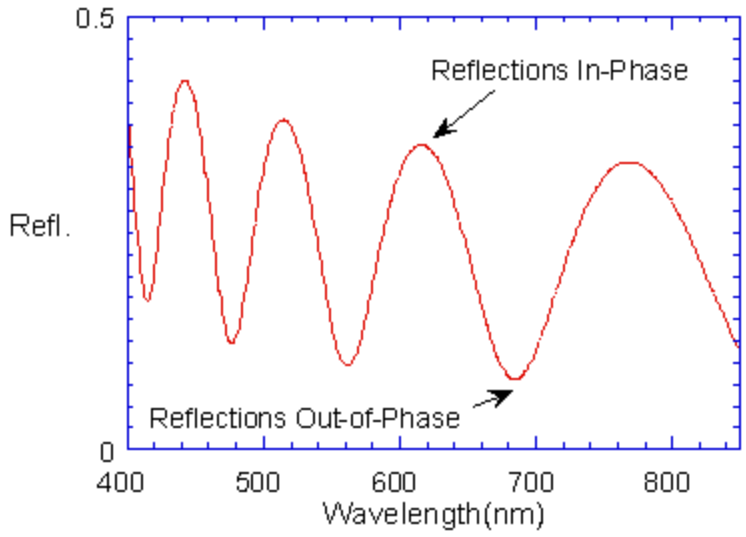
Thickness Measurement Details

Hardware Operation

The Baseline Measurement

6.1 Measurement Theory

Our instruments measure thin-film characteristics by either reflecting light from, or transmitting light through the sample, and then analyzing this light over a range of wavelengths. Because of its wave-like properties, light reflected from the top and bottom interfaces of a thin film can be in-phase so that reflections add, or out-of-phase so that reflections subtract. Whether the reflections are in- or out-of-phase (or somewhere in between) depends on the wavelength of the light, as well as the thickness and properties of the film (e.g., reflections are in-phase when $\lambda = (2*n*d)/i$, where λ is the wavelength, n is the refractive index, d is the film thickness, and i is an integer). The result is characteristic intensity oscillations in the reflectance spectrum (see Figure below). In general, the thicker the film, the more oscillations there are in a given wavelength range.



Example of reflectance spectrum with oscillations.

The amplitude of the oscillations is determined by the refractive index and extinction coefficient of the films and substrate. Therefore, by analyzing the period and amplitude of these oscillations, our instruments can determine thickness of multiple thin films.

6.2 Thickness Measurement Details

Optical thin-film thickness measurements require the successful completion of two tasks: acquisition and then analysis of an accurate reflectance spectrum. To determine film thickness, FILMeasure calculates a reflectance spectrum that matches as closely as possible the measured spectrum. FILMeasure begins with an initial guess for what the reflectance spectrum should look like theoretically, based on the user's input of a film structure for the sample. Then FILMeasure varies the parameters it is solving for until the calculated reflectance spectrum matches the measured data. Mathematically, this procedure is complicated by the fact that as the thickness of the films in the calculation is varied there can be many near matches. Therefore, an approach that simply homes in on a solution by finding successively better approximations will not work unless the starting guess for optical thickness is within approximately 1000 Å if the actual thickness.

FILMeasure avoids homing in on a false solution by searching the entire acceptable thickness range to determine the thickness that gives the best possible match between the measured and calculated spectra. The thickness range searched is determined by the initial guess for thickness together with the thickness constraint. Since the time to find a solution is proportional to the range of thicknesses being searched it is beneficial to provide an initial guess close to the thickness of the films to be measured.

6.3 F20 Hardware Operation

The F20 performs two distinct functions: data acquisition and data analysis. Data analysis specifics are discussed in **Software Overview**. In this section we describe how the F20 acquires accurate spectral reflectance data.

Light is supplied by a tungsten-halogen bulb that generates light from approximately 375 nm to 3000 nm. This light is delivered to and collected from the sample through a fiber-optic cable bundle and a lens. The intensity of the reflected light is measured at different wavelengths with a spectrometer. The F20 spectrometer uses a diffraction grating to disperse the light and a linear photodiode array to measure the light at the different wavelengths. After a user-selectable integration time, the accumulated charge in each photodiode is read by the computer. Because a longer integration time results in a larger charge, it is the integration time that determines the sensitivity of the spectrometer. Adjustment of the integration time is used to obtain the proper signal level. Too short an integration time results in a weak, noisy signal, while too long of an integration time results in a saturated signal. An optimized integration time is set in the software during the baseline process.

6.4 The Baseline Measurement

The baseline measurement allows the FILMeasure software to take into account the response inherent to the reflectance measurement hardware. It does this by measuring a reflectance standard and by taking a “dark” reading. In any optical system there are many components whose characteristics vary with wavelength (e.g., the output of the light source and the sensitivity of the spectrometer). However, when reflectance measurements are made, only variations in reflectance vs. wavelength due to the sample under test are of interest. Therefore, FILMeasure must perform a calibration to determine the spectral response of the system. This is done by making a measurement of a reflectance standard that has known reflectance characteristics. Note that it is not necessary for the reflectance standard to be the same as the substrate upon which films to be measured reside. The only purpose served by the reflectance standard is to permit calibration of the optical system. For example, it is possible to use a Si wafer as the reflectance standard and then measure films on GaAs, InP, glass, plastic, etc.

After the reference measurement is made a dark reading is taken. A non-zero dark level is due to current leakage inherent to photodiodes, which causes each photodiode in the array to slowly charge up even when no light enters the spectrometer. Thus, in order to make an accurate measurement of the light entering the spectrometer, it is necessary to subtract this “dark” current contribution. This is the purpose of the background reading, which measures the magnitude of the dark current for a given integration time. When a “dark” measurement is made, a spectrum is measured that represents the signal generated by the spectrometer when a sample of zero reflectance is measured. To simulate a sample with zero reflectance during a “dark” measurement, a specularly reflecting sample can be held at an angle with the light source turned on, or in the case of the UV light source, the shutter may simply be closed.

Due to drift in the light source and temperature of the spectrometer electronics, it is a good idea to take a baseline periodically.

How to Contact Us

We welcome suggestions from our users on ways to improve our software and hardware. Please send us any suggestions you may have for improvements in the help file or new features you would like to see in the software.

We may be reached by phone at +1-858-573-9300, by fax at +1-858-573-9400, by the Contact Us Now form on **our website**, or by e-mail at **support@filmetrics.com**.

7. Appendices

This section contains information on various topics as listed below:

Performance Specifications

How to Replace the Light Bulb

Automation and Data

Software License Agreement

7.1 Performance Specifications

	F20-UV	F20-UVX	F20	F20-EXR	F20-NIR	F20-XT
Wavelength Range:	190-1100 nm	190-1700 nm	380-1050 nm	380-1700 nm	950-1700 nm	1440-1690 nm
Thickness Measurement Range*:	1 nm-40 µm	1 nm-250 µm	15 nm-70 µm	15 nm-250 µm	100 nm-250 µm	0.2 µm-450 µm
Min. Thickness to Measure *n* and *k**:	50 nm		100 nm		500	2 µm
Accuracy*: The greater of	1 nm or 0.2%		2 nm or 0.2%		3 nm or 0.4%	5 nm or 0.4%
Precision[1]:	0.02 nm				0.1 nm	1 nm
Stability[2]:	0.05 nm				0.12 nm	1 nm
Spot Size:	Standard 1.5mm, Optional down to 20 µm					600 µm
Sample Size:	From 1mm to 300 mm diameter and up					
Light Source Lamp MTBF:	D2: 2000 hr, Halogen: 1200 hr		Halogen: 1200 hr			

* Material dependent.

[1] 1σ of 100 measurements of 500 nm SiO_2-on-Si. Average of 1σ over 20 successive days.

[2] 2σ of daily average of 100 measurements of 500 nm SiO_2-on-Si over 20 successive days.

General Requirements:

Power:

System: 100-240 VAC, 50-60 Hz, 0.3-0.1 A

External Power Supply: 24Vdc, 3.75A

Computer Interface: USB 2.0

Certifications: CE EMC and safety directives.

Computer Requirements:

Operating System:

PC: Windows XP (SP2) - Latest Windows (64-bit)

Mac: OS X Lion- Latest Mac OS running Parallels

Processor Clock Speed: 1.4 Ghz

Interface: USB 2.0

Screen Resolution: 1024x768

Specifications subject to change without notice.

7.2 System Configurations

Many accessories are available for the F20 series systems. Please refer to the following list to find different applications for the configuration provided with your system. If you would like information on purchasing additional accessories not included, please contact Filmetrics.

SS-3 Stage in Standard Spot position: The SS-3 stage is our standard stage for the F20 system. This is the most versatile stage, which is shipped with each system unless otherwise specified.

SS-3 in UV position: The SS-3 stage in the UV position is optimized for use in UV applications.

SS-3 Stage in Small Spot position: The SS-3 stage in the small spot position is used to achieve a smaller spot size for measurement. This is also useful for non-uniform or rough samples. The F20-XT system comes standard with the SS-3 in the Small Spot position.

SS-5 Stage:The SS-5 stage is an optional stage for use with F20.

UV Extended Range (UVX): UVX systems provide the maximum possible range of thickness measurements by including both UV and NIR detectors and light sources.

Contact Probe (CP-1): The contact probe is useful for measuring coatings on curved surfaces, such as ophthalmic lenses. This accessory comes standard with the F10-HC, in place of the SS-3.

Contact Probe for Cylindrical Samples (CP-C): This contact probe is used for measuring coatings on cylindrical samples.

Contact Stage Option (CS-1 Option): The contact stage option is used to measure the thickness of freestanding films and coatings on glass samples where eliminating the backside reflection of the glass is necessary. Use of the contact stage requires that the sample be flat.

Transmittance Option (T1 Option): This option is used to make transmittance measurements of flat, transparent samples.

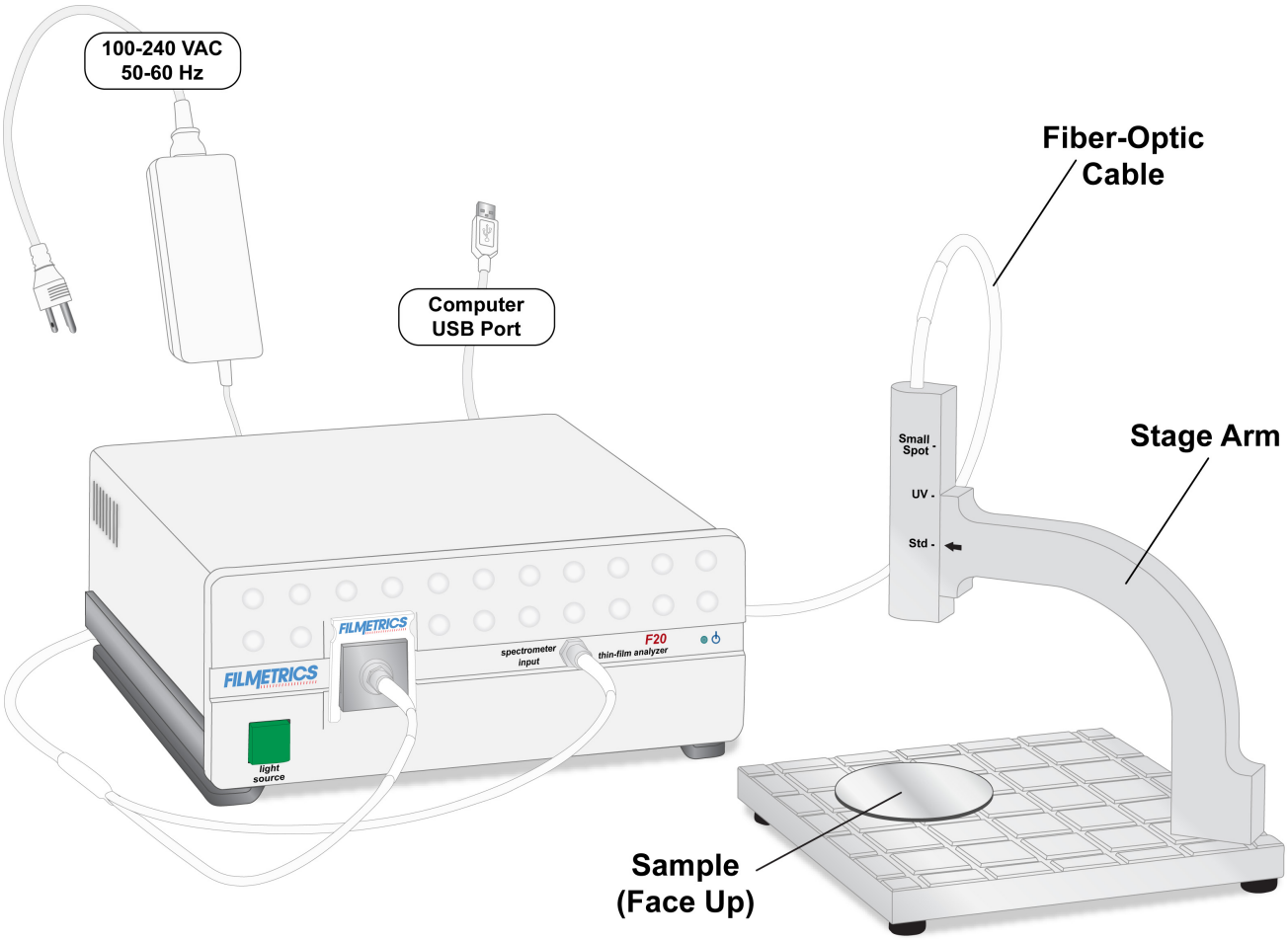
Integrating Sphere Transmittance (SS-Trans-Curved): The integrating sphere on the SS-Trans-Curved allows transmittance measurements to be taken on curved or flat surfaces.

Kinematic Mount with Lens Assembly (KM-GL25): This assembly is used for in-line and in situ applications, or when a longer working distance in required.

7.2.1 SS-3 in Standard Position

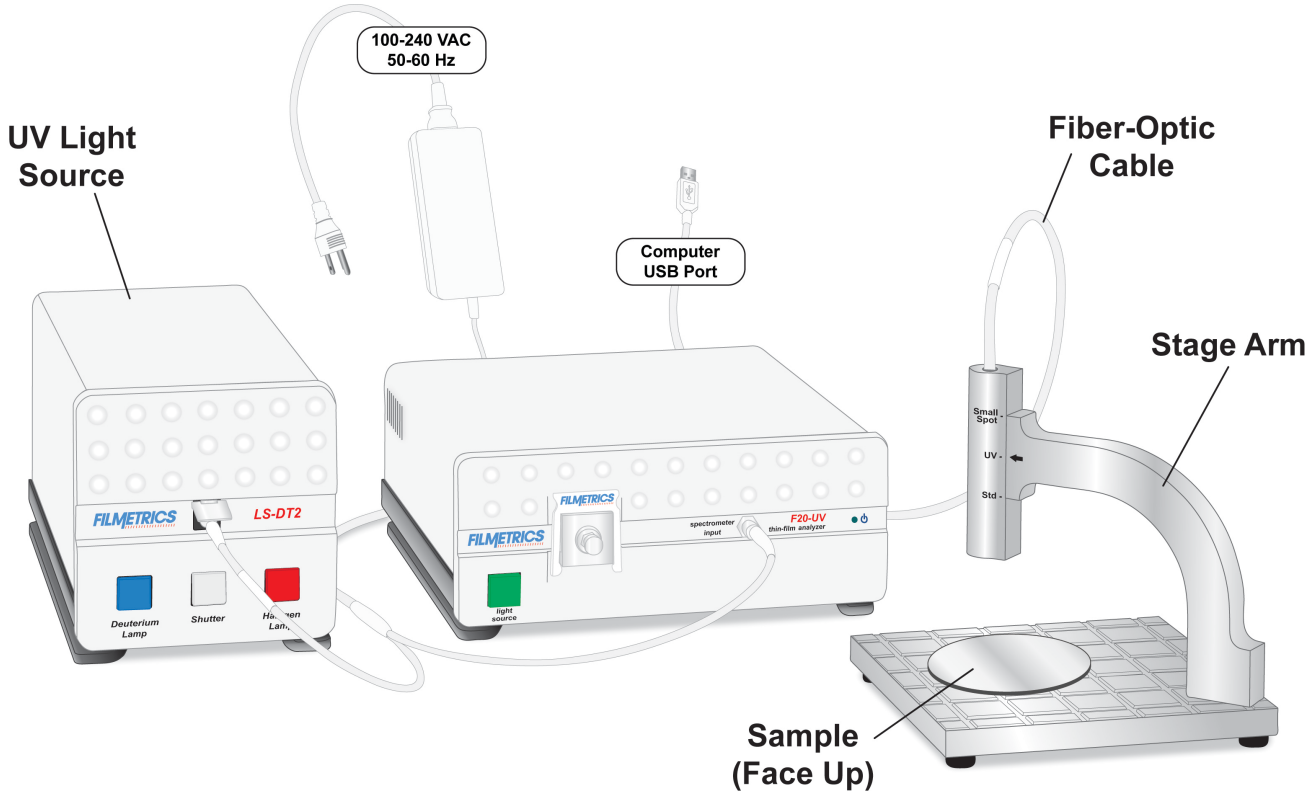
Connect the fiber-optic cable to the F20 as shown below. Fully insert the fiber-optic cable into the lens tube. Focusing may be required depending on the height of your sample, please see **Focusing your fiber-optic**.

The fiber-optic cable contains 7 fibers – a single detection fiber surrounded by 6 illumination fibers. The measurement spot size is not the illuminated area, but rather the intersection of the illuminated area with the projected image of the detection fiber. When focused for maximum signal, the measurement spot size is roughly half the diameter of the illuminated area.



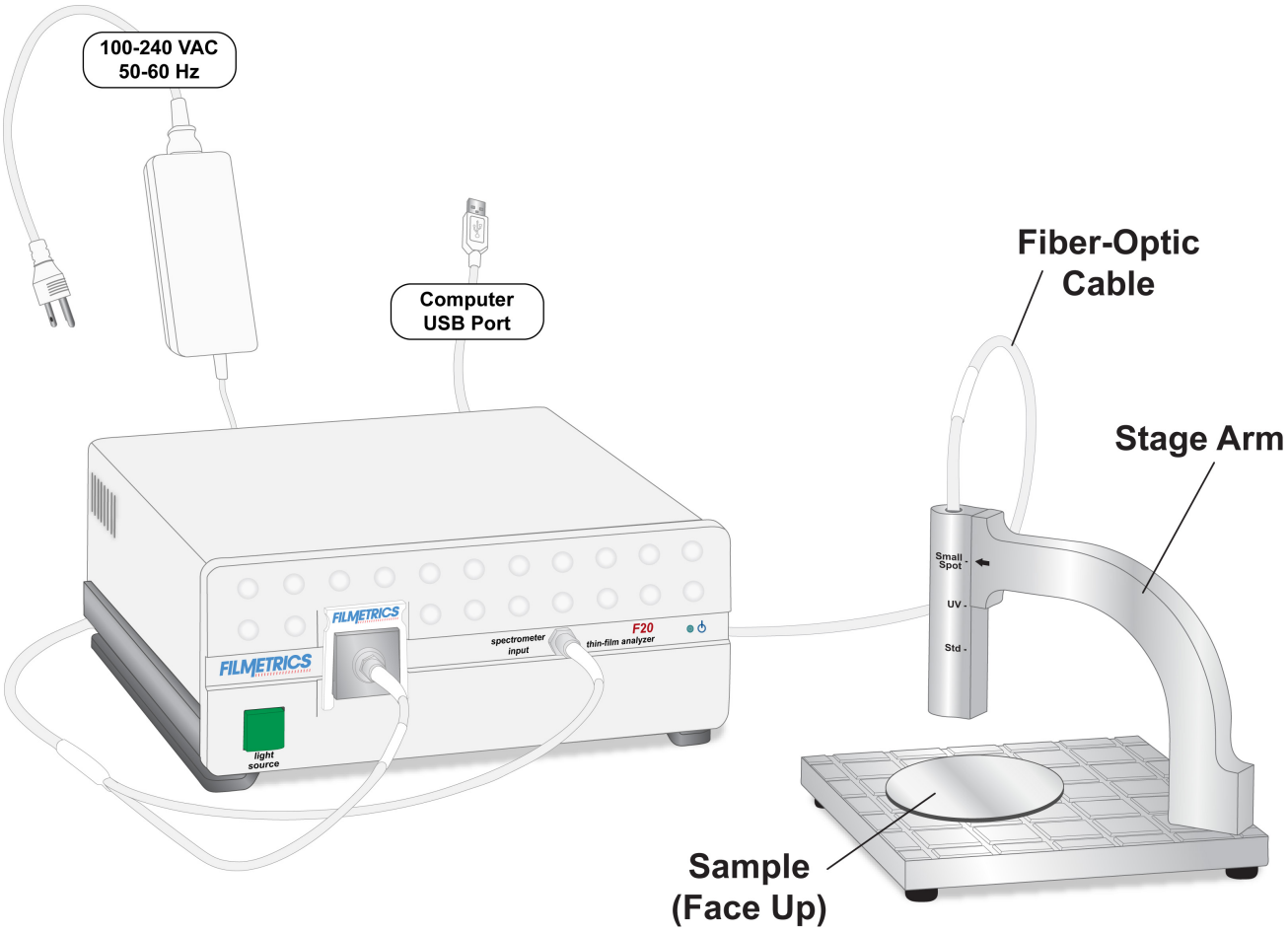
7.2.2 SS-3 Stage in UV Position

This optical setup applies to both UV and UVX measurements. If the SS-3 stage is not already in UV position, it can be repositioned by removing the 9/64" hex-socket bolt accessed on the backside of the stage arm. Next connect the fiber-optic cable between the F20, the UV light source and the stage as shown below. If the fiber does not have a white label indicating “Light Source”, then the two fiber ends are identical and can each be attached to either the light source or spectrometer. Focusing will be required before making measurements, please see **Focusing your fiber-optic**.



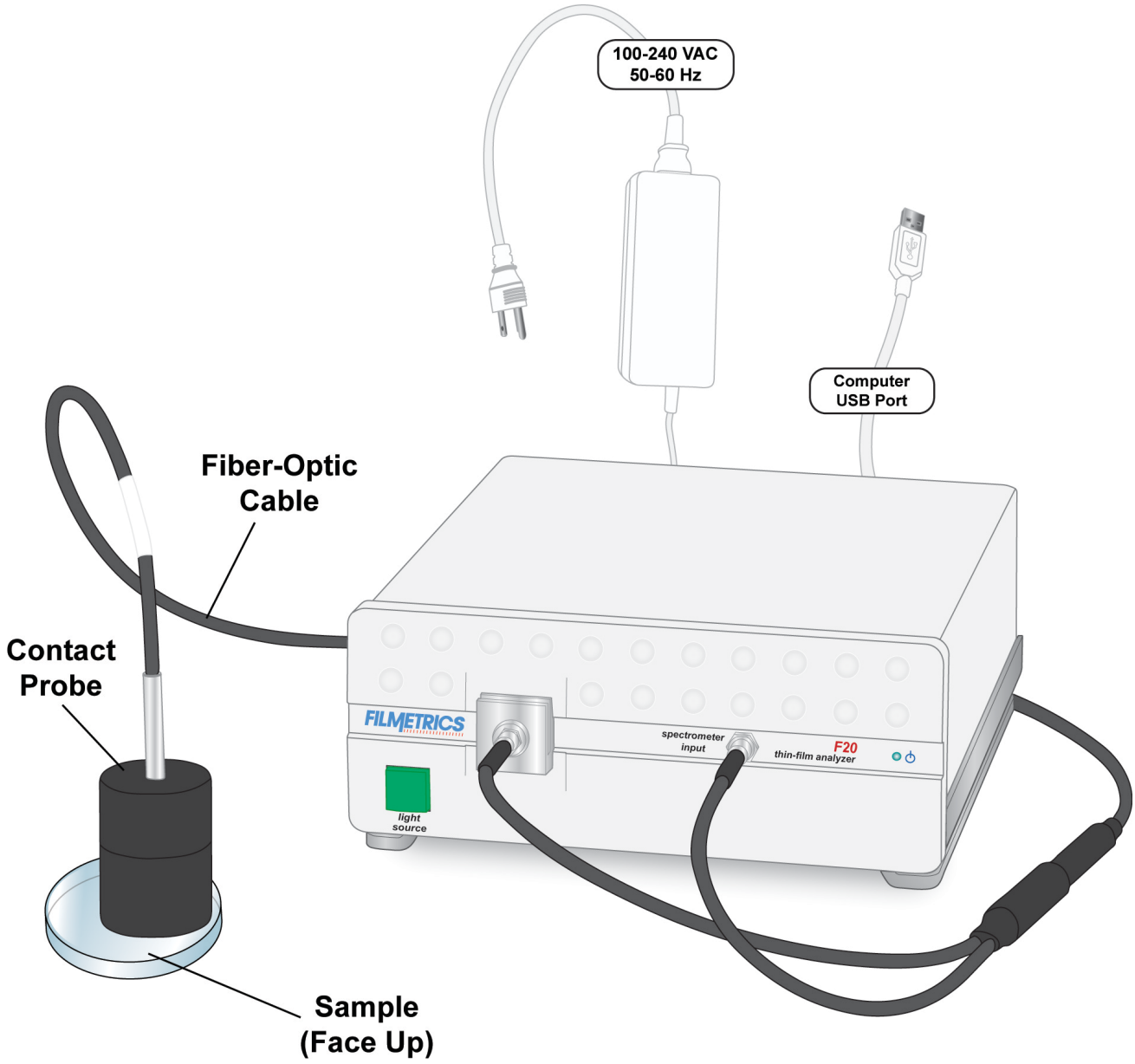
7.2.3 SS-3 in Small Spot Position

If the SS-3 stage is not already in small spot position, it can be repositioned by removing the 9/64" hex-socket bolt accessed on the backside of the stage arm. Next connect the fiber-optic cable between the F20 and the stage as shown below. If the fiber does not have a white label indicating “Light Source”, then the two fiber ends are identical and can each be attached to either the light source or spectrometer. Focusing will be required before making measurements, please see **Focusing your fiber-optic**.



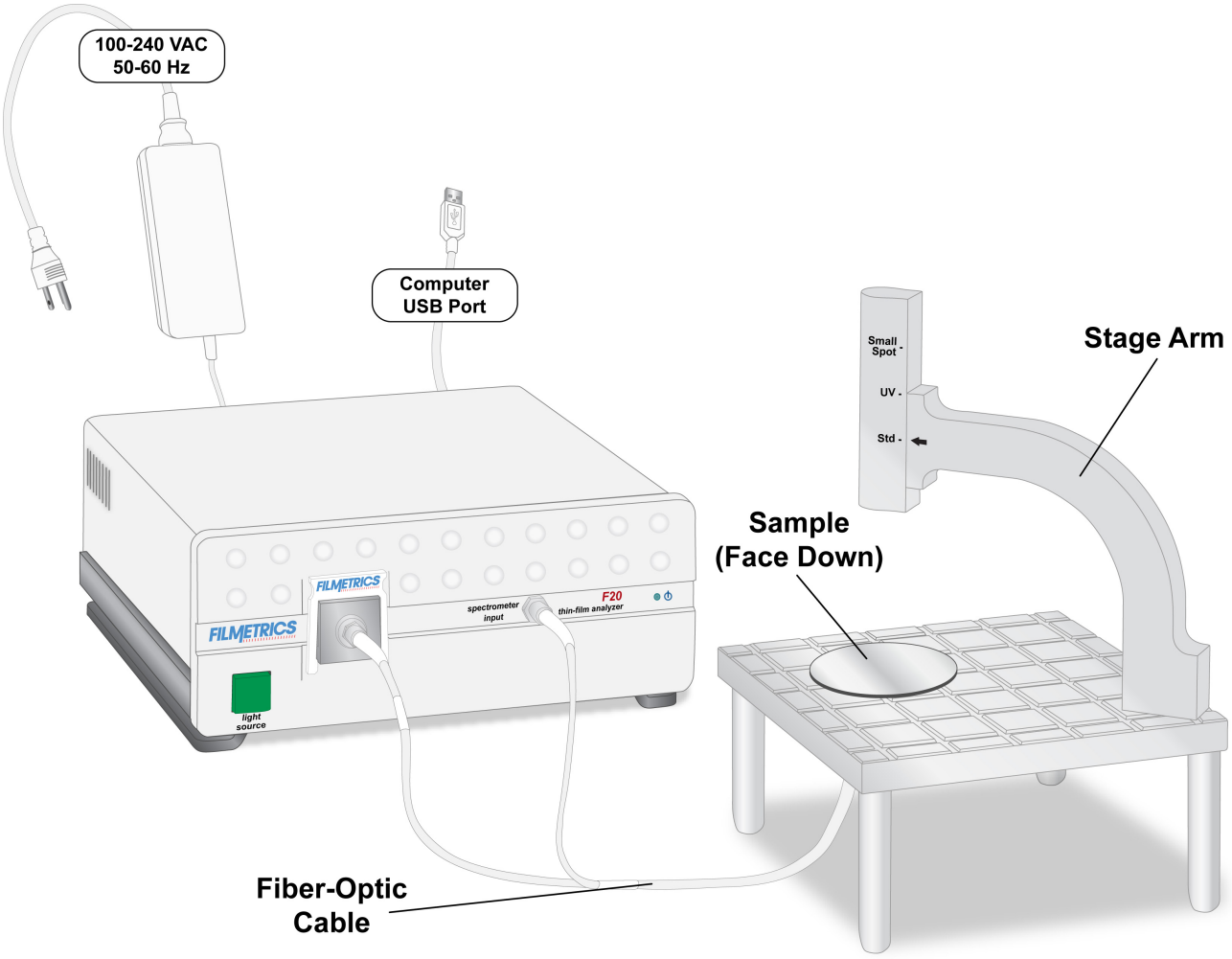
7.2.4 Contact Probe (CP-1 and CP-C)

Connect the Contact Probe to the F20 as shown below. The two fiber ends leading from the Contact Probe are identical; they can be attached to either the light source or spectrometer.



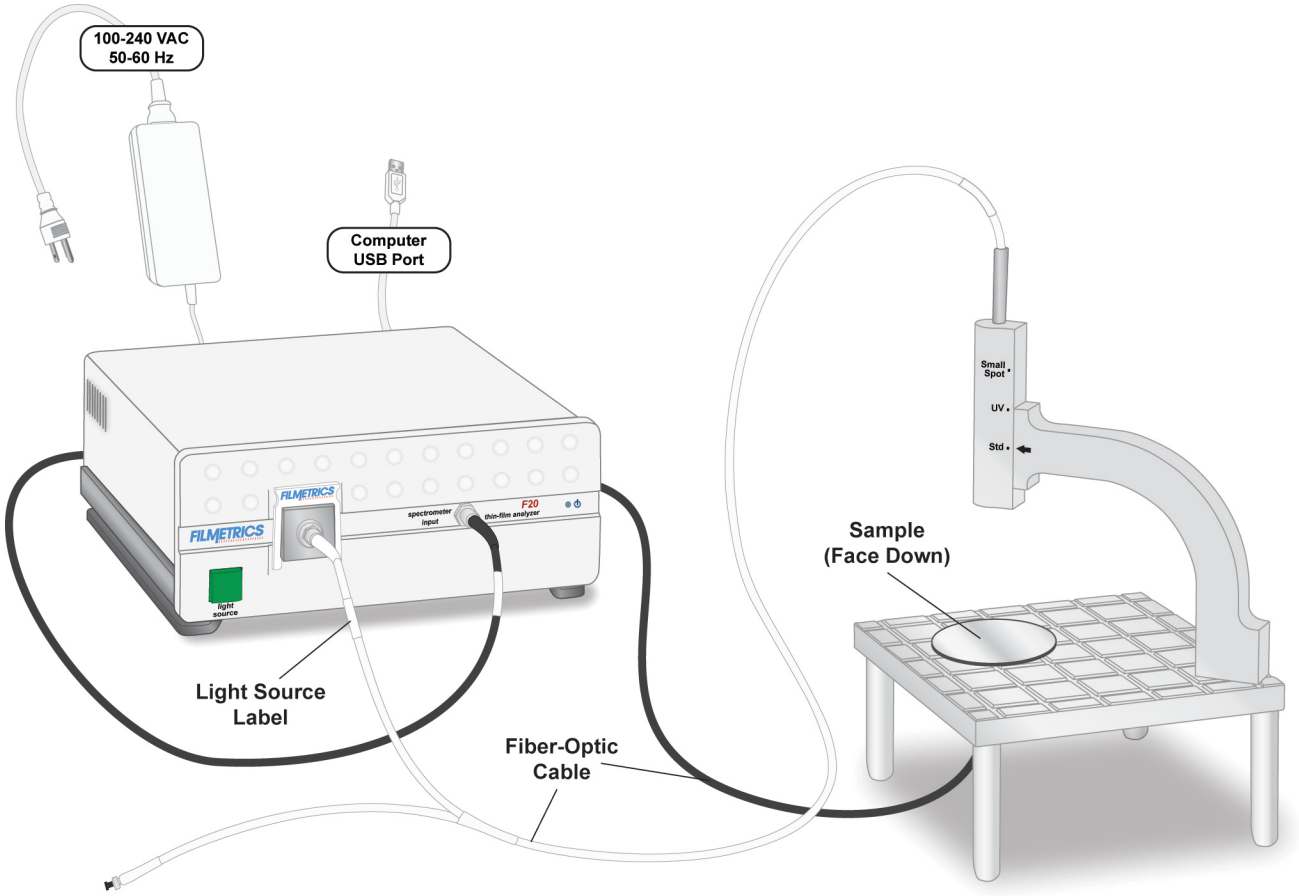
7.2.5 Contact Stage (SS-3 with CS-1 option):

Connect the contact stage fiber optic cable to the F20 as shown below. The two fiber ends leading from the common end of the fiber are identical and can each be attached to either the light source or spectrometer. Connect the four 3” legs to the bottom of the SS-3 Sample Stage. Remove the tilted mirror insert in the SS-3 stage using the supplied 1/16” hex socket driver. Screw the fiber optic cable common end into the black contact adapter, and then fully insert the contact adapter into the bottom hole in the SS-3 stage, fastening it with the 1/16” hex driver

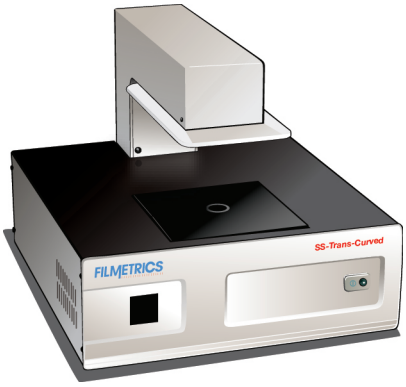


7.2.6 Transmittance Option (SS-3 with T-1 option):

Connect the fiber-optic cable between the F20 and the stage as shown below. The fiber-optic cable should be fully inserted into the arm on the stage. The screw on the end of the stage arm is actually a spring-loaded ball plunger and does not require tightening. Connect the four 3” legs to the bottom of the SS-3 Sample Stage. Remove the tilted mirror insert in the SS-3 stage using a 1/16” hex socket driver. Screw one end of the single fiber optic cable into the black adapter and fully insert the adapter into the bottom hole in the SS-3 stage, fastening it with the 1/16” hex driver. Before taking a baseline measurement, be sure to set the mode of operation to transmittance underneath the **Optics Configuration**. See **Example #4** for more information on transmittance measurements.



7.2.7 Transmittance from Curved Surfaces (SS-Trans-Curved):



The first step to taking transmittance measurements from curved surfaces is to connect the fiber optics to the **SS-Trans-Curved**. Begin by connecting the two side-by-side ends of the 605-0726 fiber to the two **External Light Source Input** ports on the rear of the **SS-Trans-Curved**. The 605-0726 fiber is "Y" shaped, with the side-by-side ends form the top of the "Y." Next, connect the other end of the 605-0726 fiber to the light source. On UV and UVX systems, connect to the UV light source. For all other systems (VIS, NIR, EXR) connect to the light source on the front of the spectrometer box. Finally, connect the 605-0517 fiber to the **Measurement Signal to Spectrometer** port on the rear of the **SS-Trans-Curved** box and to the **Spectrometer Input** port on the front of the spectrometer box.

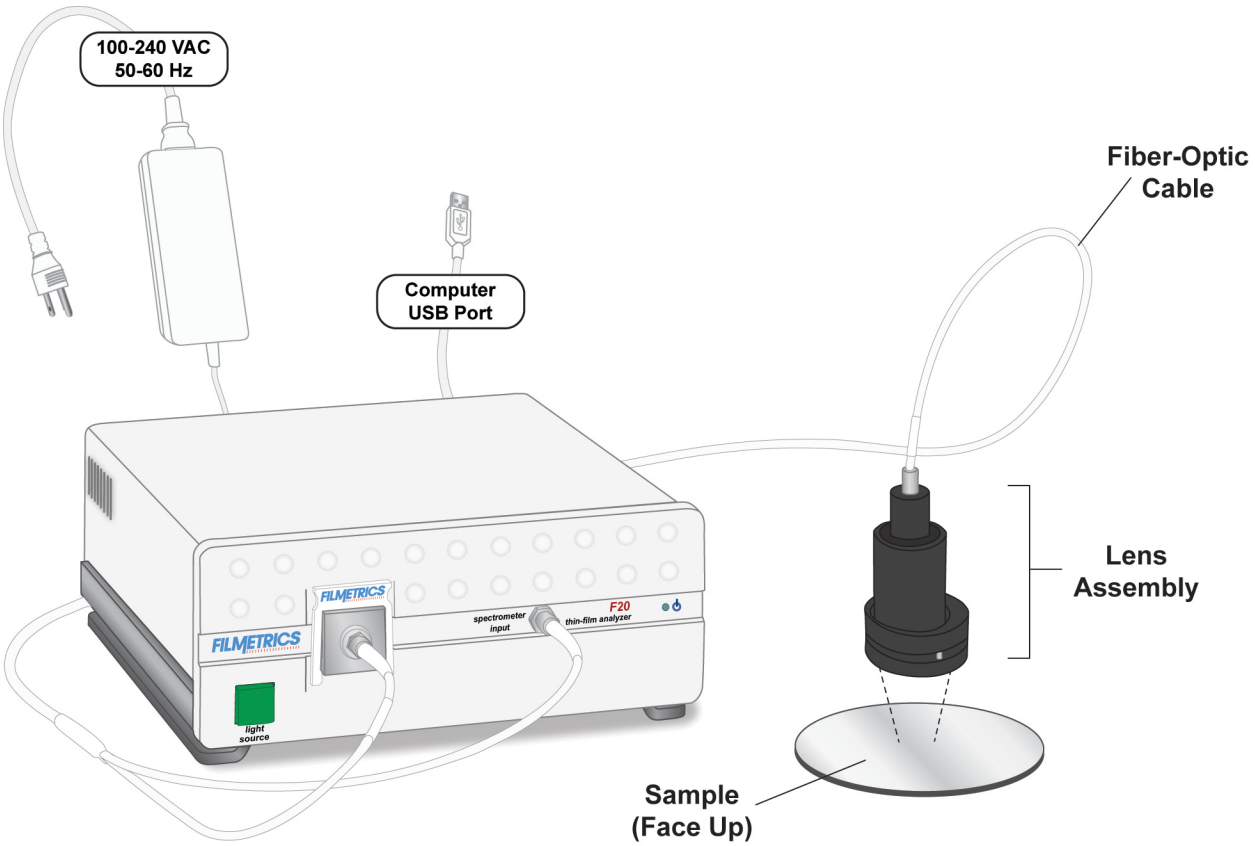
With the fibers connected, you can now connect the stage to power using the provided power cord, and the **SS-Trans-Curved** to your computer using the provided USB cable. Flick the power switch on the back panel to the on position.

With the hardware configured, you'll now need to configure FILMeasure to measure transmittance. This can be achieved in the **Edit Recipe** dialog, underneath **Acquisition Settings > Optics**. See **Example #4** for more information on transmittance measurements.

7.2.8 Kinematic Mount with Lens Assembly (KM-GL25)

Step #1: Connect Fiber-Optic Cable to the F20

Connect the fiber-optic cable between the F20 front panel and the KM-GL25 as shown below. Slide the fiber-optic cable part way into the lens assembly. The spring-loaded setscrew should be adjusted so that the fiber-optic cable is held in place, yet it may be moved in and out for focus adjustments.



Step #2: Adjust the lens assembly:

The focus and tilt of the lens assembly need to be adjusted for maximum signal and assumes that the lens assembly has been mounted on a fixture approximately 6 inches above the sample surface, and the "read" return fiber from the lens assembly to be adjusted is connected to the spectrometer input. The idea here is to adjust the tilt and focus for maximum signal. To do this,

A. Turn on the light source with the front panel switch and start FILMeasure.

B. Go to the **Setup>Raw Signal…** dialog box and set the integration time for 20 milliseconds.

C. Click **Start** on the **Setup>Raw Signal…** dialog box.

D. The fiber-optic cable contains 7 fibers – a single detection fiber surrounded by 6 illumination fibers. Do a rough focus adjustment by placing a diffuse surface (a business card works well) at the sample position and then focus the light beam on the sample position by sliding the fiber-optic cable inside the lens tube. (The diffuse reflector will allow you to see the beam.) When 6 individual spots can be seen, slide the fiber-optic cable farther into the KM-GL25 until the spots merge into the center of the illuminated area.

E. Place a reflecting sample, such as a silicon wafer, on the sample stage, under the lens assembly. You should now see a trace on the screen. Turn the tilt adjustment screws to maximize the signal. If the signal saturates, reduce the integration time (see step B).

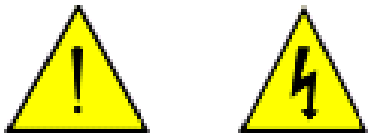
F. Optimize the focus to maximize the measured signal by adjusting the fiber-optic cable position in the lens assembly. Ideally, the best focus adjustment is when the beam is just under focused. To achieve this, focus the beam on the reflecting sample, and then move the fiber-optic cable slightly down in the lens tube until maximum signal is obtained. To see the beam, it may be necessary to place a diffuse surface, such as a business card, over the sample. When finished, tighten the flat head set screw that secures the fiber. Once again, if saturation occurs, it may be necessary to reduce the integration time.

G. Repeat steps e) and f) to make certain the adjustments are optimal.

H. The setup is now complete. Close the **Setup>Raw Signal…** dialog box.

7.3 Light Bulb Replacement

NOTE: These instructions apply to VIS, NIR, and EXR systems only. For UV and UVX systems, please see the seperate users manual for the external light source.



* * Warning: * * Hazardous voltages are present within the unit. Never attempt any maintenance without disconnecting the power cord.

Step by step:

1. Exit FILMeasure and turn off your computer.

2. Disconnect the power cable from the back of the unit.

3. Disconnect the fiber optic cables from the front of the unit.

4. Remove the two captive screws on the back panel opening the two halves of the enclosure.

* * Warning: * * Caution, avoid touching any interior components, other than the lamp, lamp base, and housing.

5. Pull the plunger behind the lamp all the way out.

6. Slide the lamp out of it holder, being careful not to touch the glass.

7. Unplug the old bulb from the lamp power connector.

8. Plug new lamp into the lamp power connector. Do not test lamp with lid removed.

9. Slide the new lamp into the holder, being careful not to touch the glass.

10. Gently release the plunger, holding the bulb in it's proper place. Do not test the bulb with the cover removed.

11. Replace the cover and tighten the screws.

12. Reconnect the power cable, and check that the new bulb illuminates.

7.4 Software Automation

Measurements can be automated and additional data analysis can be performed by user-supplied software. The mechanism by which this occurs is either by using the FIRemote class, which is part of FILMeasure.exe or by using the FIRemoteCOM dll (standard 32-bit version of FILMeasure). Accessing the FIRemote class directly using a .NET programming language such as C# offers the most flexibility and is the recommended approach. Alternatively, the FIRemoteCOM dll can be referenced from Microsoft Office programs and simple software procedures can be implemented there instead. The FIRemoteCOM dll includes fewer methods than FIRemote. Documentation for FIRemoteCOM dll is located at the end of this section.

FIRemote Class

FIRemote is exposed as a public assembly and is accessible using the Microsoft .NET programming environment. The best way to learn how to use the FIRemote class is to look at the example program and source code which are automatically installed as part of the FILMeasure software. Assuming that the FILMeasure program is installed in the default location, the example software can be found in C:\Program Files\Filmetrics\FILMeasure\SourceCode\FIRemoteTest. Additionally an executable, FIRemoteTest.exe, is located in the FILMeasure folder. All the available commands can be tested using FIRemoteTest.exe.

In order to set up a development environment for running and debugging code, the file system must be set up so that the FILMeasure.exe is located in the same folder as the client program that is being developed. Since FILMeasure expects to find its dependent files in the same directory as the executable, this means that the entire contents of the FILMeasure folder must be copied to the build output folder. For example, if the software development environment output folder is located in C:\SourceCode\TestSln\TestPrj\bin\Debug\, then all the files and subfolders from C:\Program Files\Filmetrics\FILMeasure\ must be copied to C:\SourceCode\TestSln\TestPrj\bin\Debug\.

Some of the methods listed in the following reference a SystemGuid. In the terminology used in this document, a Measurement System is an object comprising one or more measurement channels, but with operations that can be performed simultaneously on all the channels in the system. The SystemGuid is used to distinguish between multiple, simultaneously-connected measurement systems. Examples of measurement systems with multiple measurement channels include the F32. Documentation for methods applying only to such measurement systems can be found in the Less Commonly Used Methods section. In addition, some methods will accept either a SystemGuid or a ChannelGuid. Such methods thus will execute an operation either on a single channel, or simultaneously on all channels of a system.

Note that the FIMeasResults object which is returned by the Measure and AnalyzeSpectrum methods has a large number of member functions and properties. Since the names are all self-explanatory, we have not attempted to list all of the members here. The reader should use the Object Browser which is built into Microsoft Visual Studio in order to discover the available properties and methods.

FILMeasure 7 and 8 are based on the Microsoft .NET 4.0 Client Profile Framework. FILMeasure 5 and 6 were based on the .NET 2.0 Framework. Executable client programs and source code projects that were built for FILMeasure 5 or 6 will need to be recompiled, targeting the .NET 4 Client Profile Framework. The Visual Studio development environment will display a warning message if you reference an assembly that is targeting a framework different from that targeted by your development project. You must use Visual Studio 2010 or newer, because the .NET 4 Framework was released with Visual Studio 2010. Do not attempt to use Visual Studio 2008 or older because it will not work. The FIRemoteTest source code is compatible with Visual Studio 2010. The FIRemoteTest solution file can be opened with newer versions of Visual Studio, which will automatically upgrade the solution and project.

32-bit vs 64-bit Client Software

To maintain backward compatibility, the 32-bit version of FILMeasure is shipped by default. If your client software must interact with software modules which are only available in 64-bit versions, then you must use a 64-bit version of FILMeasure instead. Contact Filmetrics for access to an installer for the 64-bit version of FILMeasure.

Reference for FIRemote Class

Methods, Property and Events:

New
AcquireSpectrum
ActivateChannelDisplayTab
AnalyzeSpectrum
AuthenticateRefBac
BaselineAcquireBackgroundAfterRef
BaselineAcquireReference
BaselineAcquireReferenceT
BaselineAcquireReferenceUsingOldSampleReflectance
BaselineAcquireSpectrumFromSample
BaselineCommit
BaselineSetRefMat
BaselineShowDialog
GUIVisible
HistoryDeleteAllResults
IntegrationCycles
IntegrationTime_ms
MaximumSamplingTime
MeasurementTimingMode
MeasChannelGuid
MeasChannelGuids
MeasChannelHWName
MeasChannelHWSerialNumber
MeasChannelName
Measure
NumberOfChannels
NumberOfSpectrometers
NumberOfSubSystems
OpticsConfiguration
OpenSpectrum
OpenSpectrumFromBuffer
OperatorID
RecipeInfo
SampleID
SaveSpectrum
SaveSpectrumToBuffer
SetAnalysisWavelengthRange
SetMaterial
SetN
SetRecipe
SetRoughness
SetThickness
SetThicknessRange

SpectrometerDiagnostics
SpectrometerType

Less-Commonly-Used Methods

The following methods are only rarely used. Methods which refer to a System are used with the F32 and F37 thin film monitoring systems. Many methods accept either a measSystem or a measChannel guid. However, some methods are relevant only to a single channel. If you supply a measSystem guid as the argument for one of these methods, an exception will be thrown.

ActivateMonitorDisplayTab
BaselineHardwareAdjustInhibit
BaselineExistsAndIsAuthenticated
GeneralPurposeIOIsSupported
GeneralPurposeIOReadValue
GeneralPurposeIOSetValue
GetBaselineHardwareAdjustStatus
MaterialPath
MaterialPathCustom
MeasSystemGuid
MeasSystemGuids
MeasSystemHWName
MeasSystemName
NumberOfSystems
RecipePath
RecipePathCustom
SaveImage
SetAutoFocus
SetRecipeModeToSpectrumAnalysis
SetRecipeSpectrumAnalysis
ShutdownRequestedHandler
ShuttingDown
SystemAutoSave
SystemAutoSaveBaseFileNameAndPath
SystemMeasure
SystemStartMonitoring
SystemStopMonitoring
SystemMeasurementCompletedEventHandler

New

```
Public Sub New(ByVal showFILMeasureGUI As Boolean) (deprecated)
Public Sub New(ByVal showFILMeasureGUI As Boolean, ByRef
theConstructorWarning As ConstructorWarning, ByRef warningMessage As
String)(deprecated)
```

```
Public Sub New(ByVal guiType AsGraphicalUserInterfaceType, ByVal
guiStartupState AsGraphicalUserInterfaceStartupState, ByVal
guiWindowStartupState AsFormWindowState, ByRef theConstructorWarning
AsConstructorWarning, ByRef warningMessage As String)
```

guiType – Determines if the FIRemote user interface and associated objects will be constructed at startup. If `guiType` is set to `None`, then the startup process is accelerated and memory usage is minimized. However, the user interface cannot be displayed at a later time if it is not constructed during the startup process. If `guiType` is set to `Standard`, then the normal graphical user interface is constructed during the startup process.

guiStartupState – Determines whether or not the FIRemote user interface is displayed at startup. Note that this argument has no effect if `guiType = None`. If `guiStartupState` is set to `Hidden`, then the user interface will not be visible on screen at the completion of the startup process. If the user interface was constructed at startup, it can be displayed at a later time by setting `GUIVisible = True`.

guiWindowStartupState – If `guiStartupState` is set to `Shown` then the FIRemote User interface window will appear on the screen at the completion of the startup process. Note that this argument has no effect if `guiType = None`. The state of the window is controlled by `guiWindowStartupState`. You can set the state to `Maximized`, `Normal`, or `Minimized`.

theConstructorWarning – Returned value of type ConstructorWarning (`None` or `StartupRecipeLoadFailure`).

warningMessage – Returned warning message string. Empty if None.

Remarks: You can suppress the display of the FIRemote user interface by setting `guiType` appropriately. Typically, you would show the user interface during software development in order to aid in the testing and debugging process. However, there are some applications where it is useful to be able to show and hide the user interface while the software is running. This is accomplished with the GUIVisible Property. The constructor warns if the startup recipe has problems loading and a default recipe is loaded instead.

Return to Index...

AcquireSpectrum

```
Public Sub AcquireSpectrum(ByRef spectrumWavelengths() As Single,
ByRef spectrumData() As Single)
Public Sub AcquireSpectrum(ByRef spectrumWavelengths() As Single,
ByRef spectrumData() As Single, ByVal flushPipeline As Boolean)
```

```
Public Sub AcquireSpectrum(ByRef spectrumWavelengths() As Single,
ByRef spectrumData() As Single, ByRef secondarySpecWavelengths() As
Single, ByRef secondarySpecData() As Single, ByRef
primarySpecIsReflectance As Boolean)
Public Sub AcquireSpectrum(ByRef spectrumWavelengths() As Single,
ByRef spectrumData() As Single, ByVal secondarySpecWavelengths() As
Single, ByRef secondarySpecData() As Single, ByVal flushPipeline As
Boolean, ByRef primarySpecIsReflectance As Boolean)
Public Sub AcquireSpectrum(ByVal measChannelGuid As Guid, ByRef
spectrumWavelengths() As Single, ByRef spectrumData() As Single)
Public Sub AcquireSpectrum(ByVal measChannelGuid As Guid, ByRef
spectrumWavelengths() As Single, ByRef spectrumData() As Single,
ByVal flushPipeline As Boolean)
Public Sub AcquireSpectrum(ByVal measChannelGuid As Guid, ByRef
primarySpecIsReflectance As BooleanByRef spectrumWavelengths() As
Single, ByRef spectrumData() As Single)
Public Sub AcquireSpectrum(ByVal measChannelGuid As Guid, ByRef
spectrumWavelengths() As Single, ByRef spectrumData() As Single,
ByVal flushPipeline As Boolean, ByRef primarySpecIsReflectance As
Boolean)
```

measChannelGuid – The GUID of the measurement channel for which a spectrum should be acquired. If multiple measurement channels exist, the measChannelGuid must be specified.

spectrumWavelengths – Array to contain wavelengths of measured spectum data points.

spectrumData – Measured reflectance or transmittance data points.

secondarySpecWavelengths – For systems that can simultaneously acquire reflectance and transmittance such as the F10-RT, array to contain wavelengths of measured secondary spectum data points. For all other systems, the returned argument is Nothing.

*secondarySpecDat*a – For systems that can simultaneously acquire reflectance and transmittance such as the F10-RT, array to contain the measured secondary spectum data points. For all other systems, the returned argument is Nothing.

flushPipeline – If set to True (default), any data currently being acquired is ignored and new data is acquired. If set to False, data currently being acquired is returned when acquisition is finished.

primarySpecIsReflectance – Indicates whether the primary spectrum data is Reflectance or Transmittance.

Remarks: It is important that you catch exceptions which may occur as a result of calling this method. See the example program for details.

Return to Index...

ActivateChannelDisplayTab

```
Public Sub ActivateChannelDisplayTab(ByVal measChannelGuid As Guid)
```

measChannelGuid – The GUID of the measurement channel of the corresponding Measurement Tab to be selected.

Remarks: After execution, the active tab in the FILMeasure GUI will be the specified Measurement Tab. This method should only be executed if multiple measurement channels exist.

Return to Index...

AnalyzeSpectrum

```
Public Function AnalyzeSpectrum(ByVal formatSummaryAsRtf As Boolean)
As FIMeasResults
Public Function AnalyzeSpectrum(ByVal formatSummaryAsRtf As Boolean,
ByVal addToHistory As Boolean) As FIMeasResults
Public Function AnalyzeSpectrum(ByVal measChannelGuid As Guid, ByVal
formatSummaryAsRtf As Boolean) As FIMeasResults
Public Function AnalyzeSpectrum(ByVal measChannelGuid As Guid, ByVal
formatSummaryAsRtf As Boolean, ByVal addToHistory As Boolean) As
FIMeasResults
```

measChannelGuid – The GUID of the measurement channel for which a spectrum should be analyzed and optionally added to history. If multiple measurement channels exist, the measChannelGuid must be specified.

formatSummaryAsRtf – The measurement summary returned by the .ResultsSummary property of the FIMeasResults object can be formatted as either plain text or as rtf. Formatting the summary as rtf is advantageous if it will be displayed to the user because then the full set of information including bold font and colors will be included.

addToHistory – Normally, each analysis result is added to the collection of results displayed in the History area of FILMeasure. In some cases, it may be advantageous to suppress the addition of a result to the History area. One example where this may be needed is the case where you wish to analyze a spectrum with two different measurement recipes and then add only the best result to the History.

Remarks: The most-recently acquired spectrum can be analyzed by calling this function.

Return to Index...

AuthenticateRefBac

```
Public Sub AuthenticateRefBac()
Public Sub AuthenticateRefBac(ByVal measChannelOrSystemGuid As Guid)
```

measChannelOrSystemGuid – The GUID of the measurement channel for which a baseline should be recovered. If multiple measurement channels exist, the measChannelGuid must be specified.

Remarks: Calling this method is equivalent to clicking Baseline… to display the Baseline dialog and then clicking Recover Last Baseline… to recover an old baseline spectrum. Note that there are many situations where an exception may be generated by calling this method. See the example program for details.

Return to Index...

BaselineAcquireBackgroundAfterRef

```
Public Sub BaselineAcquireBackgroundAfterRef()
Public Sub BaselineAcquireBackgroundAfterRef(ByVal
measChannelOrSystemGuid As Guid)
```

measChannelOrSystemGuid – The GUID of the measurement channel for which a background should be acquired. If multiple measurement channels exist, the measChannelGuid must be specified.

Remarks: This method provides a means to automate what is most commonly called step #3 of the Baseline procedure (step #4 for F10-RT, “Reflection Only” or “Reflection and Transmittance”. step #2 for F10RT“Transmittance Only”). As the name of this method implies, you must have previously acquired a reference spectrum.

Return to Index...

BaselineAcquireReference

```
Public Sub BaselineAcquireReference()
Public Sub BaselineAcquireReference(ByVal measChannelOrSystemGuid As
Guid)
```

measChannelOrSystemGuid – The GUID of the measurement channel for which a reference should be acquired. If multiple measurement channels exist, the measChannelGuid must be specified.

Remarks: This method provides a means to automate what is most commonly called step #2 of the Baseline procedure. Calling BaselineAcquireReference may generate an exception. There are a number of rules required for successful execution of this method. Unless the acquisition timing mode in the recipe is set to Manual, you must first call BaselineAcquireSpectrumFromSample. Alternatively, you could call the variant of this method which accepts a user-supplied reflectance value if you know the sample reflectance and wish to skip step #1.

Return to Index...

BaselineAcquireReferenceT

```
Public Sub BaselineAcquireReferenceT()
Public Sub BaselineAcquireReferenceT(ByVal measChannelOrSystemGuid As Guid)
```

measChannelOrSystemGuid – The GUID of the measurement channel for which a Transmittance reference should be acquired. If multiple measurement channels exist, the measChannelGuid must be specified.

Remarks: This method should only be called for the F10-RT. It provides a means to automate step #3 of the Baseline procedure for “Reflectance Only” or “Reflectance and Transmittance” modes, or step #1 of the Baseline procedure for “Transmittance Only” mode. No sample should be positioned on stage when called.

Return to Index...

BaselineAcquireReferenceUsingOldSampleReflectance

```
Public Sub BaselineAcquireReferenceUsingOldSampleReflectance()
Public Sub BaselineAcquireReferenceUsingOldSampleReflectance(ByVal measChannelOrSystemGuid As Guid)
```

measChannelOrSystemGuid – The GUID of the measurement channel for which a reference should be acquired. If multiple measurement channels exist, the measChannelGuid must be specified.

Remarks: This method provides a means to automate steps 1 and 2 of the Baseline procedure by assuming that the sample reflectance is unchanged from what it was the last time a baseline procedure was performed. Note that an exception may be thrown in some situations including the case where no prior baseline spectrum exists.

Return to Index...

BaselineAcquireSpectrumFromSample

```
Public Sub BaselineAcquireSpectrumFromSample()
```

```
Public Sub BaselineAcquireSpectrumFromSample(ByVal
measChannelOrSystemGuid As Guid)
```

measChannelOrSystemGuid – The GUID of the measurement channel for which a sample spectrum should be acquired. If multiple measurement channels exist, the measChannelGuid must be specified.

Remarks: This method provides a means to automate what is most commonly called step #1 of the Baseline procedure.

Return to Index...

BaselineCommit

```
Public Sub BaselineCommit()
Public Sub BaselineCommit(ByVal measChannelOrSystemGuid As Guid)
```

measChannelOrSystemGuid – The GUID of the measurement channel for which a new baseline should be commited. If multiple measurement channels exist, the measChannelGuid must be specified.

Remarks: Call this method to finalize the baseline procedure and commit the acquired spectra for use as a new baseline measurement.

Return to Index...

BaselineSetRefMat

```
Public Sub BaselineSetRefMat(ByVal theRefMat As String)
Public Sub BaselineSetRefMat(ByVal measChannelOrSystemGuid As Guid,
ByVal theRefMat As String)
```

measChannelOrSystemGuid – The GUID of the measurement channel for which the reference material should be assigned. If multiple measurement channels exist, the measChannelGuid must be specified.

theRefMat – The reference material as it would normally appear in the list of reference materials shown in the Baseline dialog.

Remarks: This method should be called prior to acquiring a reference spectrum

Return to Index...

BaselineShowDialog

```
Public Sub BaselineShowDialog(ByRef dialogCanceled As Boolean)
```

dialogCanceled – True if user has canceled the dialog.

Remarks: Calling this method will cause the baseline dialog box to be displayed on the screen

Return to Index...

GUIVisible

```
Public Property GUIVisible As Boolean
```

Remarks: Setting this property to `True` will cause the FIRemote user interface to be displayed on the screen. Note that this will normally cause the user interface window to appear on top of any other windows. If you wish to keep another window on top of the FIRemote user interface window, it may be necessary to call .BringToFront on that other window. If the FIRemote object is constructed with `guiType` set to `None` then an exception will be thrown if `GUIVisible` is set to `True`.

Return to Index...

HistoryDeleteAllResults

```
Public Sub HistoryDeleteAllResults()
```

Remarks: All measurement results in the History tab are deleted. If the FIRemote object has no user interface (`guiType = None` in `New()`), then there is no measurement history and this function has no effect.

Return to Index...

IntegrationCycles

```
Public Property IntegrationCycles() As Integer
Public Property IntegrationCycles(ByVal measChannelGuid As Guid) As
Integer
Public Property IntegrationCycles(ByVal specIndex As
SpectrometerIndex) As Integer
Public Property IntegrationCycles(ByVal measType As
SubsystemMeasType) As Integer
Public Property IntegrationCycles(ByVal measChannelGuid As Guid,
ByVal specIndex As SpectrometerIndex) As Integer
Public Property IntegrationCycles(ByVal measChannelGuid As Guid,
Byval measType As SubsystemMeasType, specIndex As SpectrometerIndex)
As Integer
```

measChannelGuid – The GUID of the measurement channel for which to set or get IntegrationCycles.

measType – Specifies which measurement subsystem (Reflectance, Transmittance, etc.).

specIndex – Specifies which spectrometer (Short or Long Wavelengths).

Remarks: The IntegrationCycles parameter is visible in the dialog Edit Recipe->Acquisition Settings tab in the Measurement Timing group under the Use manually set parameters Radio Button. Calling the variants of this property without a measChannelGuid is acceptable unless this does not unambiguously measurement channel. If there is only a single measurement channel, then it is unnecessary to call the variants of this property that include the measChannelGuid. If a measurement channel includes multiple measTypes, but the measType is not supplied, then the Reflectance measType will be addressed. If a MeasurementType includes multiple spectrometers, but the specIndex is not supplied, then the ShortWavelength spectrometer will be addressed.

Return to Index...

IntegrationTime_ms

```
Public Property IntegrationTime_ms() As Double
Public Property IntegrationTime_ms(ByVal measChannelGuid As Guid) As
Double
Public Property IntegrationTime_ms(ByVal specIndex As
SpectrometerIndex) As Double
Public Property IntegrationTime_ms(ByVal measType As
SubsystemMeasType) As Double
Public Property IntegrationTime_ms(ByVal measChannelGuid As Guid,
ByVal specIndex As SpectrometerIndex) As Double
Public Property IntegrationTime_ms(ByVal measChannelGuid As Guid,
Byval measType As SubsystemMeasType, specIndex As SpectrometerIndex)
As Double
```

measChannelGuid – The GUID of the measurement channel for which to set or get IntegrationTime_ms.

measType – Specifies which measurement subsystem (Reflectance, Transmittance, etc.).

specIndex – Specifies which spectrometer (Short or Long Wavelengths).

Remarks: The IntegrationTime_ms parameter is visible in the dialog Edit Recipe->Acquisition Settings tab in the Measurement Timing group under the Use manually set parameters Radio Button. See discussion in **IntegrationCycles** remarks regarding unsupplied arguments.

Return to Index...

MaximumSamplingTime

```
Public Property MaximumSamplingTime() As Double
Public Property MaximumSamplingTime(ByVal measChannelOrSystemGuid As
Guid) As Double
```

measChannelOrSystemGuid – The GUID of the measurement channel or measurement system for which to set or get MaximumSamplingTime.

Remarks: The MaximumSamplingTime parameter is visible in the dialog Edit Recipe->Acquisition Settings tab in the Measurement Timing group under the Use maximum sampling time Radio Button. See discussion in **IntegrationCycles** Remarks regarding unsupplied arguments.

Return to Index...

MeasurementTimingMode

```
Public Property MeasurementTimingMode() As MeasurementTimingModes
Public Property MeasurementTimingMode(ByVal measChannelOrSystemGuid
As Guid) As MeasurementTimingModes
```

measChannelOrSystemGuid – The GUID of the measurement channel or measurement system for which to set or get MeasurementTimingMode.

Remarks: The MeasurementTimingMode parameter is visible in the dialog Edit Recipe->Acquisition Settings tab in the Measurement Timing group. Available choices are Recommended, Maximum, and Manual. See discussion in **IntegrationCycles** Remarks regarding unsupplied arguments.

Return to Index...

MeasChannelGuid

```
Public ReadOnly Property MeasChannelGuid(ByVal measChannelIndex As
Integer) As Guid
Public ReadOnly Property MeasChannelGuid(ByVal measSystemGuid As
Guid, ByVal measChannelIndex As Integer) As Guid
```

measSystemGuid – The GUID of a measurement system.

measChannelIndex – The zero-based index of the measurement channel.

Remarks: Returns the GUID of the specified channel. Note this can be used to obtain the GUID used by all methods requiring a measChannelGUID. Also note that the relationship between measChannelIndex and MeasChannelGuid is not static. Future versions of FILMeasure will dynamically reorder this list in response to an equipment connect/disconnect event. You should not rely on the order of the channels in the list when attempting to perform a command on a certain channel.

Return to Index...

MeasChannelGuids

```
Public ReadOnly Property MeasChannelGuids() As Collection(Of Guid)
```

Remarks: Returns a collection of GUID for each measurement channel that exists.

Return to Index...

MeasChannelHWName

```
Public ReadOnly Property MeasChannelHWName(ByVal measChannelGuid As
Guid) As String
```

measChannelGuid – The GUID of the measurement channel for which the channel hardware name should be returned.

Remarks: The hardware name is composed of a system name string and a serial number string separated by a colon. Example: F20:09A006.

Return to Index...

MeasChannelHWSerialNumber

```
Public ReadOnly Property MeasChannelHWSerialNumber(ByVal
measChannelGuid As Guid) As String
Public ReadOnly Property MeasChannelHWSerialNumber(ByVal
measChannelGuid As Guid, ByVal spectrometerIndex As Integer) As
String
Public ReadOnly Property MeasChannelHWSerialNumber(ByVal
measChannelGuid As Guid, ByVal subsystemIndex As Integer, ByVal
spectrometerIndex As Integer) As String
```

measChannelGuid – The GUID of the measurement channel.

spectrometerIndex – The spectrometer index (to specify which of multiple spectrometers in the channel).

subsystemIndex – The subsystem index (to select among multiple subsystems in the channel).

Remarks: Returns the controller hardware serial number of the specified spectrometer. If the spectrometer or subsystem are not specified, the default index = 0 will be used. If there are multiple subsystems, a hyphen followed by the subsystem ID is appended to the hardware serial number. Example for single subsystem: 09A006. Example for multiple subsystem: 09A007-1

Return to Index...

MeasChannelName

```
Public ReadOnly Property MeasChannelName(ByVal measChannelGuid As Guid) As String
```

measChannelGuid – The GUID of the measurement channel for which the channel name should be returned.

Remarks: By default the channel name is the empty string.

Return to Index...

Measure

```
Public Function Measure(ByVal formatSummaryAsRtf As Boolean) As FIMeasResults
Public Function Measure(ByVal formatSummaryAsRtf As Boolean, ByVal addToHistory As Boolean) As FIMeasResults
Public Function Measure(ByVal measChannelGuid As Guid, ByVal formatSummaryAsRtf As Boolean) As FIMeasResults
Public Function Measure(ByVal measChannelGuid As Guid, ByVal formatSummaryAsRtf As Boolean, ByVal addToHistory As Boolean) As FIMeasResults
```

measChannelGuid – The GUID of the measurement channel for which a spectrum should be acquired and analyzed. If multiple measurement channels exist, the measChannelGuid must be specified.

formatSummaryAsRtf – The measurement summary returned by the .ResultsSummary property of the FIMeasResults object can be formatted as either plain text or as rtf. Formatting the summary as rtf is advantageous if it will be displayed to the user because then the full set of information including bold font and colors will be included.

addToHistory – Normally, each analysis result is added to the collection of results displayed in the History area of FILMeasure. In some cases, it may be advantageous to suppress the addition of a result to the History area. One example where this may be needed is the case where you wish to analyze a spectrum with two different measurement recipes and then add only the best result to the History.

Remarks: Calling this function is equivalent to clicking the Measure button in FILMeasure. Note that if the Measure button is disabled when you call this function, an exception will be thrown.

Return to Index...

NumberOfChannels

```
Public ReadOnly Property NumberOfChannels() As Integer
```

Remarks: Number of measurement channels that exist.

Return to Index...

NumberOfSpectrometers

```
Public ReadOnly Property NumberOfSpectrometers(ByVal measChannelGuid
As Guid) As Integer
Public ReadOnly Property NumberOfSpectrometers(ByVal measChannelGuid
As Guid, ByVal subsystemIndex As Integer) As Integer
```

measChannelGuid – The GUID of the measurement channel.

subsystemIndex – Thesubsystem index (to specify which of multiple subsystems in the channel).

Remarks: The number of spectrometers in the specified subsystem. If subsystemIndex is not provided, the default index = 0 will be used.

Return to Index...

NumberOfSubSystems

```
Public ReadOnly Property NumberOfSubSystems(ByVal measChannelGuid As
Guid) As Integer
```

measChannelGuid – The GUID of the measurement channel.

Remarks: The number of subsystems that compose the specified measurement channel.

Return to Index...

OpticsConfiguration

```
Public Property OpticsConfiguration() As OpticsConfigSystemTypes
[Get/Set]
Public Property OpticsConfiguration(ByVal measChannelGuid As Guid) As
OpticsConfigSystemTypes [Get/Set]
```

measChannelGuid – The GUID of the measurement channel for which the OpticsConfiguration is to be get/set. If multiple measurement channels exist, the measChannelGuid must be specified.

Remarks: If the OpticsConfigSystemTypes is not be compatible with the hardware an exception will be thrown.

Return to Index...

OpenSpectrum

```
Public Sub OpenSpectrum(ByVal filenameAndPath As String)
Public Sub OpenSpectrum(ByVal filenameAndPath As String, ByVal
measChannelGuid As Guid)
```

filenameAndPath – The full path and filename (including extension) of the spectrum. If the file cannot be opened, an exception will be thrown.

measChannelGuid – The GUID of the measurement channel for a spectrum is to be opened. If multiple measurement channels exist, the measChannelGuid must be specified.

Remarks: Unlike the behavior when a spectrum is opened in FILMeasure, the software will not automatically analyze the spectrum after opening it. If analysis of the spectrum is desired, then the Analyze function must be called.

Return to Index...

OpenSpectrumFromBuffer

```
Public Sub OpenSpectrumFromBuffer(ByVal theBuffer As Byte())
Public Sub OpenSpectrumFromBuffer(ByVal theBuffer As Byte(), ByVal
measChannelGuid As Guid)
```

theBuffer – Byte array containing spectrum. Data in this array must have been created by SaveSpectrumToBuffer method.

measChannelGuid – The GUID of the measurement channel for a spectrum which is to be opened. If multiple measurement channels exist, the measChannelGuid must be specified.

Remarks: See remarks after **SaveSpectrumToBuffer**

Return to Index...

OperatorID

```
Public Property OperatorID() As String
```

Remarks: Get or Set the Operator ID that is visible in the user interface.

Return to Index...

RecipeInfo

```
Public ReadOnly Property RecipeInfo() As FIRecipeInfo
Public ReadOnly Property RecipeInfo(measChannelOrSystemGuid) As Guid)
As FIRecipeInfo
```

measChannelGuid - The GUID of the measurement channel associated with a measurement recipe. If multiple measurement channels exist the measChannelOrSystemGuid must be specified.

Remarks: The FIRecipeInfo object contains an assortment of information about the recipe including information about the film stack, if appropriate..

Return to Index...

SampleID

```
Public Property SampleID() As String
```

Remarks: Get or Set the Sample ID that is visible in the user interface.

Return to Index...

SaveSpectrum

```
Public Sub SaveSpectrum(ByVal filenameAndPath As String)
Public Sub SaveSpectrum(ByVal filenameAndPath As String, ByVal
ifTextFormatWriteYDataAsPercent As Boolean)
```

filenameAndPath – The full path and filename (including extension) of the spectrum. The file format is determined by the extension. Valid file extensions are the same as the extensions listed in the Save Spectrum dialog windows accessed via the File->Save Measured Spectrum menu item in FILMeasure.

ifTextFormatWriteYDataAsPercent – If this version of SaveSpectrum without this parameter is called, the result is the same as calling the version with this parameter and setting its value to True. Thus, the only reason to call the second variant of the function is if the spectrum is being saved as a text format (either .txt or .csv) and the value of this parameter is set to False

Return to Index...

SaveSpectrumToBuffer

```
Public Sub SaveSpectrumToBuffer( As Byte()
```

Remarks: This function allows the caller to rapidly retrieve the complete set of data contained in a spectrum, thereby avoiding the time that would be consumed if the spectrum were saved to a file. This may be useful in cases where the caller wishes to rapidly acquire a number of spectra and store them up for later saving to file or analysis. The contents of the buffer should be considered opaque. Any changes to the data in the buffer may render the data unusable and/or may not be compatible with future versions of FILMeasure.

Return to Index...

SetAnalysisWavelengthRange

```
Public Sub SetRecipeSetAnalysisWavelengthRange(ByVal
measChannelGuidmeasChannelOrSystemGuid As Guid, ByVal
startWavelength_nm As Single, ByVal endWavelength_nm As Single)
Public Sub SetAnalysisWavelengthRange(ByVal startWavelength_nm As
Single, ByVal endWavelength_nm As Single)
```

measChannelOrSystemGuid – The GUID of the measurement channel or system. If multiple measurement channels exist, the measChannelOrSystemGuid must be specified (since each measurement channel has its own active recipe).

startWavelength_nm – Starting wavelength for spectrum analysis.

endWavelength_nm – Ending wavelength for spectrum analysis.

Remarks: Setting the wavelength range will also switch the recipe into using a specified wavelength range instead of using the entire wavelength range visible on the spectrum graph.

Return to Index...

SetMaterial

```
Public Sub SetMaterial(ByVal measChannelOrSystemGuid As Guid, ByVal
layerNumber As Integer, ByVal materialName As String)
Public Sub SetMaterial(ByVal layerNumber As Integer, ByVal
materialName As String)
```

measChannelOrSystemGuid – The GUID of the measurement channel or system. If multiple measurement channels exist, the measChannelOrSystemGuid must be specified (since each measurement channel has its own active recipe).

layerNumber – The layer number for which to modify the material.

materialName – String containing name of the material.

Remarks: You can alter the material of a layer in the current recipe using the method. An exception will be thrown if the layer number is out-of-bounds or the material does not exist.

Return to Index...

SetN

```
Public Sub SetN(ByVal measChannelOrSystemGuid As Guid, ByVal
layerNumber As Integer, ByVal refractiveIndex As Single)
Public Sub SetN(ByVal layerNumber As Integer, ByVal refractiveIndex
As Single)
```

measChannelOrSystemGuid – The GUID of the measurement channel or system. If multiple measurement channels exist, the measChannelOrSystemGuid must be specified (since each measurement channel has its own active recipe).

layerNumber – The layer number for which to set the refractive index.

refractiveIndex – New refractive index value for layer.

Remarks: You can alter the material of a layer in the current recipe using the method. An exception will be thrown if the layer number or refractive index are out-of-bounds.

Return to Index...

SetRecipe

```
Public Sub SetRecipe(ByVal measChannelOrSystemGuid As Guid, ByVal
recipeName As String)
Public Sub SetRecipe(ByVal recipeName As String)
```

measChannelOrSystemGuid – The GUID of the measurement channel or system. If multiple measurement channels exist, the measChannelOrSystemGuid must be specified (since each measurement channel has its own active recipe).

recipeName – The recipe name to be selected must exist in the list of recipes. Note that you can select recipes contained in subfolders by specifying the relative path, using the backslash character '\' to separate folder names as is conventional in path naming.

Remarks: The current spectrum will not be automatically reanalyzed when a new recipe is selected. To reanalyze the spectrum you must call **AnalyzeSpectrum**.

Return to Index...

SetRoughness

```
Public Sub SetRoughness(ByVal measChannelOrSystemGuid As Guid, ByVal
layerNumber As Integer, ByVal roughness_nm As Single)
Public Sub SetRoughness(ByVal layerNumber As Integer, ByVal
roughness_nm As Single)
```

measChannelOrSystemGuid – The GUID of the measurement channel or system. If multiple measurement channels exist, the measChannelOrSystemGuid must be specified (since each measurement channel has its own active recipe).

layerNumber – The layer number for which to modify the roughness.

roughness_nm – Roughness value which will appear in the Layers tab of the recipe for the specified layer. Units are nanometers.

Remarks: You can alter the roughness of a layer in the current recipe using the method. An exception will be thrown if the layer number or roughness are out-of-bounds.

Return to Index...

SetThickness

```
Public Sub SetThickness(ByVal measChannelOrSystemGuid As Guid, ByVal
layerNumber As Integer , ByVal thickness_nm As Single)
Public Sub SetThickness(ByVal layerNumber As Integer, ByVal
thickness_nm As Single)
```

measChannelOrSystemGuid – The GUID of the measurement channel or system. If multiple measurement channels exist, the measChannelOrSystemGuid must be specified (since each measurement channel has its own active recipe).

layerNumber – The layer number for which to modify the thickness.

thickness_nm – Thickness value which will appear in the Layers tab of the recipe for the specified layer. Units are nanometers.

Remarks: You can alter the thickness of a layer in the current recipe using the method. An exception will be thrown if the layer number or thickness is out-of-bounds.

Return to Index...

SetThicknessRange

```
Public Sub SetThicknessRange(ByVal layerNumber As Integer, ByVal
layerNumber As Integer , ByVal thicknessRange As Double)
Public Sub SetThicknessRange(ByVal measChannelOrSystemGuid AsGuid,
ByVal layerNumber As Integer, ByVal thicknessRangePercent As Double)
```

measChannelOrSystemGuid – The GUID of the measurement channel or system. If multiple measurement channels exist, the measChannelOrSystemGuid must be specified (since each measurement channel has its own active recipe).

layerNumber – The layer number for which to modify the thickness.

thicknessRangePercent – Thickness range which will appear in the Layers tab of the recipe for the specified layer. Note that the value accepted by this function is always a percentage, regardless of the setting in the recipe editor which shows the range as either a percentage or a thickness. Minimum value is zero. There is no maximum value.

Remarks: You can alter the thickness range setting for a layer in the current recipe using the method. An exception will be thrown if the layer number or thickness range are out-of-bounds.

Return to Index...

SpectrometerDiagnostics

```
Public ReadOnly Property SpectrometerDiagnostics()
AsFISpecDiagnosticsInfo()
Public ReadOnly Property SpectrometerDiagnostics(ByVal
measChannelGuid AsGuid)As FISpecDiagnosticsInfo()
Public ReadOnly Property SpectrometerDiagnostics(ByVal
measChannelGuid As Guid, ByVal subsystemIndex As Integer) As
FISpecDiagnosticsInfo()
```

measChannelGuid – The GUID of the measurement channel.

subsystemIndex – The subsystem index (to select among multiple subsystems in the channel).

Remarks: Returns an array of objects containing information about the spectrometers. Information includes integration time, peak raw signal intensities of reference and background spectra, etc.

Return to Index.

SpectrometerType

```
Public Property SpectrometerType() As SpectrometerTypes
```

```
Public Property SpectrometerType(ByVal measChannelOrSystemGuid
AsGuid) As SpectrometerTypes
Public Property SpectrometerType(ByVal measType AsSubsystemMeasType)
As SpectrometerTypes
Public Property SpectrometerType(ByVal measChannelOrSystemGuid As
Guid, ByVal measType As SubsystemMeasType) As SpectrometerTypes
```

measChannelOrSystemGuid – The GUID of the measurement channel or measurement system for which to set or get SpectrometerType.

measType – Specifies which measurement subsystem (Reflectance, Transmittance, etc.).

Remarks: The SpectrometerType parameter is visible in the dialog Edit Recipe->Acquisition Settings tab in the Measurement Timing group. Available choices are All, ShortwavelengthsOnly, LongWavelengthsOnly, and ExtendedWavelengths. The All and ExtendedWavelengths values select the corresponding options in the Acquisition Settings in the Recipe, but these two options result in the same behavior. See discussion in **IntegrationCycles** Remarks regarding unsupplied arguments.

Return to Index.

ActivateMonitorDisplayTab

```
Public Sub ActivateMonitorDisplayTab()
Public Sub ActivateMonitorDisplayTab(ByVal measSystemGuid As Guid)
```

measSystemGuid – The GUID of the measurement System. If multiple measurement channels exist, the measChannelGuid must be specified (since each measurement channel has its own active recipe).

Remarks: This function is provided to permit synchronization of the FILMeasure user interface with the user interface of a client program.

Return to Index...

BaselineExistsAndIsAuthenticated

```
Public Function BaselineExistsAndIsAuthenticated(ByRef
authenticationIsPossible As Boolean) As Boolean
Public Function BaselineExistsAndIsAuthenticated
(measChannelOrSystemGuid As Guid, ByRef authenticationIsPossible As
Boolean) As Boolean
```

authenticationIsPossible – If baseline is not authenticated, it is possible to authenticate it when `authenticationIsPossible = True`.

Remarks: In some cases, it is desirable to check whether the baseline is already authenticated and if not whether it is possible to authenticate it. The client software has the option of automatically authenticating the baseline if that is desirable.

Return to Index...

BaselineHardwareAdjustInhibit

```
Public Property BaselineHardwareAdjustInhibit() As Boolean
Public Function BaselineHardwareAdjustInhibit(measChannelOrSystemGuid
As Guid) As Boolean
```

measChannelOrSystemGuid – The GUID of the measurement channel or system. If multiple measurement channels exist, the measChannelOrSystemGuid must be specified.

Remarks: Some measurement systems include hardware that must be periodically adjusted. The adjustment procedure happens automatically during the baseline process unless it is inhibited. The adjustment process may take several minutes to complete. This property allows the client software to control when the adjustment process is accomplished so that it can be performed at a convenient time. See **GetBaselineHardwareAdjustStatus()** for additional information.

Return to Index...

GeneralPurposeIOIsSupported

```
Public ReadOnly Property GeneralPurposeIOIsSupport() As Boolean
```

Remarks: Primary purpose of this property is to permit the example program to hide or show the General Purpose IO controls. This property will be True for equipment that has the General Purpose IO hardware.

Return to Index...

GeneralPurposeIOReadValue

```
Public Function GeneralPurposeIOReadValue() As Byte
```

Remarks: Returns a single byte containing the current values of the general purpose IO logic input signals. Bit value of 1 means that the input voltage is in the logic high state. These input signals are only available on certain types of equipment.

Return to Index...

GeneralPurposeIOSetValue

```
Public Sub GeneralPurposeIOSetValue(newValue As Byte)
```

newValue - New values for the output bits.

Remarks: Sets the output state of the general purpose IO logic output signals. Setting a bit to 1 will set the corresponding output to the logic high state. These output signals are only available on certain types of equipment.

Return to Index...

GetBaselineHardwareAdjustStatus

```
Public Function GetBaselineHardwareAdjustStatus() As
BaselineHardwareAdjustStatus
Public Function GetBaselineHardwareAdjustStatus
(measChannelOrSystemGuid As Guid) As BaselineHardwareAdjustStatus
```

measChannelOrSystemGuid – The GUID of the measurement channel or system. If multiple measurement channels exist, the measChannelOrSystemGuid must be specified.

Remarks: See **BaselineHardwareAdjustInhibit()** for additional information. Members of the GetBaselineHardwareAdjustStatus Class include:

NeedsPeriodicAdjustment (Boolean) – True if the hardware is of a type that will need to be adjusted periodically.

AdjustmentRequiredNow (Boolean) – True if adjustment is required before Baseline can be completed; if adjustment is required, but inhibited, then Baseline methods will throw an exception when called.

AdjustmentRecommendedNow (Boolean) – True if adjustment is recommended.

AdjustmentPeriodHours (Single) – time period in hours between recommended adjustments.

TimeToNextRecommendedAdjustmentHours (Single) – time until the next recommended adjustment; a negative number represents the amount of time that has passed since an adjustment has become recommended.

Return to Index...

MaterialPath

```
Public ReadOnly Property MaterialPath() As String
```

Remarks: Reports the currently effective path for Material Files.

Return to Index...

MaterialPathCustom

```
Public Property MaterialPathCustom() As String
```

Remarks: This property allows a client to set/get a custom path for Material Files. If this property is set to an empty value or an invalid path, then the effective path will be the default path.

Return to Index...

MeasSystemGuid

```
Public ReadOnly Property MeasSystemGuid(ByVal measSystemIndex As Integer) As Guid
```

measSystemIndex – The zero-based index of the system.

Remarks: Returns the GUID of the specified system. Note this can be used to obtain the GUID used by all methods requiring a measSystem GUID. Also note that the relationship between measSystemIndex and MeasSystemGuid is not static. Future versions of FILMeasure will dynamically reorder this list in response to an equipment connect/disconnect event. You should not rely on the order of the systems in the list when attempting to perform a command on a certain system.

Return to Index...

MeasSystemGuids

```
Public ReadOnly Property MeasSystemGuids() As Collection(Of Guid)
```

Remarks: Returns a collection of GUIDs containing one item for each measurement system that exists.

Return to Index...

MeasSystemHWName

```
Public ReadOnly Property MeasSystemHWName(ByVal systemGuid As Guid) As String
```

systemGuid – The GUID of a measurement system.

Remarks: The hardware name is composed of a system name string and a serial number string separated by a colon. Example: F37:09A006

Return to Index...

MeasSystemName

```
Public ReadOnly Property MeasSystemName(ByVal systemGuid As Guid) As String
```

systemGuid – The GUID of a measurement system.

Remarks: The default system name is the empty string.

Return to Index...

NumberOfSystems

```
Public ReadOnly Property NumberOfSystems() As Integer
```

Remarks: Number of measurement systems that exist.

Return to Index...

RecipePath

```
Public ReadOnly Property RecipePath() As String
```

Remarks: Reports the currently effective path for Measurement Recipes.

Return to Index...

RecipePathCustom

```
Public Property RecipePathCustom() As String
```

Remarks: This property allows a client to set/get a custom path for Measurement Recipes. If this property is set to an empty value or an invalid path, then the effective path will be the default path.

Return to Index...

SaveImage

```
Public Sub SaveImage(ByVal filenameAndPath As String)
```

Remarks: Save image from most-recently acquired spectrum as a separate file.

Return to Index...

SetAutoFocus

```
Public Property SetAutoFocus As Boolean
```

Remarks: Get or set the state of the Use AutoFocus checkbox in the Acquisition Settings tab of the current recipe.

Return to Index...

SetRecipeModeToSpectrumAnalysis

```
Public Sub SetRecipeModeToSpectrumAnalysis()
Public Sub SetRecipeModeToSpectrumAnalysis(ByVal
measChannelOrSystemGuid As Guid)
Public Sub SetRecipeModeToThickness()
Public Sub SetRecipeModeToThickness(ByVal measChannelOrSystemGuid As
Guid)
```

measChannelOrSystemGuid – The GUID of one of the measurement channels in a muliple channel system or the GUID of the system. If multiple measurement channels exist, the measChannelGuid must be specified.

Remarks: If Spectrum Analysis recipes are supported by the hardware or add-on licenses, then the Measure tab in the user interface will display a tab control that allows the user to switch modes. The tab associated with the Spectrum Analysis recipe mode is labeled “AR” or “Spectrum”. The other tab is labeled “HC” or “Thickness” or “Thickness, n, k, r”. Calling SetRecipeModeToSpectrumAnalysis or SetRecipeModeToThickness is the same as clicking on the Spectrum or Thickness tab in the user interface.

Return to Index...

ShutdownRequestedHandler

```
Public Delegate SubShutdownRequestedHandler(ByVal e As
Windows.Forms.FormClosingEventArgs)
Public Event ShutdownRequested As ShutdownRequestedHandler
```

e – The standard argument type for a form about to close.

Remarks: The client software can prohibit closing of the software by setting `e.Cancel` to `True`. Alternatively, the client software may wish to perform certain actions prior to allowing the FIRemote class to shut down.

Return to Index...

SetRecipeSpectrumAnalysis

```
Public Sub SetRecipeSpectrumAnalysis(ByVal recipeName As String)
Public Sub SetRecipeSpectrumAnalysis(ByVal measChannelOrSystemGuid As
Guid, ByVal  recipeName As String)
Public Sub SetRecipeThickness()
Public Sub SetRecipeThickness(ByVal measChannelOrSystemGuid As Guid,
ByVal  recipeName As String)
```

measChannelOrSystemGuid – The GUID of one of the measurement channels in a multiple channel system or the GUID of the system. If multiple measurement channels exist, the measChannelGuid must be specified.

recipeName – The recipe name to be selected must exist in the list of recipes. Note that you can select recipes contained in subfolders by specifying the relative path, using the backslash character '\' to separate folder names as is conventional in path naming.

Remarks: Same as **SetRecipeModeToSpectrumAnalysis** or **SetRecipeModeToThickness**, but caller specifies name of recipe to load during recipe mode switch.

Return to Index...

ShuttingDown

```
Public Event ShuttingDown()
```

Remarks: Client software can add an event handler for this event in order to facilitate performing any necessary shut-down actions. This event is primarily used in order to permit client software to react to closing of the FIRemote user interface.

Return to Index...

SystemAutoSave

```
Public Property SystemAutoSave() As Boolean
Public Property SystemAutoSave(ByVal systemGuid As Guid) As Boolean
```

systemGuid – The GUID of a measurement system.

Remarks: You can set or get the state of the AutoSave setting using this property. Note that setting a new state for SystemAutoSave will not affect an on-going monitor process, but will take effect the next time a new set of data starts being collected.

Return to Index...

SystemAutoSaveBaseFileNameAndPath

```
Public Property SystemAutoSaveBaseFileNameAndPath() As String
Public Property SystemAutoSaveBaseFileNameAndPath(ByVal systemGuid As
Guid) As String
```

systemGuid – The GUID of a measurement system.

Remarks: You can set or get the base filename and path for the AutoSave file using this property. The full filename is generated by appending a date and time stamp to the base filename. Note that setting a new filename will not affect an on-going monitor process, but will take effect the next time a new set of data starts being collected.

Return to Index...

SystemMeasure

```
Public Function SystemMeasure() As FIMeasResultValuesOnly
Public Function SystemMeasure(ByVal
deleteAccumulatedDataBeforeMeasure As Boolean) As
FIMeasResultValuesOnly
Public Function SystemMeasure(ByVal systemGuid As Guid) As
FIMeasResultValuesOnly
Public Function SystemMeasure(ByVal systemGuid As Guid, ByVal
deleteAccumulatedDataBeforeMeasure As Boolean) As
FIMeasResultValuesOnly
```

deleteAccumulatedDataBeforeMeasure – There is no analog for this parameter in the user interface. Setting this parameter to True causes any existing accumulated data points to be deleted before the measurement is performed.

systemGuid – The GUID of a measurement system.

Remarks: Acquire and measure on all channels in a system. Measurement is performed simultaneously on all channels. Measured results are contained in an FIMeasResultValuesOnly object.

Return to Index...

SystemMeasurementCompletedEventHandler

```
Public Delegate Sub SystemMeasurementCompletedEventHandler(ByVal
systemGuid As Guid, ByVal theResults As FIMeasResultValuesOnly)
Public Event SystemMeasurementCompleted As
SystemMeasurementCompletedEventHandler
```

systemGuid – The GUID of a measurement system.

Remarks: This event is raised upon completion of a SystemMeasure task and after each measurement during a SystemMonitor process. You should avoid performing lengthy operations (such as writing data to a file) in the handler for this event because the monitor process will be halted until the handler function exits. If execution of a lengthy operation is needed, you should invoke a different thread to perform the operation. If the event is raised as a result of completing a SystemMeasure task, then you will need to exercise care to avoid creating a threading deadlock. A deadlock will occur if you invoke the thread which called SystemMeasure in the handler for this event because that thread will not return from SystemMeasure until all the event handlers have finished executing. See the FIRemoteTest example program to see one way to avoid the deadlock problem.

Return to Index...

SystemStartMonitoring

```
Public Sub SystemStartMonitoring()
Public Sub SystemStartMonitoring(ByVal systemGuid As Guid)
```

systemGuid – The GUID of a measurement system.

Remarks: This function starts a monitoring process

Return to Index...

SystemStopMonitoring

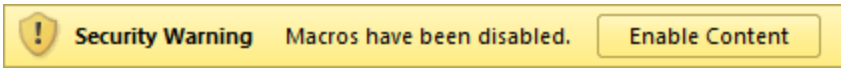
```
Public Sub SystemStopMonitoring()
Public Sub SystemStopMonitoring(ByVal systemGuid As Guid)
```

Remarks: This function stops a monitoring process.

Return to Index...

FIRemoteCOM dll

The dll is automatically included whenever FILMeasureis installed. Since the dll references the corresponding .exe file it is located inside the FILMeasure folder in the Program Files (or Program Files (x86) on 64-bit systems) path. The best way to learn how to use the dll is to examine the example Excel workbook that is included with the software. The workbook has the same name as the dll, but has the extension .xlsm and is located in the same folder as the dll. When you open the . xlsm file for the first time, you will see a warning banner as shown below.



You must click the Enable Content button in order to allow the code associated with the buttons on the worksheets to run. If you do not see the security warning when you open the spreadsheet, then you may need to modify settings in Excel using the Trust Center. The exact procedure varies, depending on which version of Excel you are using.

To see the example source code, click the Developer tab on the ribbon. Then click Visual Basic and double-click the ThisWorkbook object.

Reference for FIRemote Class in FIRemoteCOM dll:

Methods, Properties, and Events:

```
Public Sub AcquireSpectrum(ByRef spectrumWavelengths() As Single,
ByRef spectrumData() As Single)
```

arguments – Same as FIRemote Class

Remarks: See remarks for **AcquireSpectrum()** in FIRemote Class.

```
Public Sub AuthenticateRefBac()
```

arguments – Same as FIRemote Class

Remarks: See remarks for **AuthenticateRefBac()** in FIRemote Class.

```
Public Sub BaselineAcquireBackgroundAfterRef()
```

arguments – Same as FIRemote Class

Remarks: See remarks for **BaselineAcquireBackgroundAfterRef()** in FIRemote Class.

```
Public Sub BaselineAcquireReference()
```

arguments – Same as FIRemote Class

Remarks: See remarks for **BaselineAcquireReference()** in FIRemote Class.

```
Public Sub BaselineAcquireSpectrumFromSample()
```
arguments – Same as FIRemote Class

Remarks: See remarks for **BaselineAcquireSpectrumFromSample()** in FIRemote Class.

```
Public Sub BaselineCommit()
```
arguments – Same as FIRemote Class

Remarks: See remarks for **BaselineCommit()** in FIRemote Class.

```
Public Sub BaselineSetRefMat()
```
arguments – Same as FIRemote Class

Remarks: See remarks for **BaselineSetRefMat()** in FIRemote Class.

```
Public Sub Measure(ByVal formatSummaryAsRtf As Boolean, ByRef
measLayerThickness() As Single, ByRef measGOF As Single, ByRef
summary As String)
```

formatSummaryAsRtf – The measurement summary can be formatted as either plain text or as rtf. Formatting the summary as rtf is advantageous if it will be displayed to the user because then the full set of information including bold font and colors will be included.

measLayerThickness – Array containing thickness of all layers. Thickness of Layer 1 is in index 1 in the array.

measGOF – Measured Goodness Of Fit parameter.

Summary – Same as text which appears in Results area in Measure tab.
Remarks: See remarks for **Measure()** in FIRemote Class.

```
Public Sub SetRecipe(recipeName As String)
```

Arguments and Remarks – See documentation for **SetRecipe** in FIRemote Class.

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